OECD

OECD ECONOMIC SURVEYS

1998

SPECIAL FEATURE
Public enterprise reform

GREECE

© OECD, 1999.
© Software: 1987-1996, Acrobat is a trademark of ADOBE.
All rights reserved. OECD grants you the right to use one copy of this Program for your personal use only. Unauthorised reproduction, lending, hiring, transmission or distribution of any data or software is prohibited. You must treat the Program and associated materials and any elements thereof like any other copyrighted material.

All requests should be made to:

Head of Publications Service,
OECD Publications Service,
2, rue André-Pascal, 75775 Paris
Cedex 16, France.

OECD ECONOMIC SURVEYS

1997-1998

GREECE

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

Pursuant to Article 1 of the Convention signed in Paris on 14th December 1960, and which came into force on 30th September 1961, the Organisation for Economic Co-operation and Development (OECD) shall promote policies designed:

- to achieve the highest sustainable economic growth and employment and a rising standard of living in Member countries, while maintaining financial stability, and thus to contribute to the development of the world economy;
- to contribute to sound economic expansion in Member as well as non-member countries in the process of economic development; and
- to contribute to the expansion of world trade on a multilateral, non-discriminatory basis in accordance with international obligations.

The original Member countries of the OECD are Austria, Belgium, Canada, Denmark, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, the United Kingdom and the United States. The following countries became Members subsequently through accession at the dates indicated hereafter: Japan (28th April 1964), Finland (28th January 1969), Australia (7th June 1971), New Zealand (29th May 1973), Mexico (18th May 1994), the Czech Republic (21st December 1995), Hungary (7th May 1996), Poland (22nd November 1996) and Korea (12th December 1996). The Commission of the European Communities takes part in the work of the OECD (Article 13 of the OECD Convention).

Publié également en français.

© OECD 1998
Permission to reproduce a portion of this work for non-commercial purposes or classroom use should be obtained through the Centre français d'exploitation du droit de copie (CFC), 20, rue des Grands-Augustins, 75006 Paris, France, Tel. (33-1) 44 07 47 70, Fax (33-1) 46 34 67 19, for every country except the United States. In the United States permission should be obtained through the Copyright Clearance Center, Customer Service, (508)750-8400, 222 Rosewood Drive, Danvers, MA 01923 USA, or CCC Online: http://www.copyright.com/. All other applications for permission to reproduce or translate all or part of this book should be made to OECD Publications, 2, rue André-Pascal, 75775 Paris Cedex 16, France.

Table of contents

Assessment and recommendations 1

I. Recent trends and short-term prospects 17

Buoyant in 1997, domestic demand has slowed in 1998 18
Output growth has been more balanced 23
Employment has been slow to respond to the recovery 23
Wages have moderated but disinflation was slowed by the exchange rate adjustment 25
Trade performance and the balance of payments 27
Prospects for 1999 and 2000 37

II. Macroeconomic policies 41

Monetary policy: meeting the Maastricht inflation criterion 42
Fiscal policy: continued adjustment is necessary to improve the policy mix further 48
An evaluation of the anti-inflationary policy options 61
The medium-term fiscal strategy: an assessment 63

III. Implementing structural reform: a review of progress 67

Debate on broad-based labour market reform 69
A small reform of the social security system 78
Health reform: progress has been slow in introducing the 1997 health care bill 81
Financial sector reform: restructuring and consolidation propel the sector 85
A strategy for accelerating and improving the process of structural reform 91

IV. Reforming the public enterprise sector 93
Key features of the public enterprise sector 94
Reform of specific sectors 112
Quantification of the effects of liberalisation 149
Assessment 152

Notes 160

Glossary of acronyms 176

Bibliography 177

Annexes
I. The Aiginger Index and revealed comparative advantages 178
II. Calendar of main economic events 181

Boxes

1. The structure of the current account and the level of the exchange rate 32
2. Structural reform initiatives of the 1997 and 1998 Budgets 53

Tables

Text

1. Public investment: key features 21
2. Labour market developments 24
3. Prices and wages 26
4. Balance of payments 28
5. Trade indicators 31
6. External debt and debt service 37
7. The short-term outlook 38
8. General government budget adjustment between 1996 and 1997 49
9. Public sector debt 51

10. General government accounts 56
11. The 1998-2001 Convergence Programme 58
12. Budget plans and outcomes 60
13. Sensitivity analysis on debt dynamics 64
14. Implementing the *OECD Jobs Strategy* – an overview 74
15. A cross-country comparison of commercial bank indicators 89
16. Main public enterprises by sector 95
17. Public enterprise accounts 97
18. Debt assumptions from public enterprises 99
19. Weight of public enterprises in the CPI basket 103
20. Average compensation per employee 106
21. Main privatisations of public enterprises 111
22. An international comparison of telecommunications sector indicators 125
23. A comparison of telephone charges 126
24. An international comparison of airline performance indicators 136
25. An international comparison of the rail transport sector 141
26. An international comparison of postal service indicators 146
27. Assumptions and effects of sectoral deregulation 150

Annex

A1. Detailed structure of Greek exports, 1995 179

Figures

Text

1. Conjunctural and leading indicators 19
2. Contributions to real GDP growth 20
3. Prices and wages 30
4. Revealed comparative advantages and quality *versus* price competition 33
5. Competitiveness and export performance 34
6. The exchange rate and private sector credit growth 45
7. Interest rates, spreads, and yield curves 47
8. Fighting inflation: nominal anchors and macroeconomic policies 62

9. GDP per capita developments in Greece, Ireland, Portugal and Spain 67
10. Active labour market policies and the labour force 72
11. Public enterprise prices and underlying inflation 102
12. Public enterprise employment 104
13. Institutional structure of the oil sector 114
14. Institutional structure of the natural gas sector 117
15. Electricity sector: prices and performance 120
16. Tariff rebalancing in telecommunications 127
17. Water: prices and performance 131
18. Air transport: prices and performance 134
19. Rail transport: prices and performance 140
20. Urban transport: prices and performance 144
21. Postal services: prices and performance 148

BASIC STATISTICS OF GREECE (1997)

THE LAND

Area (sq. km)	131 957
Cultivated area (sq. km)	39 290

Major urban areas, 1991 census (thousand inhabitants)	
Greater Athens (including Piraeus)	3 073
Greater Thessaloniki	978

THE PEOPLE

Population (thousands)	10 476
Number of inhabitants per sq. km	79.4
Net natural increase (thousands), 1995	1.0

Total employment (thousands)	3 854.1
By sector (per cent):	
Agriculture	19.8
Industry and construction	21.0
Other activities, mainly services	59.2

PRODUCTION

Gross national product, GNP (billion Dr)	32 941.7
GNP per head (US$)	11 514.1

Gross fixed investment	
Per cent of GNP	19.9
Per head (US$)	2 294.2

THE GOVERNMENT

General government (per cent of GNP)	
Current expenditures	40.4
Current revenue	38.6

Composition of Parliament (number of seats):	300
Panhellenic Socialist Movement (PASOK)	159
New Democraty (ND)	102
Communist Party (KKE)	11
Others	28

Last general elections: October 1996

THE CURRENCY

Monetary unit: Drachma

Currency units per US$, average of daily figures:	
Year 1997	273.1
October 1998	281.7

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Greece by the Economic and Development Review Committee on 30 October 1998.

•

After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 23th November 1998.

•

The previous Survey of Greece was issued in August 1997.

Assessment and recommendations

The Greek economy makes solid progress towards the objective of joining EMU by 2001, but considerable challenges remain

Economic conditions have improved steadily since the turn of the decade. Inflation has been brought down from near 20 per cent to around 5 per cent in 1998, and output growth has gradually increased past the EU average in recent years. The improved economic performance is due, in large measure, to the steady progress in correcting the large fiscal imbalance; the deficit has been reduced from over 15 per cent of GDP in 1990 to a level which is expected to be below the Maastricht ceiling in 1998. Following the considerable nominal convergence, Greece intends to join the European single currency area at the beginning of 2001, which would require qualifying by mid-2000. The challenges for macroeconomic policies remain considerable since some of the Maastricht criteria have not been satisfied yet – most notably on inflation but also on a sufficiently diminishing debt to GDP ratio. The Government acknowledges that meeting them will require an additional policy effort, and commitments are contained in a new convergence programme, including a further tightening of fiscal policy and a timetable for implementing a new structural reform programme. However, given the structural problems identified in this *Survey*, attaining the ultimate economic objective of convergence of per capita income towards the EU average will need to be underpinned by an even bolder and wider-ranging structural reform effort. The large transfers from EU structural funds provide an opportunity for structural change, but also pose policy challenges regarding their inflationary impact.

An exchange rate adjustment in March 1998...

The macroeconomic environment changed from March 1998. The drachma entered the ERM at a central parity that represented a 12.3 per cent depreciation of the currency. The adjustment was necessary after the prolonged use of a stable exchange rate to support inflation reduction. The associated loss of competitiveness had slowed exports to the main EU market, and weakened the current account. The credibility of the previous exchange rate regime, combined with a wide interest-rate differential, also had induced some foreign borrowing by the public and private sectors, which raised certain concerns in financial markets. This vulnerability was exacerbated by the economic developments in Asia, which started to put heavy pressure on the exchange rate in October 1997, and finally resulted in the exchange rate adjustment. The market reaction to ERM entry was very positive, and Greek financial markets, though dented by the more recent turbulence in international markets, have remained up for the year as a whole.

... temporarily reversed the downward trend in inflation

The devaluation together with moderate wage settlements has slowed domestic demand and, as a result, output growth is likely to be slowing somewhat, though it will likely be about 3 per cent in 1998 compared with 3.2 per cent in 1997. Private consumption is decelerating markedly, as disposable income slows and the devaluation reduces households' purchasing power. Investment has remained the main source of demand growth, led by a booming public-sector investment programme. Business investment is also strong, despite higher interest rates and a heavier tax burden, as firms upgrade their capital equipment and capacity utilisation is high. Sharp wage increases have threatened the disinflation process during the past several years, but should be considerably lower in 1998. Civil service cost-of-living adjustments were set to increase by less than expected inflation, following a surge in civil service wages of about 30 per cent during 1996-97, and a Pact between

the social partners on labour market reform has helped, *inter alia*, to secure a relatively moderate two-year accord in the private sector. Despite slowing demand and labour costs, the devaluation led to higher inflation. The disinflation gains of the preceding year were reversed, and the underlying 12-month inflation rate rose from a low of 4.4 per cent in February, to a peak of 6.1 per cent in July 1998 before falling back to 5.3 per cent in November (the same rate as one year earlier). The headline inflation rate, though one percentage point lower than the underlying rate in November, was about 2 percentage points above the Maastricht limit. The current account is set to improve modestly in 1998, mainly due to lower import demand, but is benefiting also from a large drop in international commodity prices and increased transfers from the EU.

Central bank independence and entry into the ERM have strengthened the monetary policy framework

The monetary policy framework has been strengthened in 1998. First, Parliament approved central bank independence at end-1997. The Bank of Greece's mandate is to achieve price stability (currently set as inflation below 2 per cent by end-1999). Second, the drachma entered the ERM at the time of the March exchange rate adjustment. The new framework should make it easier to conduct a high interest rate policy. There is less focus on the stability of the exchange rate, and thus less likelihood of attracting large capital inflows, while the central parity plays the role of an anchor. This strategy worked during the period to late-August, when financial markets viewed the high Greek interest rates as a last opportunity to profit from a convergence of interest rates to the euro-area level. During the March to August period, the drachma appreciated – about 6 per cent, on average, above the central rate (effectively halving the initial depreciation) – and, despite the subsequent financial market volatility, remained about 4 to 6 per cent above the central rate through November. Higher real

interest rates and this exchange rate appreciation have worked towards offsetting the loosening of monetary conditions resulting from the March devaluation.

The budget deficit should be below the Maastricht target in 1998, yet fiscal policy is not sufficiently restrictive

Fiscal policy has achieved a significant reduction in the general government deficit. It is projected by the OECD Secretariat to fall from 7½ per cent in 1996 to below 3 per cent of GDP in 1998 (as compared with the Government's estimate of 2.2 per cent of GDP). The considerable fiscal adjustment reflects commendable persistence in efforts to broaden the tax base and to make the tax system more equitable, for instance, by reducing tax evasion and tax expenditures. It also represents a sharp reduction in interest payments (by 2½ per cent of GDP). In contrast, current primary expenditures have remained broadly unchanged as a percentage of GDP, mainly due to outlays on personnel, while the public investment programme has increased at a very rapid pace. Despite the sizeable deficit reduction, the underlying deficit nevertheless remains high, as several expenditure items have been shifted off-budget or to past budgets (with these transactions being in accordance with Eurostat rules). The fiscal stance is effectively looser than the 1997 and 1998 deficit outturns suggest because of these transactions (about 1½ per cent of GDP per year) and because investment is financed by a rapidly increasing amount of EU capital transfers, which pump liquidity into the economy.

Looking beyond 1998, the convergence programme projects further budget deficit reduction based on the convergence of interest rates. A broadly unchanged primary balance between 1998 and 2001 is predicated on relatively high output growth, while restraint on current primary expenditure is expected to be offset by a further sharp rise in investment. The Government views the latter as critical to sustaining rapid output growth, as well as for supporting

structural change. For 1999, the announced Budget was tighter than originally envisaged, despite the advent of higher interest rates. The budget deficit target will be 1.9 per cent of GDP – below the convergence programme's target of 2.1 per cent of GDP – with more emphasis placed on containing personnel expenditures, while indirect tax cuts are used as an anti-inflationary measure. The OECD Secretariat projects a slightly higher deficit reflecting mostly higher interest payments than projected by the authorities.

The OECD projects strong output growth and a gradual decline in inflation

The OECD Secretariat projects output growth to regain momentum in 1999-2000, though in view of the international uncertainty, the risks surrounding the projections are greater than usual. The main force driving domestic demand will remain investment. Public investment will continue to be the engine of growth, increasing by about 10 per cent in real terms in both 1999 and 2000. Private investment should also remain strong, as profits should be reinforced by moderating labour costs, lower imported input costs, and declining interest rates. The intangible positive effect of moving toward EMU is expected to boost investor and consumer confidence. The external sector will continue to restrain output growth as the slowdown in world activity partly offsets the positive impact of the devaluation on exports, and the pick up in domestic demand draws in more imports. Overall, output is projected to grow by 3 to 3½ per cent in 1999-2000. Despite the moderation in labour costs and strong output growth, the decline in the unemployment rate will be limited – falling to about 9½ per cent – as firms continue to aim for productivity gains. With unit labour costs decelerating, import prices expected to remain moderate, and monetary policy tight, inflation is projected to fall to below 3 per cent by 2000, implying the need for an additional policy effort in 1999 to realise the Maastricht target.

The two main risks to the projections concern inflationary pressures and a greater deceleration in world output. In the first case, demand pressures may be stronger if the ambitious public investment programme is successfully completed (average annual increases of 18 per cent in real terms are contained in the convergence programme). Cost pressures would arise if enterprises do not resist wage claims in an environment of an improving business climate, healthy profits, and rigid labour markets. On the other hand, a sharper international downturn would hurt exports, especially since a dynamic share goes to countries of Eastern and Central Europe which may be vulnerable to financial contagion effects. This would slow demand and inflation, but also raise the external imbalance.

The main risks are to the pace of disinflation and to the maintenance of confidence, and thus, fiscal policy should be tightened, and structural reform accelerated...

In view of the potential volatility of international capital markets in general, and the focus that is likely to be placed on Greece in particular before joining EMU, the authorities will have to be sensitive to the signals engendered by their policy decisions. Concerns that inflation may not be falling fast enough to meet the Maastricht limit, or that the pace of structural reform was too slow, could weaken confidence and, thus, the credibility of the central parity. These risks suggest erring on the side of caution and taking additional measures now to dampen inflation and wage pressures; this is the present policy stance of the Government and of the Bank of Greece. The recent indirect tax cuts will have an initial dampening one-off effect on prices. However, a more restrictive macroeconomic policy stance would appear necessary for a sustainable reduction in inflation. Monetary policy will certainly have to remain tight, and to focus on developments in core inflation. However, it will inevitably become looser prior to joining monetary union as interest rates converge, and the exchange rate may move back towards the central rate. A tighter fiscal policy is thus needed, and it may be more effective than a high interest

rate policy in reducing demand in the more sheltered – and inflation prone – sectors, where the risk of wage slippage is the greatest. Thus, 1999 fiscal policy should be tightened further, for instance by: *i)* containing the wage bill for public servants; *ii)* refraining from off-budget operations which fuel demand; and *iii)* delaying less-pressing investment projects. The need for further consolidation, concurrent with the pressing need for infrastructure development poses a fiscal policy dilemma which may require cuts in other current discretionary expenditure in the short-term, but in the longer run will likely require reforms of the entitlement programmes. As an additional pillar of the adjustment strategy, rapid pursuit of a structural reform programme will signal efforts to enhance efficiency and cost competitiveness, thus allowing the economy to cope with an appreciation of the real exchange rate, which could otherwise undermine the central parity.

... which would also be necessary for solid output growth over the medium term

Structural rigidities have hampered output growth in the past, and would be even more onerous once inside EMU. The authorities' programme has identified the critical areas where structural reform is needed – budget administration, the labour market and the social security system, and the reform of public enterprises and state-owned banks. Though progress has been made, only a short period remains for Greece to prepare for the attainment of its ambitious objective to join the EMU. To this end, the current pace of structural reform will need to be accelerated significantly.

Further fiscal adjustment is also needed to reduce the debt to GDP ratio more rapidly

Over the medium-term, fiscal consolidation needs to be more ambitious, since a flexible budgetary policy (which requires a low level of debt) will be the main instrument for demand management inside the single currency area. Even under the favourable assumptions of the Government's convergence programme, reducing the debt burden from

over 100 per cent of GDP to below the Maastricht Treaty's limit of 60 per cent of GDP will require a decade and a half. Furthermore, the ageing process will add to pension and health outlays, and the example of other countries suggests that decentralisation of Government competencies could add to expenditure. A strategy for significant further adjustment will need to focus on personnel costs and the social security system, which together comprise 80 per cent of current primary expenditures. Regarding personnel management, reforms should be sweeping, including the use of more flexible employment and wage contracts. Other measures which would work to preserve the integrity of the investment programme, and obtain budgetary savings, include transferring the responsibility for investment projects to the private sector, along with the concomitant EU co-financing. A more ambitious privatisation programme would also help reduce the debt.

The ageing of the population requires the pension system to become less generous...

The authorities envisage a major overhaul of the social security system, taking place most probably after the year 2000, following changes to pension arrangements in 1999 (which will be mostly of an organisational nature) while some earlier announced cost cutting measures have not yet been implemented. The significant reform, which has yet to be announced, will need to address the system's non-viability. It is one of the most generous and inequitable in the OECD, due to low contributions and to large differences in pension benefits between pension funds, and has contingent liabilities as a percentage of GDP which are among the highest in the OECD. Two main problems of the state-run pay-as-you-go system are short contribution periods and early retirements. Options for addressing these issues should include eliminating early retirement for several privileged groups and replacing the high minimum pension by a means-tested benefit (as the former promotes contribution evasion by reducing incentives to contribute

past the minimum period required for eligibility). The objective of prolonging contribution periods should include lowering the statutory replacement rates in many cases (they would exceed 100 per cent of a last salary in the event of 35 years of contributions – a level far in excess of actuarial fairness). Less generous publicly-financed pensions may also result in additional national savings by encouraging private pensions and may form part of a strategy of shifting to a partly capitalised private system. The announcement of a specific plan to overhaul the system would also have an important impact on market expectations regarding the pace of structural reform.

... and a correction of the poor incentive structure in the health care system

Progress in the implementation of the 1997 health care legislation has been slow. Delays have been incurred in the introduction of professional managers, global budgets, and reimbursement differentiated according to diagnostic category, which aim to improve public hospital management. On the other hand, expenditure on pharmaceuticals was reduced through the introduction of a positive list of medicines and measures to contain prices. Looking forward, the Government contemplates separating health funds from their parent pension funds and unifying them into a single health fund. As a first step, a service responsible for the co-ordination of public funds' health expenditure will be created in 1999. This enhancement of the role of third party payers, in combination with the Government's intention to gradually raise hospital fees to cover variable costs fully, would provide better incentives for raising efficiency in hospitals and for containing pharmaceutical expenditure. Nevertheless, the greatest weakness of the health system remains the poor state of primary care, and the network of country-wide primary care providers, envisaged by the 1997 health reform bill, has still to be put in place.

Employment creation requires much bolder labour market reform...

Labour market reform takes on additional importance in view of the need to raise the retirement age and to reduce public-sector employment. The Government placed a high priority on achieving labour market reform in 1998. After an intensive dialogue with social partners, an agreement was reached on an outline for a structural reform package, which covers a wide range of issues. The resulting legislation led to a slight increase in the flexibility of working hours (which will reduce overtime costs), a more flexible use of part-time work, and the option for employers to opt out of regional collective bargaining in areas of high unemployment (and hire at the national minimum wage). Though these reforms represent a step forward, giving them top priority has delayed reforms in other areas, and, in the event, an opportunity to address some of the key rigidities was missed.

... to address the many remaining rigidities

The Greek labour market is notable for its high degree of segmentation and low turnover. This is partly due to strict employment protection, which forces employers to resort to severance payments far in excess of the statutory ones. Furthermore, a national minimum wage, which stands at 60 per cent of the average blue-collar wage, contributes to the low employment rate for more susceptible individuals (mostly women and youth). High non-wage labour costs are another important deterrent to employment. Though basic payroll contributions currently cannot be reduced as they are critical to the financing of the pension system, non-wage labour costs also include several *ad hoc* add-ons which should be abolished (*e.g.* the largest, which is linked to financing housing subsidies). Problems of several social groups in getting a foothold in the labour market could, to some extent, be addressed by improving active labour market polices (ALMPs), and the experience of other countries suggests the need for a well-functioning public employment service (PES). In view of the large amounts of EU

funds earmarked for ALMPs, care should be taken in their design and their operation. To this end, the objective to improve the efficiency of Greece's PES (OAED) is important. Regarding programme design, vocational training should be aimed at younger individuals, and employment subsidies, which the Greek system favours, should be targeted. The National Action Plan for Employment contains some steps in this direction. However, the need for an additional and more far-reaching reform to address these remaining structural rigidities is pressing, as a well-functioning labour market will be an important shock absorber, which would be critical for strong economic performance following entry into EMU.

Reforms have transformed the financial sector quite rapidly...

Successful product market reform would improve productivity in key sectors of the economy, and thus provide essential inputs at lower cost to the economy as a whole. The experience of other countries suggests that this process will boost overall output growth substantially, and ultimately generate employment. In recognition of their large drag on the economy, as well as the burden on the budget, the Government has embarked on a programme to revitalise inefficient public enterprises and state-controlled banks. So far, more progress has been achieved in restructuring banks than public enterprises. The National Bank of Greece and the Hellenic Industrial Development Bank have both nearly completed a clean-up of their portfolios and now meet minimum capital requirements, while the Agricultural Bank, which had been a critical exception, has begun to make some progress under new management. Nevertheless, the state-controlled banks continue to have high operating costs, due mostly to over-staffing, and these are reflected in a very high intermediation spread, and thereby a high cost of capital. To raise competition in this sector, the Government sold four small state-controlled banks to the private sector during 1997-98, and is in the process of selling the

mid-sized Ionian Bank. Though their influence is declining, the market will, nevertheless, continue to be dominated by state-controlled banks. Indications are that the Government may proceed with additional sales and will eventually keep only the largest commercial bank (the National Bank of Greece) and the Agricultural Bank under state control. A rapid completion of the planned sales would help the privatised banks' survival in a rapidly changing international market, and provide a more dynamic domestic financial market, which is critical for business development. Private sector capital, both domestic and foreign, could also help the National Bank of Greece to expand its capital base and to acquire international operational know-how. Its full independence from state influence should be achieved.

... while public enterprise reform has been slow, and resulted in a large drag on the economy

Products of public enterprises are often expensive (*e.g.* telecommunications and energy products) or heavily subsidised (*e.g.* transportation, postal, and water services). Over the past one and a half decade, the state-enterprise sector has received an increasing amount of financial assistance from the EU and national sources (equivalent to about 3½ per cent of GDP in 1998) and has also accumulated a large amount of liabilities. OECD Secretariat illustrative calculations, contained in this *Survey*, suggest that a far-reaching reform of public enterprises might yield eventual output gains of at least 10 per cent of GDP. The Government's strategy regarding public enterprises places much emphasis on the need for improved management and operating efficiency. The Government has appointed professional managers, and has begun introducing performance-based management contracts. To be effective, the new managers will need to be independent of the Government (within the context of the management contract), and be responsible for key financial decisions, especially hiring and remuneration policy – as high labour costs and low productivity largely account for the poor financial performance of public enter-

prises. The Government believes that these contracts are a first step towards improved management, and will pave the way for a more ambitious reform effort. International experience, however, has been disappointing regarding such contracts, as they have failed to prevent conflicts of interest between management and short-term government considerations. As a result, most countries have opened markets and privatised these enterprises to improve competition, with only few enterprises remaining in public hands. To this end, Greece should follow the example of most other EU countries, and liberalise markets at a faster pace than the timetable prescribed by the EC Directives (and their derogation periods). Privatisation can be facilitated by following successful experiments in tendering public services and infrastructure construction. The Government's decision to seek strategic investors and sell minority equity shares in some public enterprises are steps in this direction, as is the use of build-operate-transfer transactions to finance large infrastructure projects. However, leaving majority ownership in the hands of the public sector, which is current policy for all enterprises providing public services, weakens investor interest. Further involvement of the private sector, and the concomitant capital it would bring, would provide significant relief to the budget and reduce the debt. It would also create the appropriate incentives for infrastructure management. In any case, independent regulators (sector specific as well as a general competition watch-dog) should be rapidly developed as they play essential roles in ensuring efficient pricing and a competition-friendly environment. In Greece, the effectiveness of existing regulators needs to be strengthened significantly.

Enhanced competition is needed in the electricity sector...

International experience also indicates that it is not easy to introduce a contestable environment, especially in sectors containing elements of natural monopoly, such as Greece's electricity sector. Though the envisaged electricity reform is expected to be in line with the corresponding EC directive, several factors suggest that this is an opportunity to introduce a bolder reform. Geography implies that the Greek market is relatively isolated from imports, and, therefore, care should be taken to develop competition in the domestic market. Currently, non-transparent and distorted access and pricing for key inputs (natural gas and lignite), as well as recourse to public resources appear to be highly advantageous to the incumbent monopolist, the public power corporation, and thus provide barriers to entry. Competition between energy sectors is also undermined by cross-sectoral financial interests. A bolder reform would sever all financial links between these sectors so as to enhance competition between the upstream sectors (natural gas, lignite, and petroleum), which provide inputs to the electricity sector. Fair access to inputs and customers could then attract private-sector players to generation. To promote competition further, and following the example of other countries, one option is the legal separation of the electricity company's generation, high-voltage distribution network, and low-voltage supply operations, while the separation of the generation capacity of the incumbent into several competing firms would level the playing field further. Implementing these strategies would permit all customers to shop for the cheapest source of electricity. A successful liberalisation of this market should yield positive results similar to those of the petroleum sector, which was liberalised in 1992. Establishing a more competitive environment in the nascent natural gas sector requires setting clear and uniform rules on access pricing, and allowing private investors a majority ownership in distribution companies. To this end, the distribution of the high cost of

building the pipeline should not discriminate among market participants.

... while other sectors will need to shape up to survive in an inevitably more competitive environment

For other sectors where public ownership still dominates, enhancing competition is more straightforward. Two prime examples in Greece are air transport and telecommunications. Soon after the partial liberalisation of the sectors (affecting domestic flights and mobile phones), new entrants have provided alternatives to the public monopolist and have stimulated service improvements and/or forced a decline in price. Looking forward, with EC derogations soon ending in both sectors, the incumbent enterprise, especially in the case of Olympic Airways, will need to restructure quickly, and compete with dynamic new entrants or face bankruptcy. Survival will require, at a minimum, a strategic alliance with a major international carrier. To spur competition for the highly profitable telephone company, and thus achieve more rapid price reductions to levels of other OECD countries, the market for basic telephony should be opened immediately to more competitors.

The Government must make public service requirements transparent

Almost all public enterprises have been burdened with uncompensated and ill-designed public service obligations, and these are partly to blame for the poor financial condition of some of the largest loss makers (*e.g.* rail services, urban transport, air transport, water and sewerage services, and postal services). When restructuring these firms, the Government must decide the extent of the public service obligation, and should take this decision on a full cost evaluation. In many cases, the current budgetary constraints might call for a more limited public service obligation and should include halting the most costly operations (cases to be considered are the highly under-used trains in the southern Peloponnese region, the under-utilised postal offices in rural areas, long-distance flights to countries with Greek immigrants, and the extensive public television organisa-

tion which serves a very small share of the viewing audience). A more efficient use of scarce budgetary resources would be achieved through the use of targeting in the determination of social obligations, for instance through need-based mechanisms. Once transparency concerning the public service obligation has been achieved, tenders to private operators for a pre-determined duration could lead to further efficiency gains.

Summing-up

The Government's attention is currently focused on meeting the conditions for entry into the European single currency area by 1 January 2001. Substantial policy efforts, sustained over several years, have put this objective within reach. A further push on tightening macroeconomic policies, especially fiscal policy, should hasten the deceleration of inflation and the decline of the debt to GDP ratio. To support the exchange rate during the period to qualification for entry, and to assure a good macroeconomic performance following entry, the Government needs to accelerate the structural reforms it has correctly identified as necessary, notwithstanding the resistance from various vested interests. With the macroeconomic imbalances being brought under control, structural reform will remain the key tool for supporting a convergence of per capita incomes to the level of other EU countries.

I. Recent trends and short-term prospects

Macroeconomic conditions have improved considerably since the recession in 1993. Output accelerated steadily and outstripped the EU average in 1996 and 1997, while inflation receded markedly. The upswing was driven by investment, originating largely in the wider public sector, and with substantial support from EU structural funds. Private business investment has also been strong and oriented towards capital deepening, underpinning the ongoing restructuring process in the industrial sector. Private consumption was boosted by sharp increases in real wages, but these hindered employment creation. The steady decline in inflation was helped by productivity gains and subdued import prices, but it was not rapid enough to stop the deterioration in Greece's international competitive position. However, the strong recovery in Greece's major trading partners partly offset the sizeable loss in Greece's export market share and, combined with the sharp fall in oil and other international commodity prices, this has served to keep the current account deficit in check.

In 1998, two major factors have shaped macroeconomic developments, and should permit a more balanced output growth. First, there was a step adjustment in the exchange rate in March 1998 and, second, social partners reached a relatively moderate wage agreement for 1998-99. Nevertheless, domestic demand continues to be led by public investment. Business investment also remains buoyant, supported by the confidence effect arising from the improved prospect of joining the European monetary union, and has partly offset the slowdown of household demand resulting from a more restrained growth in labour income. Overall, output growth is likely to be slowing somewhat in 1998, though it will likely remain about 3 per cent. In an economy containing various structural rigidities – manifest in low output growth over the past couple of decades – inflation pressures will be difficult to assuage, especially as labour costs are set to rise faster than in the euro area over the next couple of years (albeit with a

narrowing differential), and the depreciation has set back the process of reducing inflation expectations. The downturn in international markets has dampened output projections for 1999-2000, *inter alia*, due to the fact that the dynamic component of exports is linked to economies of Central and Eastern Europe, which are more vulnerable to contagion effects.

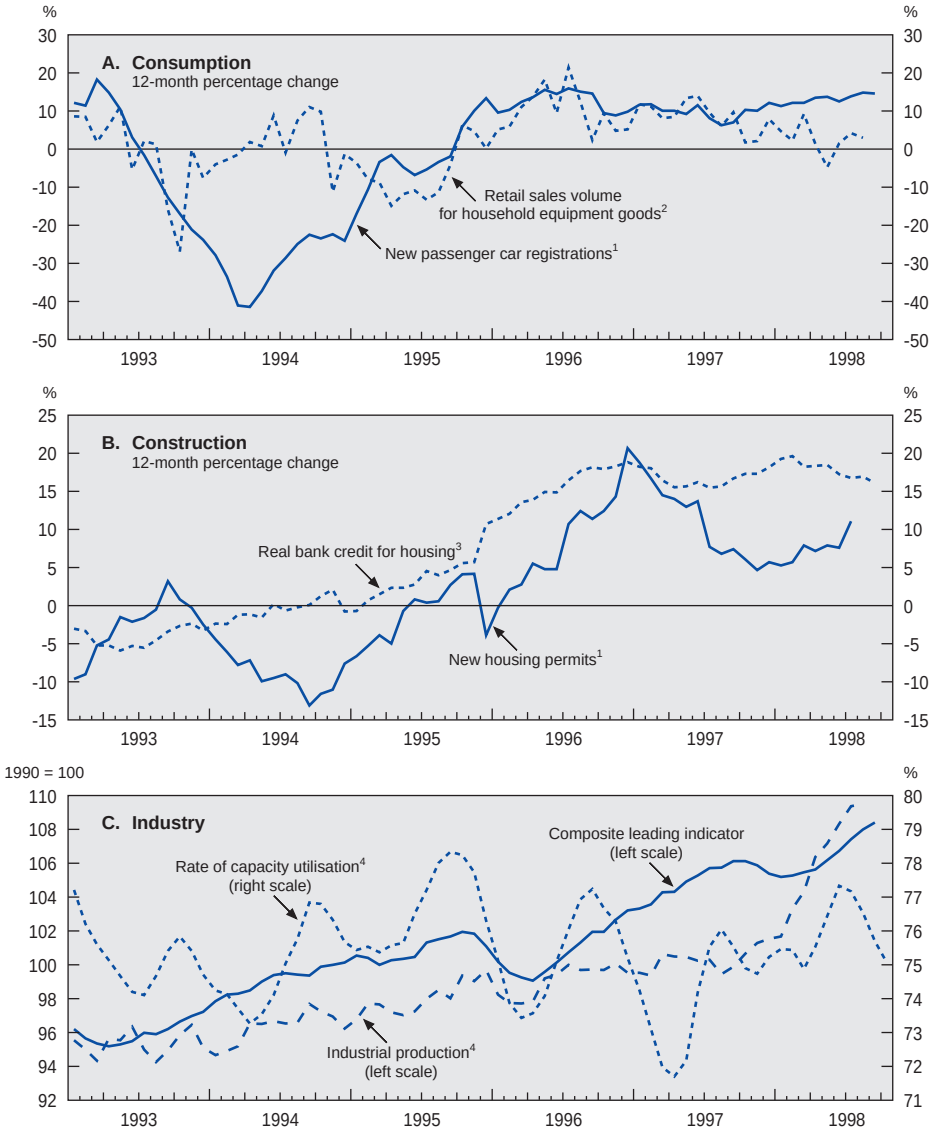
Buoyant in 1997, domestic demand has slowed in 1998

In 1997, household demand continued to gather momentum, prompted by substantial real wage gains for the fourth year in a row. Though national accounts data for households do not exist for this period, proxies for real disposable income suggest that it increased by nearly 5 per cent in 1997. Additional impetus to household demand was provided by the decline in interest rates, until financial turbulence in Asia spilled over to the Greek financial market in October 1997. Both housing investment and consumer durable purchases grew at a solid pace for the year as a whole, and this was paralleled by a boom in mortgage and consumer loans (Figure 1). The very rapid increase in housing investment also reflected lower costs, especially due to the intensive use of illegal immigrant workers in construction activity.

Household demand has weakened in 1998. The substantial deceleration in real wage and pension growth combined with a step up in the effective tax burden have resulted in a slower growth in household disposable income.[1] Evidence of a slowdown in consumption can be seen in the deceleration in retail sales volumes for household equipment. Moreover, as international financial turmoil forced an increase in interest rates, the pace of issuance of construction permits is set to be lower than the one in 1997. Housing construction may also be curtailed by the introduction of a property tax in the 1998 Budget, and could abate further if the recent registering of illegal immigrants results in a significant increase in construction costs.

Total investment accelerated in 1997 and remained strong in 1998, mostly driven by the public sector (Figure 2). Overall, public investment (general government and public enterprises) grew by over 7 per cent in volume terms in 1997 and is set for an even stronger performance in 1998. It would thus account over one quarter of total investment in 1998 and represent nearly 6 per cent of GDP (Table 1). The brisk acceleration in public investment is due to the maturing of a

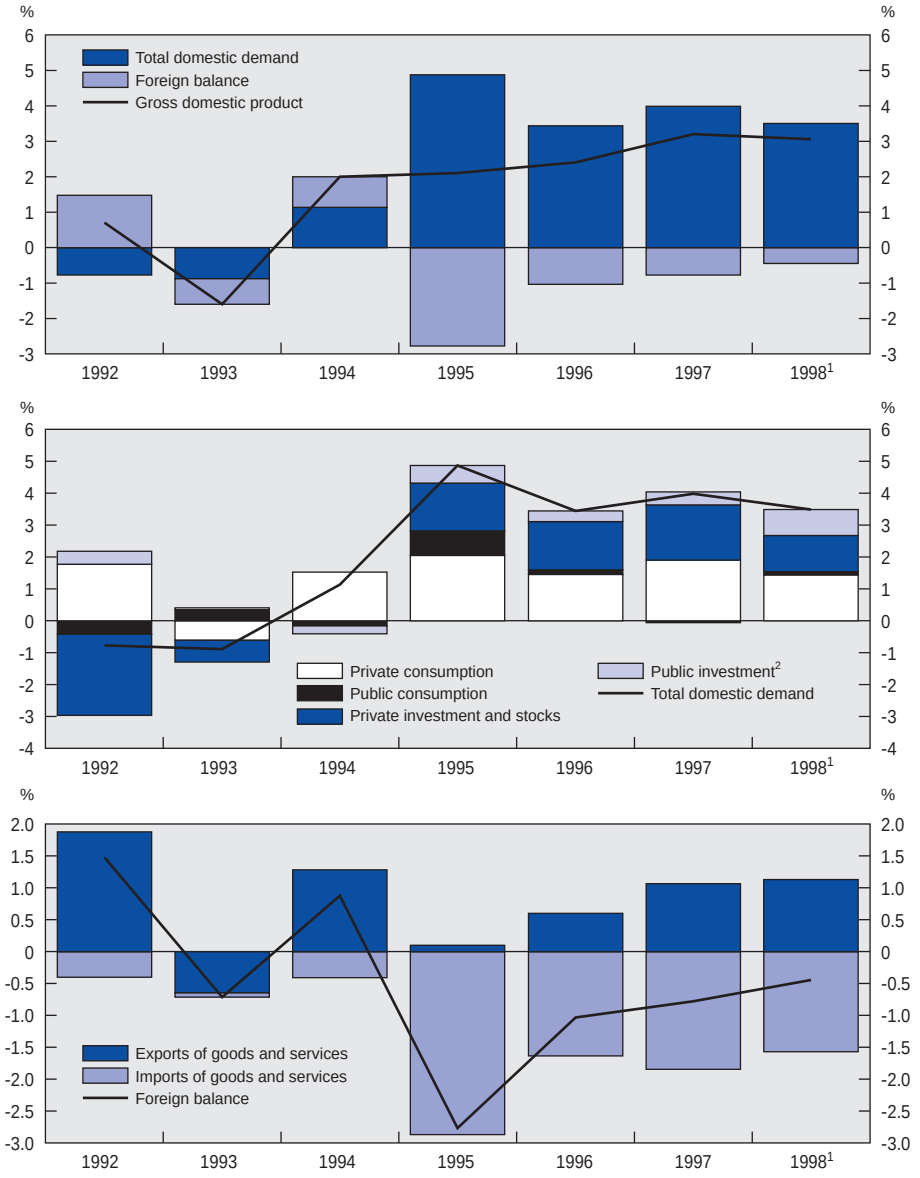
Figure 1. **Conjunctural and leading indicators**



1. 12 month moving average.
2. Break in series in 1996.
3. Deflated by the consumer price index.
4. 3 month moving average.

Source: Bank of Greece and OECD, *Main Economic Indicators.*

Figure 2. **Contributions to real GDP growth**



1. OECD Secretariat projections.
2. General government and public enterprise investment.

Source: Ministry of National Economy and OECD Secretariat.

Table 1. **Public investment: key features**

Percentages

	1996	1997	1998 Official
Public fixed investment as a share of GDP [1]	5.1	5.2	5.8
Public fixed investment (as a share of total fixed investment) [1]	26.2	25.8	26.7
General government	17.1	16.7	16.6
Public enterprises	9.1	9.1	10.1
Share of public enterprises in business fixed investment [1]	16.6	16.7	18.5
Public investment budget (in billions of drachmas) [2]	1 095	1 625	1 875
As a share of total			
Road transportation	19.1	19.2	17.0
Regional government public works	25.1	16.4	13.8
Education	8.4	7.4	8.2
Health	2.3	2.5	3.7
Railroad	3.0	6.9	9.3
Energy	. .	5.8	5.9
Agriculture	1.1	5.5	6.4
Industry	12.0	11.0	9.7
Environment	1.6	3.0	3.7
Other	27.4	22.3	22.3
Public investment co-financed by the EU (as a share of total public investment) [2]	71.4	66.4	73.3
EU funds directed at public investment (as a share of total public investment) [2]	50.4	43.0	47.5
Memorandum items:			
Subsidies to business sector investment (as a share of total business investment) [3]	5.5	5.0	n.a.
Value of business investment projects covered by incentive legislation (as a share of total business investment)	13.1	11.5	n.a.
Average subsidy (as a share of the investment value) [3]	42.3	43.2	n.a.
Rate of absorption of annual target for EU structural funds	84	67	99

1. On the basis of national accounts data.
2. On the basis of budget data. Includes fixed capital investment and capital transfers to public and private entities.
3. Excluding subsidies granted through tax deductions.

Source: Ministry of National Economy.

large number of projects under the Second Community Support Framework which envisages the disbursement of ECU 29.7 billion (at 1994 prices) over the six-year period 1994-99. These include several large infrastructure projects, and investment by public enterprises in the process of upgrading and enlarging their capacity prior to the deregulation and opening to competition of key sectors, such as the energy and telecommunication sectors (see Chapter IV).

Though less buoyant than in the public sector, private sector business investment (excluding public enterprises) continued to exhibit strong growth in 1997, spurred by healthy profits. Earnings rose because of higher sales rather than through improved margins, as higher wage costs were balanced against lower financial and imported-input costs. Overall, the return on assets (measured by the ratio of profits to own assets or the net return to total assets) fell marginally in 1997 compared to 1996, and remained at a relatively low level compared to the yield on government securities.[2] In the context of substantial wage increases combined with growing international competitive pressures for several years, private business investment in 1997 continued to be largely oriented toward capital deepening.

In 1998, several offsetting forces have been influencing private business investment. On the one hand, capacity utilisation remains at high levels, and profits have been boosted by a sharp slowdown in wages and the support of public funds through various channels. The latter include a new investment subsidy law,[3] capital transfers from the budget to finance various projects, and new employment subsidy programmes (see Chapter III). On the other hand, high and variable interest rates appear to be leading to a downward revision of the rate of growth of investment plans, in particular by firms which had borrowed heavily in foreign currencies or import the bulk of their raw materials and other intermediate inputs. Cash flow has been further squeezed by the rise in the effective tax burden, due to the advancement of corporate tax payments. On balance, private business investment (excluding public enterprises) appears to have continued to grow at high single digit rates in 1998. It has been accompanied by a brisk revival in industrial production, which partly reflects buoyant public enterprise activity.

An underlying current of household and business confidence may be dampening a slowdown in overall demand. Adding to a general feeling of an improving economic conjuncture, Greece's enhanced prospects to join monetary union following the entry into the ERM (at the new exchange rate parity) were hailed by domestic and international investors. Long-term interest rates dropped by about 300 basis points and the Greek stock market increased by 70 per cent during the first 7 months of 1998. The euphoria appeared to trickle down to the economy more broadly. This intangible factor is susceptible, however, to changes in expectations, and in late-August 1998 market confidence was challenged in the

wake of widespread international market turmoil, with the stock market losing one-half its yearly gains by early October 1998 but stabilising thereafter and long-term interest rates rising by about 100 basis points. The aftershocks have created some uncertainty for consumers.

Output growth has been more balanced

Overall, domestic demand picked up to a 3½ per cent rate in 1997, before decelerating in 1998 due to slowing household demand. Output growth appears to be more balanced in 1998, as the negative contribution of net exports abates. Imports have slackened in 1998 as domestic demand has slowed and the devaluation raised import prices sharply, while exports have been sustained by the relatively strong recovery in activity in Greece's main trading partners. Overall, the negative contribution of net exports to GDP growth has lessened, from 1 percentage points in 1996, to an estimated 0.4 points in 1998. Output growth should be about 3 per cent in 1998 after reaching 3¼ in 1997.[4]

Employment has been slow to respond to the recovery

The acceleration in output growth was accompanied by sharp productivity gains, as total employment declined in 1997.[5] Job losses were concentrated in the manufacturing and construction sectors, with a contraction in employment of 3 and 1 per cent, respectively. The number of workers employed in textile industries, for example, was cut by 6.5 per cent in 1997, prolonging the severe restructuring under way since the mid-1980s in this important export sector (the number of workers employed in this sector has almost halved between 1989 and 1997). In 1997, the sluggishness in employment in traditionally dynamic sectors (banking and other service sectors) came on top of the long-standing restructuring process in the industrial sector. Dependent employment increased by ½ per cent, in part due to general government hirings. Nevertheless, the unemployment rate remained stable, as the drop in employment was fully offset by a decrease in the labour force which, for the most part, reflected the ageing of the population (Table 2). Withdrawals from the labour force by youth and discouraged unemployed also played a role, and the participation rate fell slightly.

Table 2. **Labour market developments**

Annual percentage changes[1]

	1994	1995	1996	1997	1998 projections[2]
Civilian labour force	1.8	1.3	1.6	–0.6	1.2
Participation rate[3]	59.6	60.1	61.0	60.8	61.3
Unemployment rate	9.6	10.0	10.3	10.3	10.0
Total employment	1.9	0.9	1.3	–0.5	1.4
Agriculture	–0.5	–1.0	0.5	–2.7	. .
Industry	–0.5	–0.9	–0.1	–2.3	. .
Services	3.9	2.4	2.1	1.1	. .
Dependent employment	1.9	2.1	2.0	0.5	1.6
Private sector	2.1	2.2	1.9	0.3	. .
Public sector[4]	1.5	2.7	1.5	0.1	. .
of which: General government[4, 5]	0.3	3.6	2.5	0.6	. .
of which: Education[4]	3.5	2.1	2.6	2.5	. .
Public enterprises and state-controlled banks[4]	4.4	0.3	–0.9	–1.2	. .

1. Except for the participation rate and the unemployment rate.
2. OECD.
3. For ages 15-64.
4. End of period.
5. On an annual average basis, employment in general government increased by an estimated 0.7 per cent in 1994, 2.0 per cent in 1995, 3.0 per cent in 1996, and 1.7 per cent in 1997.

Source: Data submitted by the national authorities and OECD estimates.

The strength of industrial activity suggests a boost to employment in 1998, though there is little solid data on employment developments.[6] The number of registered unemployed has accelerated very rapidly (by 34 per cent over the first 7 months of 1998), but this development is deemed to arise principally from increased participation in new training programmes, which depends on registration with the public employment organisation. Further complicating the analysis of employment/unemployment data is the existence of many illegal workers in Greece. Since the early 1990s, a large number of immigrants from the former communist block have entered the country – an estimated 500 to 700 thousand persons, representing over 10 per cent of the labour force. This large number of immigrants, most of whom work without permits and for wages which are below the minimum wage, have helped keep down growth in labour costs in Greece during the past several years, thus providing a boost to economic activity, particularly in the construction and agriculture sectors. In 1998, the Government

introduced a registration programme for these workers, and through June about 375 thousand illegal workers had registered with the labour authorities.[7]

Wages have moderated but disinflation was slowed by the exchange rate adjustment

Over the last few years, inflation has fallen substantially. From a double-digit rate in early 1995, consumer price inflation dropped to 4.3 per cent in February 1998, hitting a 25-year low. Disinflation was largely achieved by the hard drachma policy, relatively small increases in indirect taxes, and subdued energy and raw material prices.[8] A restrictive macroeconomic policy environment was also important, which included a further tightening of monetary policy to stem capital outflows during the second half of 1997.[9] Labour cost pressures, on the other hand, remained strong. In 1997, the increase in the minimum wage arising from the national collective agreement amounted to 8 per cent, and the overall increase in private sector compensation per employee rose by about 9 per cent and for civil servants by 13 per cent; 3¼ and 7½ percentage points, respectively, above the rate of inflation (Table 3).[10] Despite sizeable productivity gains, unit labour costs exceeded inflation in 1997, and at 8 per cent still grew 7 percentage points faster than in the EU on average.

Following two consecutive years of strong wage increases which threatened the disinflation process, 1998 wage developments were influenced by a Pact negotiated between social partners on a strategy for labour market reform (see Chapter III) and a tight incomes policy for civil servants. The Pact included, *inter alia*, an agreement by social partners to keep real wage increases in line with productivity developments and to take into account the need to support employment. The 1998 Budget set cost of living increases for civil servants at 2.5 per cent, which is well below the inflation target. The government's strategy to moderate wages appears to have been successful, as in May 1998 trade unions and private employers reached a 2-year agreement which can be viewed as moderate when set against developments over the past four years. For 1998, the agreement contains minimum wage increases of 4.5 per cent (2.5 per cent in January and 2 per cent in July), one percentage point higher than the official target for end-year inflation (but it also includes an inflation catch-up clause).[11] These wage developments, accompanied by productivity gains which would be

Table 3. **Prices and wages**

Percentage changes

	1994	1995	1996	1997	1998[1]
Prices					
Price deflators					
GDP deflator	11.2	9.8	7.9	6.9	4.8
Private consumption deflator	11.0	8.8	8.3	5.5	4.7
Consumer prices					
Period average	10.9	8.9	8.2	5.5	4.9[2]
End of period	10.6	7.9	7.3	4.7	4.7[2]
Underlying inflation[3]					
Period average	11.0	9.4	8.1	6.0	5.3[2]
End of period	10.5	8.6	7.4	5.1	5.5[2]
Wholesale prices, total (period average)	8.7	7.8	6.1	3.6	3.8[4]
of which:					
Exports	8.7	10.2	5.6	4.6	4.6[4]
Imports	9.2	7.0	1.6	1.8	5.3[4]
Industry (for domestic consumption)	7.4	8.1	7.1	4.8	3.8[4]
Wages					
Total wage bill (national accounts)	12.9	15.3	14.0	11.5	7.4
Unit labour cost (national accounts)[5]	10.7	12.9	11.3	8.0	3.8
Manufacturing, blue collar[6]	13.2	13.3	8.8	8.8	6.5
Civil servants	9.4	12.3	14.9	13.5	5.4
Banks[7]	17.5	13.1	12.7	9.7	6.3[8]
Public enterprises	18.8	10.9	14.1	11.0	7.2
Minimum wage, blue collar	12.6	9.4	7.8	8.0	5.4
Memorandum items:					
Effective exchange rate	–3.2	0.0	–1.5	–1.7	–3.3[9]
Real effective exchange rate[10]	3.5	7.8	2.6	2.4	0.1[9]

1. Official estimates, unless where otherwise indicated. OECD projections are included in Table 7.
2. October 1998. The period average covers the 12-month period to October 1998.
3. Excluding energy and fresh food products. Based on 1988 weights prior to 1995, and 1994 weights thereafter.
4. The 12-month period to September 1998.
5. Total compensation of salaried employees divided by GDP in constant price terms.
6. Weekly earnings.
7. October on October.
8. Estimate of average annual increase.
9. Based on the technical assumption that the exchange rate remains constant from end-October to end-December 1998.
10. Unit labour cost weighted, OECD calculation.

Source: Bank of Greece and Greek National Statistical Service.

large by historical standards, though lower than in 1997, should permit a deceleration in unit labour costs to about 4 per cent in 1998. Thus, for the first time since 1994, unit labour costs will have increased by less than price inflation.

The March 1998 devaluation fed through quickly to retail prices, pushed up by an acceleration of import price rises to 7½ per cent in April 1998. The

underlying 12-month rate of inflation (*i.e.* excluding fuel and fresh fruits and vegetables) increased from 4½ per cent in February before peaking near 6 per cent and subsequently falling to 5.3 per cent in November. By this time, the headline inflation rate had already returned to a broadly declining path and fallen to 4.2 per cent, as both energy and fresh food price increases were very moderate and indirect tax cuts were undertaken in late September on gasoline and mid-October on heating oil (Figure 3, see page 30). Inflation pressures are strongest in the service sector, where labour intensity is high and competitive pressures low (and where inflation has remained near 6 per cent). Nevertheless, inflation is set to continue on a downward course during the remainder of 1998 as monetary policy has been tightened, and the drachma has appreciated since the March exchange rate adjustment; in the period March to October 1998, the drachma, on average, was about 6 per cent above the central parity established in March. Moreover, the indirect tax on cars was cut in mid-November and the VAT on electricity was lowered on January 1, 1999. The four cuts in indirect taxes are expected to have a cumulative direct impact on inflation of about 1 percentage point (see Chapter II). Finally, the Government is negotiating ''gentleman's agreements'' with key producers to keep the price of their goods in check.

Trade performance and the balance of payments

In 1997, the current account was buffered from losses in competitiveness by exogenous developments

The current account deficit – based on OECD estimates using an amalgamation of customs and settlements data – remained broadly unchanged in 1997, despite a deterioration in the non-oil trade balance (Table 4).[12] The main offsetting factors were the fall in international oil prices, and a recovery in activity in the OECD area, which is a critical market for Greek exports as well as the main source of tourist arrivals and remittances. The deterioration in the non-oil trade balance reflected a continued loss of export market share, as the exchange rate continued to appreciate in real terms, and more buoyant imports as domestic activity accelerated. The loss in market share was concentrated in Greece's main market, the EU, where Greece's exports of goods dropped by

Table 4. **Balance of payments**

Billion drachmas

	1994	1995	1996	1997	1997 (Jan.-June)	1998 (Jan.-June)
A. Current account [1]						
Trade balance, custom basis	**–2 772**	**–3 268**	**–3 607**	**–3 994**	**–2 074**	**–2 322**
(per cent of GDP)	–11.6	–12.2	–12.1	–12.2	–6.3	–6.6
Oil balance	–274	–268	–455	–288	–156	–183
Non oil balance	–2 499	–3 000	–3 152	–3 706	–1 918	–2 139
Exports (f.o.b.)	2 053	2 374	2 483	2 730	1 399	1 447
Imports (c.i.f.)	4 551	5 374	5 635	6 435	3 317	3 586
Invisible balance, settlement basis	**3 065**	**3 141**	**3 128**	**3 438**	**1 485**	**1 635**
Credits [2]	4 370	4 644	4 727	5 230	2 306	2 702
of which:						
Tourism	935	950	891	1 041	378	468
Shipping	472	506	543	574	274	318
Remittances	640	709	719	796	352	390
Net EU transfers [3]	1 050	1 163	1 217	1 247	537	643
Debits	1 305	1 503	1 599	1 793	822	1 068
of which: Interest payments, dividends and profits	512	621	726	680	323	397
Current balance [1]	**298**	**–125**	**–478**	**–556**	**–590**	**–687**
(per cent of GDP)	1.2	–0.5	–1.6	–1.7	–1.8	–1.9
Memorandum items:						
Current balance, settlement basis (per cent of GDP)	–0.1	–2.5	–3.8	–4.2	–2.9	–2.6
Exports of goods						
To EU countries, as a share of total (custom basis)	56.6	60.6	52.5	46.7	. .	. .
To Balkan countries, CEECs and former USSR, as a share of total (custom basis)	13.3	15.0	18.2	19.8	. .	. .
Ratio of settlements to custom basis	55.3	52.7	51.1	49.2	. .	. .
Terms of trade (percentage change) [4]	–0.5	3.0	3.9	2.8	2.8	–0.5

approximately 2 per cent in drachma terms, despite economic growth gathering steam in the EU in 1997. This poor export performance was partly offset by stronger ones in other OECD countries and central and eastern European countries (CEECs). There was also an apparent loss of competitiveness in the export of services to the EU as the number of tourists arriving from EU countries rose by only 3.8 per cent, and an overall good tourist season was mostly due to arrivals from former Yugoslavia and Poland. Regarding factor services, interest

Table 4. **Balance of payments** *(cont.)*

Billion drachmas

	1994	1995	1996	1997	1997 (Jan.-June)	1998 (Jan.-June)
B. Capital account						
Capital account (net)	**1 633**	**708**	**2 025**	**–85**	**190**	**2 865**
Private sector	900	534	1 714	–1 308	375	849
of which:						
Entrepreneurial capital adjusted, net[5]	240	328	343	265	3	757
Real estate investment	230	240	251	263	122	107
Other capital[6]	356	–155	1 057	–1 967	181	–65
Public sector	733	174	311	1 223	–185	2 016
Long-term borrowing	521	–27	1 045	880	–202	2 315
Government	1 122	909	2 332	2 634	681	3 222
Public enterprises	233	143	209	83	53	152
Amortisation	–839	–1 079	–1 496	–1 836	–936	–1 060
Short-term borrowing (net)	212	201	–734	342	18	–298
Errors and omissions	–52	–19	92	143	180	–100
Valuation adjustment	33	22	26	332	330	346
Adjustment[7]	–369	–559	–657	–803	–367	–276
(per cent of GDP)	–1.5	–2.1	–2.2	–2.5	–1.1	–0.8
Overall balance	**1 538**	**25**	**1 007**	**–970**	**–256**	**2 148**
Memorandum items:						
Loans by domestic banks in foreign currencies[8]	267.9	530	811	405	614	–15
Change in deposits in foreign currencies at Bank of Greece	14.9	–486	–149	–936	–202	394

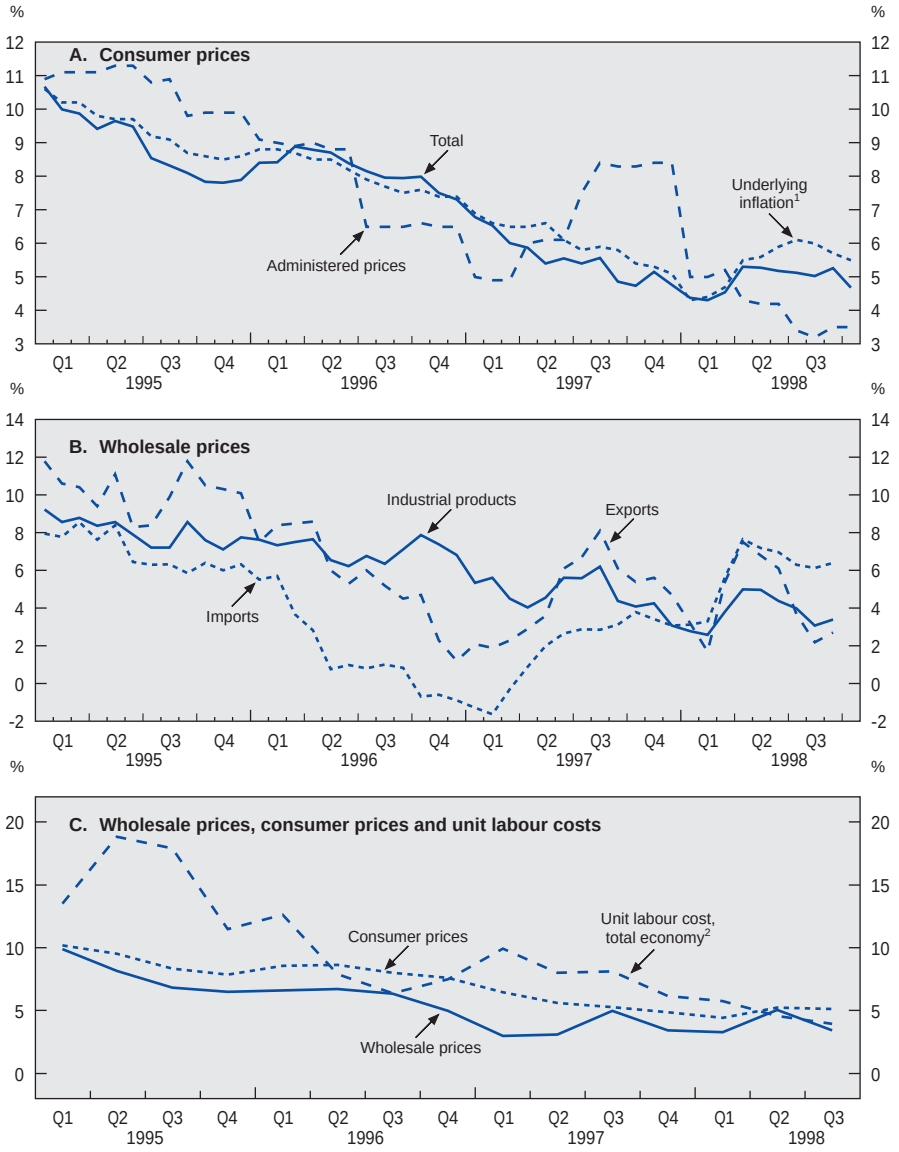
1. The current account presented in this table is an OECD approximation based on a mixture of customs and settlements data. This presentation attempts to reflect current account developments which are not captured by the settlements data due to their under-recording of exports.
2. Includes an adjustment to the item ''withdrawals out of non-residents' convertible drachma accounts''. It is assumed that 25 per cent of this item consists of exports of goods.
3. Including capital transfers which amounted to Dr 288 billion, Dr 322 billion, Dr 548 billion and Dr 700 billion in 1994, 1995, 1996, and 1997 respectively.
4. Proxied by wholesale price indices for exports and imports.
5. Excluding loans in foreign currencies extended by domestic banks.
6. Including the difference between the change in foreign currency deposits by commercial banks at the Bank of Greece and commercial banks' lending in foreign currencies.
7. Includes the difference between the trade balance in the settlement and customs basis accounts and the correction for the drachma withdrawals of non-residents' item which corresponds to exports of goods.
8. Change in outstanding balances (end of year).

Source: Bank of Greece, *National Statistical Service* and OECD Secretariat.

payments declined in 1997 from a high 1996 level, due both to the large capital outflows experienced in the second half of the year as well as the non-renewal of two large government credits.

Figure 3. **Prices and wages**

Change over corresponding period of previous year



1. Consumer price index, excluding energy and fresh food products.
2. Quarterly patterns are based on unit labour cost in manufacturing, 1998 data are OECD estimates.

Source: Bank of Greece and OECD Secretariat.

The devaluation has had mixed impacts on the 1998 current account

The March 1998 devaluation has not been fully reflected in improved competitiveness of exports, both because some exporters initially rebuilt margins which were squeezed in 1996-97, and because by November the exchange rate had appreciated significantly compared with the amount of the original adjustment. Poor price competitiveness has continued to dampen merchandise exports to EU markets in 1998 (they fell by about 3½ per cent in drachma terms over the first 6 months compared with the corresponding period in 1997) reflecting their concentration on goods which compete on price rather than on quality (see Box 1). In tourism, only a small impact is expected from the devaluation in 1998, since most payment agreements with tour operators had already been concluded far in advance of the devaluation. Nevertheless, the continued strong economic expansion in the industrial countries appears to have produced a second bountiful year for tourism. Other invisible receipts have also been improving. Net EU transfers have increased, reflecting both the change in the drachma parity toward the ECU and the scheduled rise in net EU transfers in ECU terms. The drachma devaluation also gave a boost to inflows of Greek workers' remittances.[13]

Table 5. **Trade indicators**

	Greece 1980	Greece 1990	Greece	Spain	Portugal	Italy	Germany
			1997				
Exposure to foreign competition[1]	26.2	28.5	27.0	37.2	45.8	34.2	40.0
Import penetration[2]	17.1	20.5	19.0	21.4	29.1	17.2	20.7
Ratio of goods exports to total exports	56.8	55.0	54.1	70.3	76.5	75.8	86.1
Ratio of manufacturing exports to total goods exports	50.8	55.1	56.1	80.7	87.2	91.2	89.4
Ratio of agricultural exports to total goods exports[3]	25.8	29.6	26.1	15.3	7.2	6.4	4.7
Ratio of textile exports to total goods exports[3]	9.0	22.1	15.4	3.4	23.5	9.6	1.7
Ratio of tourism receipts to total exports[4]	19.0	17.8	18.8	18.1	14.6	9.5	2.8

1. Exposure = X/GDP + (1 – X/GDP) × M/D where X and M represent the exports and the imports of goods respectively and D the total domestic demand. Custom basis.
2. Penetration = M/(GDP – X + M). Custom basis.
3. 1996 for Spain and Portugal.
4. Settlements basis for tourist receipts, customs basis for total goods in the case of Greece.

Source: OECD, *Foreign Trade Statistics and Analytical Database.*

Box 1. **The structure of the current account and the level of the exchange rate**

Evidence suggests that Greek exports are concentrated in goods which compete on price rather than on quality. Specifically, agricultural products account for over one quarter of total exports of goods, which is a high share by international standards (Table 5). Similarly, for the manufactured products for which Greece has a significant revealed comparative advantage (RCA), price is the main determinant of international competitiveness, rather than quality (Figure 4).[14] Moreover, exports are heavily concentrated on a few products (*e.g.* textiles and footwear), as only 9 products (at a SITC 3 digit level) account for 50 per cent of Greece's total manufactured exports. This is a much lower number than for other countries of a similar size; *e.g.* 28 products in Spain, 15 in Portugal, and 24 in Denmark.

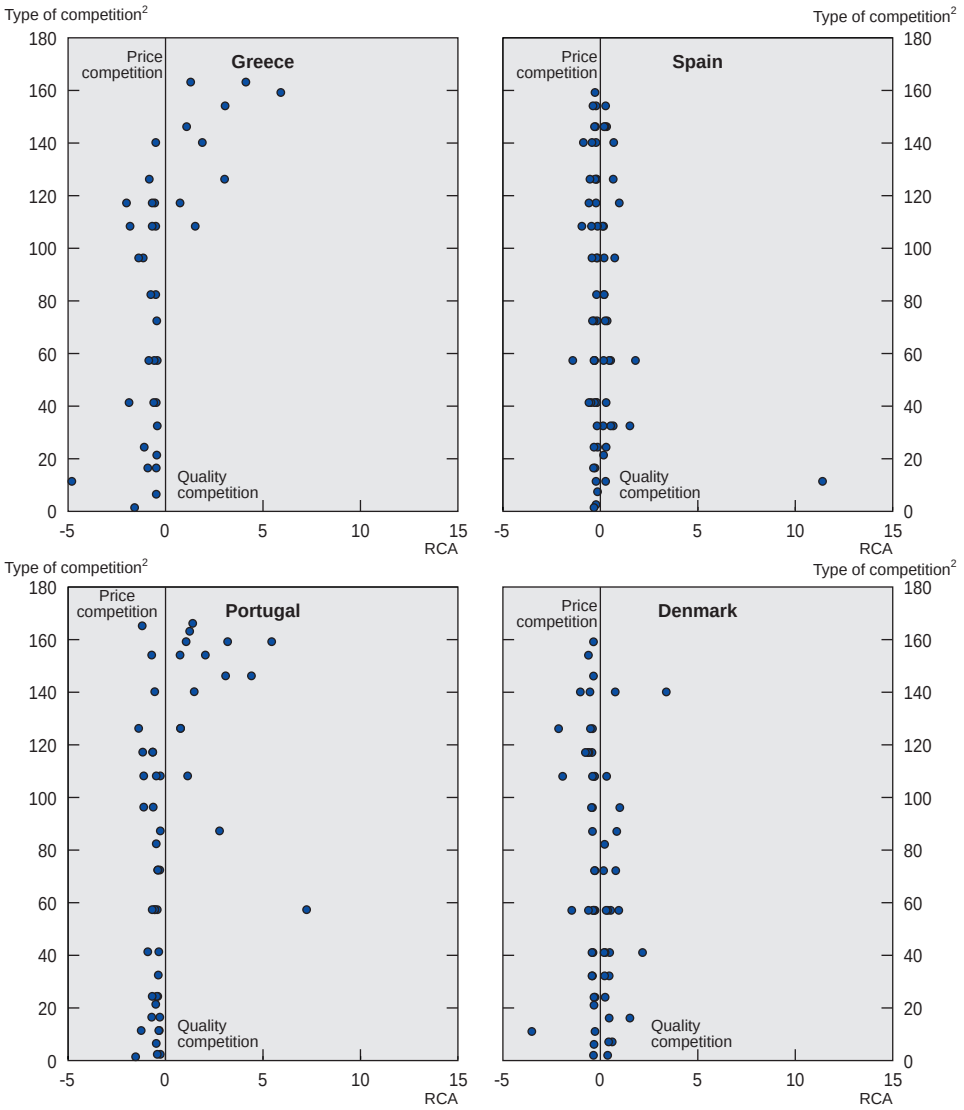
The deterioration in competitiveness in recent years, measured by the real effective appreciation of the drachma, has been associated with a large loss in export market share to EU countries. Cumulatively from 1993 to 1997, the nominal effective exchange rate depreciated by 6 per cent, while the real effective exchange rate (in terms of relative unit labour costs) appreciated by 17 per cent according to OECD calculations and somewhat more according to Bank of Greece calculations which include changes in social security contribution rates (Figure 5). The ULC-based indicator may overstate the loss of competitiveness since it does not take into account the change in the price of other factor inputs, such as the decline in interest rates. In the event, during the same period, Greek exports suffered an approximately 20 per cent decline in market share for total goods. This loss in competitiveness is also observed in the developments of some of Greece's traditional manufacturing sectors. Production in footwear and clothing industries was cut by almost one third between 1993 and 1997, and the textile industry has shifted production to low-wage countries, such as Bulgaria and Albania.[15]

The Greek tourism industry is another sector where price competitiveness is important. Prior to a reported freeze in prices to tour operators during the past few years, Greece lost significant market shares during 1995 and 1996, though experiencing a strong recovery in 1997 and 1998. Overall, tourist receipts in 1997, adjusted for CPI inflation, stood below their 1994 level, while Spain and Italy experienced increases of 27 and 15 per cent, respectively during the same period.[16]

In the event, the magnitude of the current account deficit would help throw light on the extent to which the exchange rate is misaligned. However, data problems with Greek balance of payments data result in mixed evidence. On a settlements basis, the current account deficit was near 4 per cent of GDP in 1997, compared with near balance in 1994. Using customs data for merchandise exports and imports – which are generally accepted to present a more accurate representation of trade developments – the current account deficit would be about 1½ per cent of GDP in 1997 compared with a surplus of 1½ per cent of GDP in 1994.[17] Though the level of the current account differs markedly

(continued on next page)

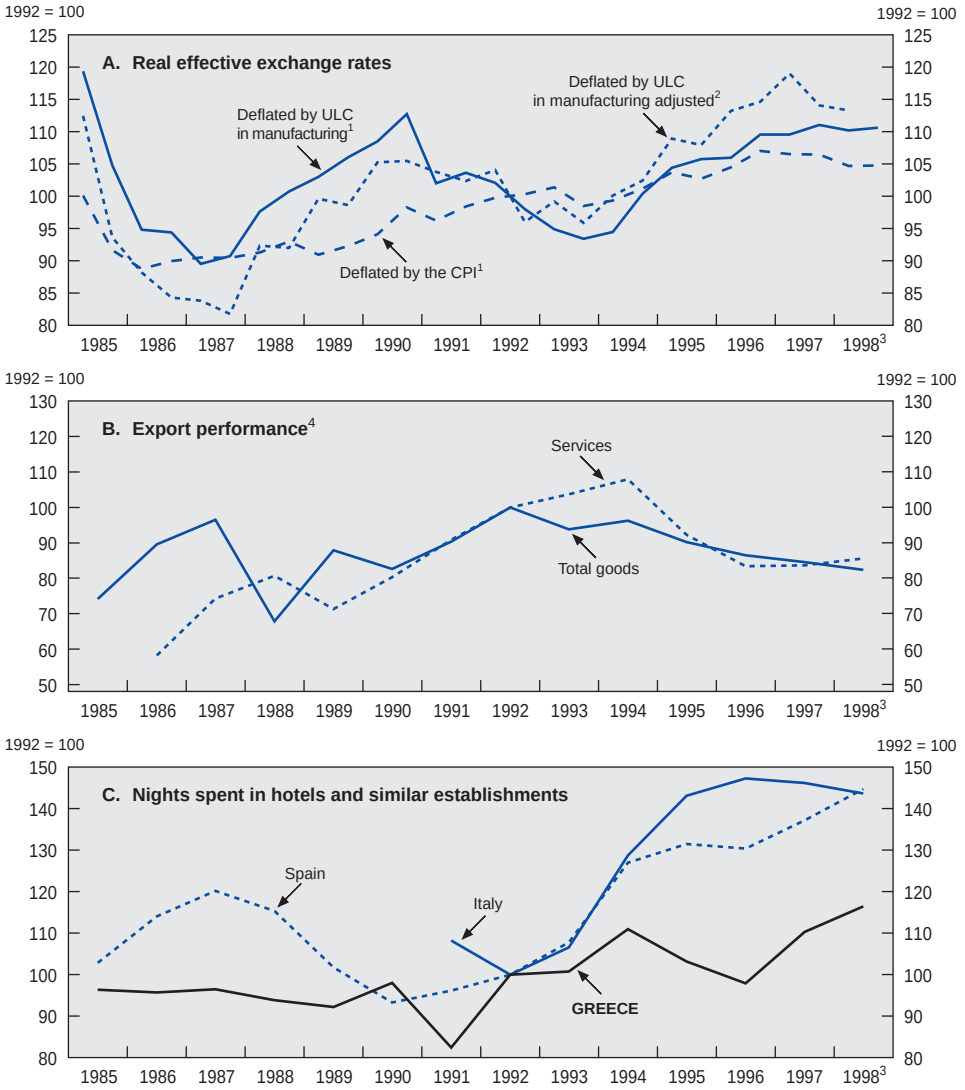
Figure 4. **Revealed comparative advantages and quality versus price competition**[1]



1. Revealed comparative advantage indicator = (Xi/X-Mi/M)*100 where Xi (Mi) and X (M) are exports (imports) of manufactured good i and exports (imports) of total manufactured goods respectively.
2. Ranking of the SITC 3-digit manufacturing industries according to price versus quality competition for the top export (import) products accounting for 50 per cent of the country's total exports (imports) of manufactured goods. The highest quality competition (the lowest price competition) is represented by 1, the lowest quality competition (highest price competition) is represented by 166 (see Aiginger 1995).

Source: OECD *Foreign Trade Statistics* and Aiginger.

Figure 5. **Competitiveness and export performance**



1. OECD calculation.
2. Including social security contributions, Bank of Greece calculation.
3. OECD estimates. For the exchange rates, the technical assumption is that they remain constant from end-October 1998 until the end of the year.
4. Export performance is the ratio between export volumes on a customs basis and export markets.

Source: Bank of Greece and OECD Secretariat.

(continued)

depending on the data source, the magnitude of deterioration in the current account has been of broadly similar magnitudes during 1994 to 1997 (3 to 4 per cent of GDP).

It is obviously difficult to judge the appropriate level of the nominal exchange rate. While traditional Greek exports which compete on price will undoubtedly benefit from the March devaluation, the Greek current account also contains several large items whose demand or supply is broadly insensitive to such relative price changes. These items comprise one-third of exports of goods and non-factor services, and include transfers from the EU (both agricultural support and structural funds), transfers from Greeks residing abroad, and revenues from the shipping sector. Their existence has probably resulted in an appreciation of the ''equilibrium'' exchange rate, and been the root cause, along with the development of the tourism sector, of a trade deficit of the order of 12 per cent of nominal GDP in 1997. The recent surge in EU structural funds, much of which is used for large construction projects which use essentially domestic (rather than imported) products, may be exacerbating this phenomenon.

The appropriate level for the exchange rate may also be affected by the positive trade shock of strong exports of goods to, and tourism from, Central and Eastern European Countries (CEECs). Exports to non-EU European countries, in particular to CEECs (Bulgaria, Poland, Romania, Russia), surged by 40 per cent in 1997, and exports to Balkan and Central and Eastern European countries accounted for 20 per cent of Greece's total exports of goods in 1997, up from 13 per cent in 1994. Developments in tourism receipts have been broadly similar to those in exports of goods, with an increase in arrivals of tourists originating from CEECs by 50 per cent in 1997. The increase in exports to these countries has masked the loss in market share to EU countries. However, the positive contribution to the current account on a net basis is smaller (albeit rising), as imports from these same countries have also increased rapidly.

Import growth in 1998 has mirrored developments in private consumption and equipment investment, with only a small initial response to the change in relative prices.[18] The narrow domestic supply base for key products may limit substitution possibilities, and thus exacerbate the ''J-curve effect'' from the devaluation. Specifically, Greece's reliance on imports for the provision of capital goods and motor vehicles is high by international standards.[19] Furthermore, regarding imports for which there exist supply alternatives domestically, high capacity utilisation may have hindered a rapid response to growing demand. Finally, the demand for oil imports is relatively inelastic and these account for 12 per cent of total merchandise imports in 1997.

Capital account developments have reflected international financial turbulence

The capital account has experienced sharp swings, following the international financial market turbulence originating in Asia. Beginning in the second half of 1997 and continuing until the March 1998 devaluation, Greece suffered sizeable capital outflows, mostly of a speculative nature. In addition, there was less borrowing by Greek firms and individuals in foreign exchange (for which Greek domestic banks usually acted as intermediaries).[20] Despite a substantial increase in the external borrowing by the Government to lower the cost of financing the deficit, official foreign exchange reserves fell markedly in the second half of 1997. The increased external borrowing from the public sector drove its external debt to 26 per cent of GDP at end-year 1997 but did not impede a decline in its interest service ratio to 9 per cent (see Table 6).[21] Regarding private sector external borrowing, Bank of Greece estimates suggest that the net asset position of the commercial banking sector improved by about US$6 billion in 1997 to about US$2 billion, basically as a result of the reduction in required reserve requirements on foreign currency deposits, which allowed banks to increase their investment abroad. The net asset position of the private non-bank sector is estimated to have deteriorated by US$5 billion, to a net asset position of US$6 billion at end-1997, most likely due to a reduction in synthetic swaps, which had been used extensively in 1996, and involved cross-border deposits.

Official foreign exchange reserves more than fully recovered the preceding year's losses following the March devaluation, which combined with the announcement of fiscal measures and a privatisation agenda, resulted in ''convergence trading''. Specifically, the improved prospect of Greece joining monetary union induced private investors to pour large amounts of foreign exchange into Greece's bond and equity markets. Moreover, in the first 6 months of 1998, the Government borrowed US$6 billion in international markets. As a result, foreign exchange reserves rose to over US$20 billion in mid-1998, from US$11 billion in October 1997. The August bout of speculation resulted in only a small loss in foreign exchange reserves, and these stood at US$17.7 billion at end-September 1998, equivalent to approximately 7 months of estimated 1998 imports of goods and non-factor services.

Table 6. **External debt and debt service**

Million US dollars

	1992	1993	1994	1995	1996	1997
	External debt					
Total	27 971	30 361	37 193	39 301	39 373	40 706
General government	19 848	22 948	28 275	30 589	30 780	31 627
of which:						
Long-term	16 595	18 473	22 498	23 558	27 180	28 110
Short-term	454	1 446	2 272	3 150	160	410
Military debt	2 799	3 029	3 505	3 881	3 440	3 107
Business sector [1]	8 123	7 413	8 918	8 712	8 593	9 079
of which: Banks and public enterprises	6 803	6 645	8 321	7 915	7 457	7 644
	External debt service					
General government [2]	5 298	4 268	4 316	5 944	6 783	7 132
Interest payments	1 823	1 566	1 617	2 047	2 193	1 896
Amortisation	3 475	2 702	2 699	3 897	4 590	5 236
Business sector	973	973	997	1 045	1 032	1 522
Interest payments	335	238	240	186	266	192
Amortisation	638	735	757	859	766	1 330
Total debt service	6 271	5 241	5 313	6 989	7 815	8 654
	Percentages					
Memorandum items:						
Debt to GDP ratio	28.4	32.9	37.6	33.8	31.7	33.6
of which: General government	20.1	24.9	28.6	26.3	24.8	26.1
Interest service ratio [2, 3]	11.3	10.0	9.4	10.3	11.6	10.1
of which: General government	9.5	8.7	8.2	9.4	10.3	9.2

1. External debt of banks and public enterprises and private non-guaranteed suppliers' credits. Excludes a large share of private sector external borrowing.
2. Covers foreign bonds and long-term loans.
3. Calculated as interest payments as a percentage of the sum of exports of goods (on a settlement basis) and invisible receipts (excluding net EU transfers).

Source: Bank of Greece.

Prospects for 1999 and 2000

Turbulence on financial markets world-wide, the risks for contagion, and slowing growth in Greece's main trading partners increase the risks pertaining to Greece's economic prospects. In this more uncertain environment, output growth is projected to regain momentum after slackening in 1998. The main force

driving domestic demand is set to remain business and public investment (Table 7). Private investment will be underpinned by the growing need to build capacity, as utilisation of installed capacity is already high, and by entrepreneurs' expectations of strong prospects over the medium term. Order books have been

Table 7. **The short-term outlook**

	1997	1998	1999	2000
	Percentage changes			
Output				
Private consumption	2.5	1.9	2.1	2.3
Government consumption	–0.4	0.8	0.6	0.8
Gross fixed investment	9.6	8.0	8.7	9.1
of which:				
Private business investment	10.3	7.8	8.7	10.2
Public investment [1]	7.0	9.5	11.5	9.1
Stockbuilding and statistical discrepancy [2]	–0.1	0.0	0.0	0.0
Total domestic demand	3.5	3.1	3.4	3.7
Exports of goods and services	5.3	5.5	5.0	5.8
Imports of goods and services	5.4	4.5	5.0	6.0
Foreign balance [2]	–0.8	–0.4	–0.7	–0.9
GDP at constant prices	3.2	3.0	3.2	3.4
Prices and wages				
GDP deflator	6.9	4.5	3.1	2.8
Private consumption deflator	5.5	4.8	3.0	2.7
Unit labour costs, total economy	8.0	4.5	3.2	2.7
Labour market				
Total employment	–0.5	1.4	1.5	1.6
Unemployment rate (per cent)	10.3	10.0	9.8	9.4
Labour force	–0.6	1.2	1.2	1.2
	Per cent of GDP			
General government				
Net primary balance	5.7	6.8	6.7	6.5
Net lending	–4.0	–2.7	–2.5	–2.1
Current account balance [3]	–1.7	–1.6	–1.7	–1.9
(Settlements basis)	–4.1	–4.0	–4.1	–4.3
Memorandum item:				
Net official transfers to current account	3.9	4.6	4.7	4.8

1. Includes general government and public enterprises.
2. Contribution to changes in GDP.
3. Based on OECD calculations, which combine customs and settlements data (see Table 4).

Source: Ministry of National Economy and OECD projections.

filling, both in the domestic and external market, and the composite leading indicator is pointing to sustained output growth. Furthermore, business profitability will be boosted by lower labour costs, as wage increases should remain relatively moderate, and by small import price rises as the exchange rate remains stable and international prices are set to rise very slowly from low levels. At the same time, investment by the public sector (including public enterprises) is being boosted by greater disbursements of EU structural funds, as the disbursement pattern of the Second Community Support Framework for the period 1994-99 is back-loaded. Improved utilisation of the structural funds should also be facilitated by the fact that most investment projects are maturing. Household demand is set to rebound only slightly since slower increases in real wages will dampen the growth in household's disposable income. However, the decline in the unemployment rate and the intangible effect of moving towards joining the EMU is expected to give a boost to consumer confidence.

Net exports are projected to continue to restrain growth, partly reflecting Greece's business cycle position, which is stronger than that of its main trading partners. Export volume growth is expected to stabilise at about 5 to 5¾ per cent, in the face of offsetting forces. On the demand side, the full impact of the devaluation on Greece's international competitive position is expected to materialise. However, the initial devaluation is not expected to be fully reflected in gains of market shares, since the drachma has subsequently appreciated, some exporters have recouped part of their squeezed margins by raising their prices, and unit labour costs, though receding, are projected to continue to grow faster than elsewhere. Import growth should increase gradually, as the effect of the devaluation diminishes, activity strengthens, and large infrastructure projects mature to the stage which requires the installation of equipment.

Accelerating output growth should create jobs, thereby allowing the unemployment rate to fall to about 9½ per cent in 2000, though employment growth will be moderate as firms continue to strive for productivity improvements. In spite of an apparent tightening of a relatively rigid labour market, wage pressures should remain moderate through 2000. The two-year agreement signed early in 1998 should lead to a further deceleration in nominal wage increases for 1999, and sustained productivity gains are also expected to contribute to keep down increases in unit labour costs for the business sector to near 2½ per cent in 2000. As cost pressures remain moderate, entrepreneurs should revise their inflation

expectations steadily downwards, and some further progress in lowering inflation is expected. To this end, the investment-led growth should help to alleviate capacity constraints on output growth. Inflation reduction is expected to be supported by a tight monetary policy, as well as a gradual further tightening of the fiscal stance. Based on announced policies, which include the measures contained in the 1999 Budget and the reductions in indirect taxes that have occurred through January 1999, inflation is projected to fall to below 3 per cent by 2000.

Risks surround the central projections, which show steady progress in reducing inflation coupled with robust output growth. The main domestic risk to the projections pertains to the path of inflation reduction. The Government is trying to meet an ambitious investment programme which, if successful, may add significantly to demand pressures. In addition, enterprises may not resist wage claims in an environment of an improving business climate and healthy profits. If the Bank of Greece responds to stubborn inflation pressures by tightening monetary conditions, which may lead to an appreciation of the exchange rate, the tradeable sector would suffer. The other main risk to the projections comes from the possibility that the international downturn is larger than expected. With a dynamic share of exports to countries of Central and Eastern Europe, which are vulnerable to contagion effects, the impact on Greece's exports could be large and output and inflation would slow. However under this unfavourable scenario, the current account would also widen, and this could result in pressures on interest rates in the run up to EMU entry. The likelihood of such a scenario would rise if there were slippage in the agenda for structural reforms or fiscal consolidation.

II. Macroeconomic policies

Macroeconomic policies continue to make steady and creditable progress in reducing the large fiscal disequilibrium that existed at the beginning of the decade and in bringing down inflation from double digits. However, the Government has acknowledged that to attain the objective of meeting the Maastricht criteria by 2000 will require further measures, because the inflation rate remains about 3 percentage points above the EU average and the public debt, which exceeds 100 per cent of GDP, has remained very high, albeit diminishing. Reducing inflation to below 2 per cent by end-1999 (the official target) will be a challenge, as output growth is robust and the March exchange rate devaluation loosened monetary conditions and has had an adverse impact on inflation expectations. However, monetary policy credibility has been strengthened by the introduction of a new framework, comprising inflation targeting, central bank independence and the drachma's entry into the ERM. With less emphasis now placed on the short-term stability of the exchange rate, it will be easier for the Bank of Greece to tighten monetary policy without attracting large capital inflows. Attaining the inflation objective will, nevertheless, need a further amelioration of the policy mix because long-term interest rates may be converging to the lower ones in the euro area and a tightening of monetary conditions, which would entail a large appreciation of the exchange rate, may adversely impact the current account, and in turn, market expectations regarding the sustainability of the central parity.

Fiscal policy has continued to gradually tighten and this has helped reduce inflation pressures. In 1998, the fiscal deficit should decline to below 3 per cent of GDP, which will reinforce the reversal in the debt dynamics first achieved in 1997. However, the underlying deficit remains high and will need to be reduced further in order to achieve a more rapid pace of disinflation, especially as an increasingly larger size of the deficit is financed through capital transfers from

the EU, which provide an additional boost to domestic demand. As a first step, the Greek Government has tightened fiscal policy in 1999 compared with the targets set in the 1998-2001 convergence programme. However, further fiscal adjustment appears necessary for the attainment of the short-term inflation objective. This fiscal adjustment will be difficult since current expenditure is concentrated on a few items which are difficult to cut in the short term (personnel outlays, interest payments, pensions, and health care), and the Government is committed to expand public investment tied to the Community Support Framework II (CSF II). Regarding revenues, base broadening has made steady progress over the past several years, but further yield from measures against tax evasion will probably diminish, while indirect taxes are being used to influence inflation expectations.

Over the medium term, the task of fiscal policy will be even more difficult as the already large primary surplus must be increased if debt reduction is to be attained over a reasonable period, while current commitments will continuously add to expenditure. An eventual entry into the single currency area will require further fiscal adjustment, since budgetary policy would become the main instrument for demand management, and price stability will be more important for output and employment growth if recourse to exchange rate adjustments is to be foregone. Active budgetary policy will require more flexibility, which implies the need for a lower debt and, once that has been achieved, a target of a broadly balanced budget in structural terms. Such budgetary reform will form part of a broad-ranging structural reform programme, as a large share of the budget is committed to entitlement programmes. Making progress on these issues now will provide important signals to markets about future inflation, and the sustainability of the level of the exchange rate, during the crucial period to 2000.

Monetary policy: meeting the Maastricht inflation criterion

Since Parliament approved central bank independence in December 1997, the Bank of Greece has conducted monetary policy in a new framework.[22] Henceforth, the primary objective of the Bank of Greece will be the attainment of price stability. The central bank can support other Government policies to the extent this primary objective is met. Though price stability was not defined by legislation, the Bank of Greece announced that the target for CPI-inflation is a

rate below 2 per cent by the end of 1999. The Bank of Greece considers that this objective will be consistent with meeting the Maastricht criterion for inflation in the first months of 2000, as well as the one on long-term interest rates.[23] The law granting central bank independence also gave the Bank of Greece control over the conduct of exchange rate policy within a framework agreed with the Government.

The Bank of Greece's ability to meet its objectives was further facilitated by the ERM entry of the drachma in mid-March 1998 at a central rate of 357 drachma to the ECU, implying a devaluation by 12.3 per cent (foreign currency per drachma). The setting of a credible central parity should help Greece observe the Maastricht Treaty clause on exchange rate stability for two years prior to qualification for EMU. It will also strengthen the hand of the central bank in the conduct of monetary policy. Nevertheless, in its first monetary policy report (April 1998), the Bank of Greece stated that it would not set an inflation target for end-1998 in view of the lags with which monetary policy affects inflation, as well as the uncertain impact of the devaluation on inflation. Its intermediate target would be to maintain a stable exchange rate, defined as an average annual exchange rate within 2½ per cent of the central rate. However, in view of the primary emphasis placed on the inflation target, the drachma would be allowed to appreciate above the levels implied by this narrow band.[24] The main instruments of central bank intervention were also changed so as to improve the conduct of monetary policy. Henceforth, Bank of Greece interventions in the interbank market rely more heavily on auctions of short-term repos (especially repurchase and deposit agreements of up to 14 days). Earlier, the overnight window was the main instrument. In addition, the Bank of Greece abolished its foreign exchange ''fixing'' window and relies now on forex interventions to influence the foreign exchange markets.

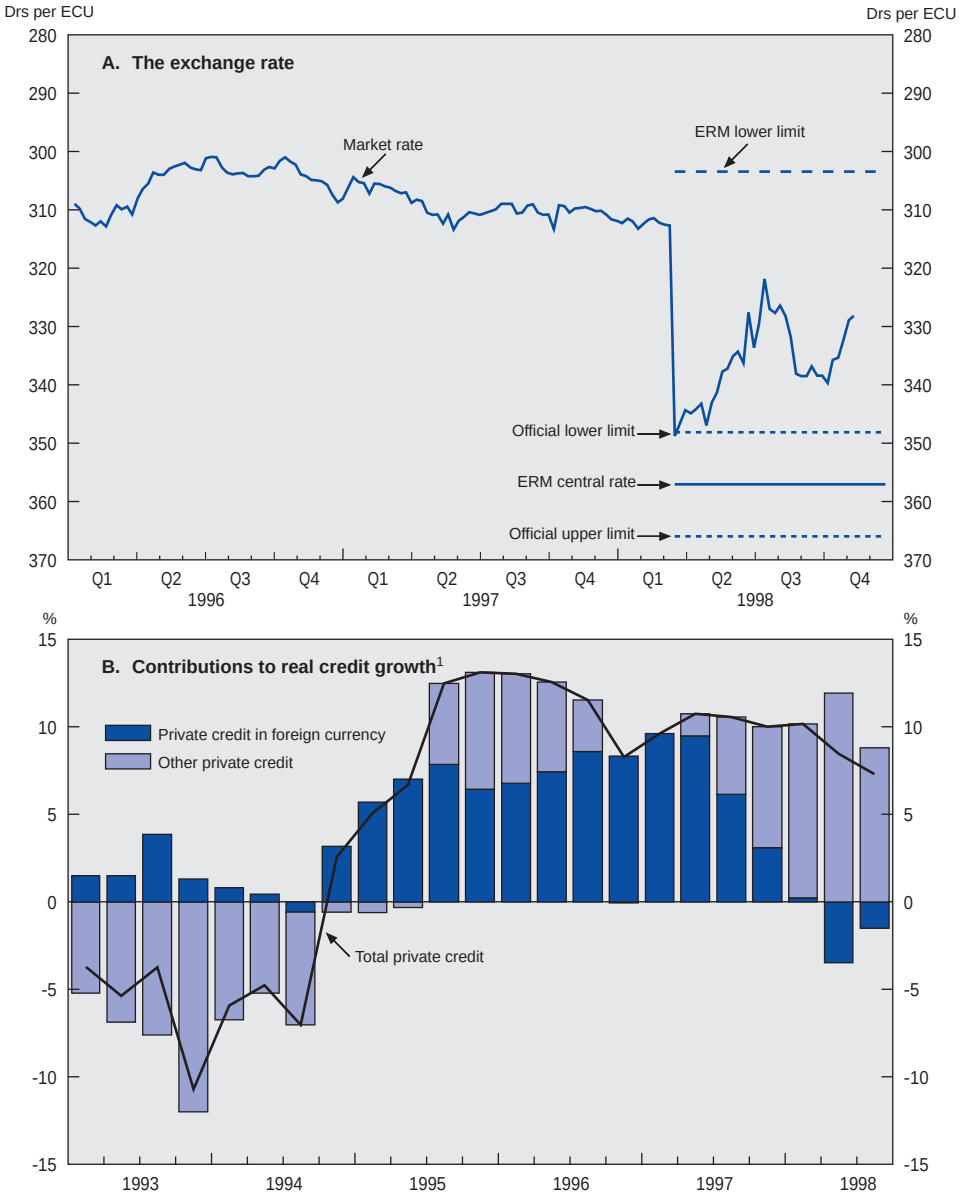
Within the new framework, the central bank follows a new variant of the ''hard drachma'' policy, a policy already pursued during the past several years. This policy has been crucial to the disinflation process and the 1997 end-year inflation target of 4.5 per cent was nearly attained. However, during most of the past three years, the central bank has had to sterilise huge amounts of imported liquidity arising from a tight monetary policy combined with a credible exchange rate commitment. Neutralising the resulting liquidity required recourse to sterilisation, and to raising reserve requirements in 1995 and 1996 to bring them to

12 per cent (and expanding the base on which they are levied). These responses were not sufficient to stem inflows and, in mid-1997, a two-tier window was introduced in the overnight interbank market (with the lower rate applying to deposits exceeding a pre-determined maximum level). This instrument did have some success in discouraging capital inflows, as did greater exchange rate fluctuations.[25]

The new monetary framework has facilitated the implementation of a tight monetary policy with less incentives to attract large capital inflows, which have plagued monetary policy in the past. After the March 1998 devaluation Greece was again initially swamped by capital inflows, as there was a very positive reaction by international capital markets, and Greece was viewed as the last opportunity for a ''convergence play''. The central bank had initially to sterilise as much as Dr 1.7 trillion (equivalent to 5 per cent of GDP), though the amount tapered off to below 800 billion after August 1998, in part due to the sharp appreciation of the drachma above the central rate. The exchange rate was allowed to appreciate by nearly 10 per cent above the central rate during the period to late August 1998 (Figure 6, Panel A). By creating a sufficiently large depreciation risk, the Bank of Greece kept domestic interest rates at the higher level necessary to combat inflation without attracting large capital inflows. Following a dip in the exchange rate in September and October, the exchange rate again appreciated in November after international capital market turbulence abated.

Though Greece has faced capital inflows for most of the past few years, there have been two recent periods of capital outflows which have complicated the task of monetary policy – October 1997 to March 1998 and August to September 1998. The first was triggered by the turbulence in international capital markets originating from the economic crises in Asia. At end October 1997, overnight intervention rates had to be briefly raised to 170 per cent (on marginal financing from the central bank) to stem large capital outflows. Interest rates did not return quickly to pre-crises levels, with the one-month interbank rate remaining above 15 per cent until the mid-March depreciation, as temporary episodes of outflows recurred. The volatility in Greek financial markets arising from the ''flight to quality'' was exacerbated because of domestic considerations. Most importantly, the competitiveness of the Greek economy had fallen considerably because of the prolonged use of the exchange rate as the main anti-inflationary

Figure 6. **The exchange rate and private sector credit growth**

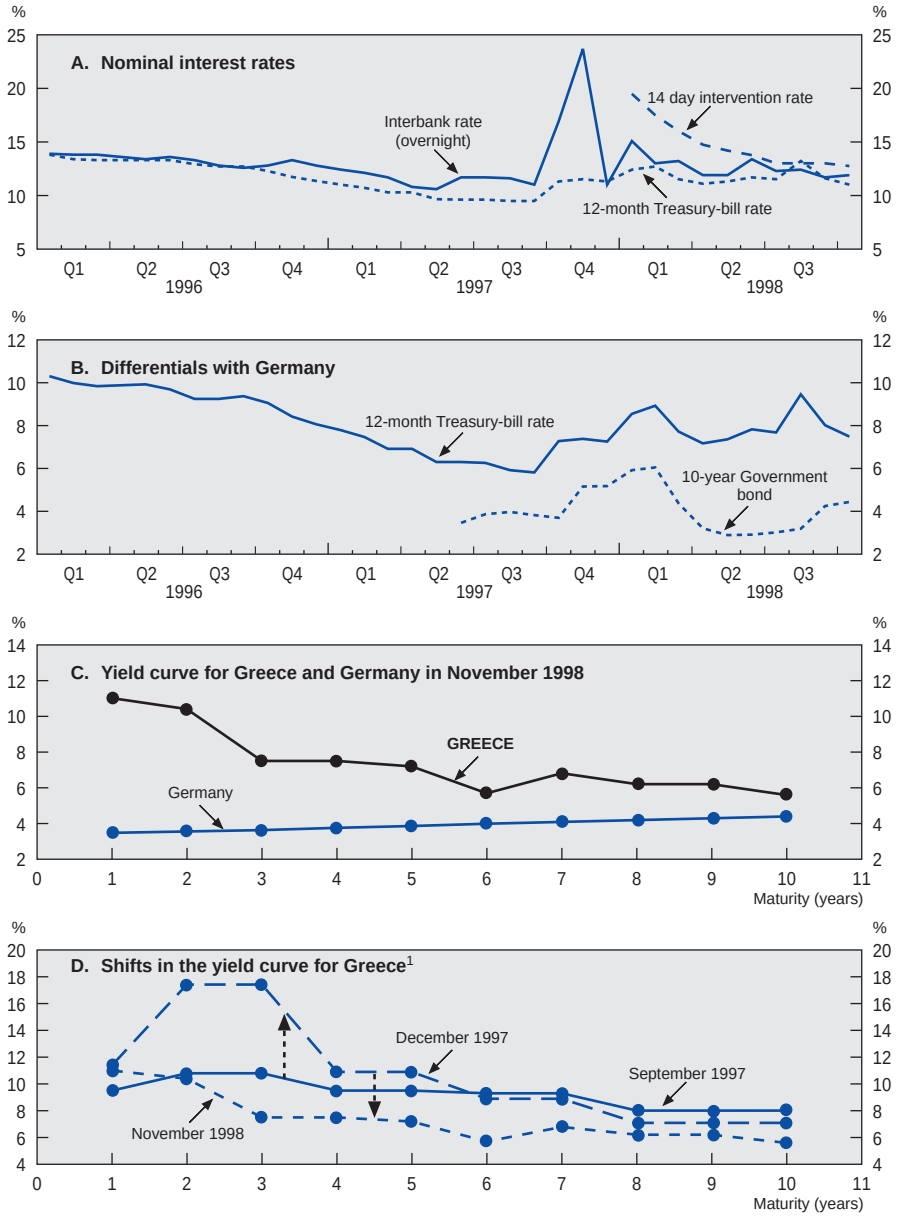


1. Deflated by the consumer price index.
Source: Bank of Greece.

policy tool, without sufficient support from income and structural policies to attenuate the appreciation of the real effective exchange rate. This situation reinforced concerns about the sustainability of the current account, which had widened to 4 per cent of GDP in 1997 (settlements basis), and of the Greek external debt which had increased rapidly. Another important factor that contributed to the eventual depreciation of the drachma in March was the financial markets' expectation that Greece would need to devalue the drachma in 1998 so as to then be in a position to meet the Maastricht treaty obligation to keep the exchange rate stable for two years prior to entry into the EMU. The second period of capital outflows was of a brief duration. The speculation beginning at the end of August 1998 originated primarily in international markets, and quickly abated as ERM entry in conjunction with the devaluation have lessened the concerns about the external disequilibrium, and appear to have situated the Greek economy closer to that of the euro-area countries in the eyes of foreign investors.

In addition to the volatility arising from the external sector, developments in the domestic economy are likely to pose difficulties for the anti-inflationary policy, though the situation has improved in 1998 compared with 1997. Private-sector credit demand is strong, but decelerated slightly to 7½ per cent in real terms in September 1998 (including valuation effects of about 1 percentage point from the devaluation) from 10½ per cent at end-December 1997, including an important reduction in foreign-currency credits (Figure 6, Panel B). The public sector financing requirement has also been quite restrictive compared with 1997. In 1997, only 25 per cent of the Government's net borrowing requirement was financed through the placement of securities with the domestic non-bank public, mostly through the primary market as the secondary market was not yet very active (compared to an 80 per cent share in 1996 and a 70 per cent share in 1995).[26] The low level of non-bank financing was partly linked to the interest rate policy followed for government securities, as the Government's aggressive reduction of the benchmark interest rate on Treasury bills until September 1997 resulted in a yield of 3 per cent (after tax) in real terms (not much different from the return on time deposits). Through August 1998, the domestic non-bank public has covered 100 per cent of the public financing requirement in the primary market, but less than 10 per cent if one takes into account transactions in the budding secondary market.[27]

Figure 7. **Interest rates, spreads and yield curves**



1. The 12-month Treasury bill rate was used for year one of the yield curve.
Source: Bank of Greece, Deutsche Bundesbank and OECD.

Other indicators of a renewed inflation slowdown are mixed. The investment-led recovery may be reducing capacity constraints, but enterprise capacity utilisation is high and underlying inflation was slow to reverse; from a peak of 6 per cent in August 1998, it was 5.3 per cent in November compared with a low of 4½ per cent prior to the March devaluation. The Bank of Greece has signalled that inflation reduction will require support from fiscal, incomes, and structural policies and that it will maintain its intervention rate as high as necessary to reduce inflation. To this end, the benchmark 14-day deposit rate was maintained at 12.75 per cent until December 9, 1998 when it was reduced to 12.25 per cent, and the yield curve remained inverted, with the implied 10-year forward one-year interest rate about 500 basis points below the current one-year interest rate. In comparison to DM denominated instruments, the one year spread was about 8 percentage points and the spread between yields on 10-year securities was about 400 basis points (Figure 7). The yield curve indicates that market expectations are placing an increasing probability on Greece's entry into the EMU as there is a significant kink in the yield curve corresponding to the date of prospective entry.

Fiscal policy: continued adjustment is necessary to improve the policy mix further

The 1997 Budget outturn: a reduction in the deficit but the fiscal stance remains expansionary

Deficit reduction in 1997 was somewhat sharper than programmed. The general government budget deficit fell from 7.5 per cent in 1996 to 4.0 per cent of GDP, compared to a target of 4.2 per cent of GDP. However, a much lower than expected share of the adjustment came from the current primary balance (Table 8). The overall target was achieved mainly by a large decline in interest payments,[28] and the use of equity acquisitions in lieu of capital transfers to public enterprises. The latter operation effectively moves public investment off budget, thus eliminating its negative impact on the deficit, and in fact improves the budget balance.[29] The fiscal stance, thus, remained expansionary after taking into account the equity acquisitions (1.3 per cent of GDP) as well as the expenditures which have been credited to past budgets but were paid in 1997 (0.2 per cent of GDP)[30] – with both adjustments in line with Eurostat rules. Another reason why

Table 8. **General government budget adjustment between 1996 and 1997**

As a per cent of GDP[1]

	Budget[2]	Outturn	Difference
Total	**3.2**	**3.5**	**0.3**
Interest payments	**1.1**	**2.5**	**1.4**
Primary balance	**2.1**	**1.0**	**–1.1**
Current account	**2.2**	**0.4**	**–1.8**
Central and local governments	1.6	0.4	–1.2
Revenue	1.6	0.8	–0.8
Expenditure	0.0	–0.4	–0.4
Social security	0.6	0.0	–0.6
Capital account (net)	**–0.1**	**0.6**	**0.7**

1. Totals may not add up due to rounding.
2. There is only a budget for the central government, with Ministry of Finance projections for the corresponding general government budget (comprising central government, social security and local authorities).

Source: Ministry of National Economy.

the budget is more expansionary than the overall deficit numbers indicate is that the increasing investment programme is being financed to a great extent by EU capital transfers (equivalent to about 2 per cent of GDP in 1997). These funds raise national income and push up domestic demand. Judged by the change in the cyclically-adjusted primary balance, the stance of fiscal policy was broadly neutral in 1997 (maintaining an unchanged surplus after including the adjustments for equity acquisitions and shifts of expenditures to past budgets).

The 1997 Budget made significant progress in broadening the tax base without increasing tax rates. Overall, revenues increased by an impressive 12½ per cent, but fell short of the ambitious target of an increase of 18 per cent, mainly due to the performance of indirect taxes. This is partly explained by fuel price movements arising from the fall in international prices and a cut in the tax on heating fuel (with the latter designed to dampen inflation). Nevertheless, substantial progress was made in the fight against VAT evasion, and the elasticity of total indirect taxes with respect to private consumption was greater than one (despite there having been hardly any increase in excise taxes). In contrast, direct taxes increased by the budgeted amount (14 per cent in real terms). The main factors driving their substantial increase were cuts in tax expenditures, the non-indexation of tax brackets, and the increase in real wages. On the expenditure

side, the authorities failed in their budgetary objective to contain primary expenditures. The main area of slippage was the wage bill, as wages increased by 7½ per cent in real terms (leading to a two year cumulative increase of 13½ per cent in real terms), while general government employment rose by 1½ per cent despite the commitment not to replace all retiring civil servants.[31] Investment expenditure and capital transfers – two-thirds co-financed by the EU's structural funds – increased by about 10 per cent (national accounts basis) to 5 per cent of GDP. However, their impact on the deficit was more than fully offset by the budgetary impact of equity acquisitions.

The fiscal adjustment was sufficient to permit a turn-around in the debt dynamics, with the debt to GDP ratio declining from 112 per cent of GDP at end-1996 to 109 per cent of GDP at end-1997 (Table 9). The recorded primary surplus of 5.5 per cent of GDP would have resulted in a much larger fall in the debt ratio but for several ''quasi-fiscal'' financial operations, which added about Dr 0.8 trillion to the debt (equivalent to about 2½ per cent of GDP).[32] Nevertheless, progress has been made in containing the amount of newly-issued government-guaranteed debt, and thus the potential for future take-overs by the Government. At end-1997, the stock had fallen to 6.7 per cent of GDP from 8.5 per cent at end-1995, with new issues equivalent to 1.1 per cent of GDP; for 1998, the stock is estimated to have fallen to 6.1 per cent of GDP with new issues equivalent to 0.6 per cent of GDP.

The Government's emphasis on improving debt management forms a key part of its fiscal strategy. To deepen the bond market and take advantage of declining interest rate expectations, several new medium and long-term bonds were introduced in 1997 and 1998 (fixed-interest bonds of 3, 5, 7, 10 and 15 year maturities, two-year zero-coupon bonds and 5 and 10 year inflation-indexed bonds). These lengthened the maturity of the newly issued drachma-denominated public debt from 3.3 years in 1996 to an estimated 4.2 years in 1998, reduced the sensitivity to interest rate fluctuations, and established a yield curve. In addition, the government bond market's liquidity and price-setting properties have also been greatly enhanced by the introduction of an electronic secondary market (HDAT), and a network of primary dealers through which all securities have henceforth been placed using an auction (while previously, prices had often been administratively set).[33] The 1997 financing of the deficit, however, was achieved mostly from external sources, partly due to slow take-up by domestic investors

Table 9. **Public sector debt**

Billion drachmas

	1993	1994	1995	1996	1997	1998[1]
Central government	24 078	27 910	31 781	36 043	38 837	41 041
Domestic	19 596	22 522	26 109	29 666	31 384	32 698
Treasury bills	5 766	7 533	8 422	10 012	6 800	5 106
Government bonds	9 551	10 438	12 804	14 748	19 563	22 373
of which:						
Restructuring	5 397	5 179	5 291	4 736	4 171	. .
Foreign-currency denominated	1 702	1 879	1 574	239	151	. .
Bank of Greece	3 444	3 622	3 866	3 949	4 023	4 186
Military	835	929	1 017	956	998	1 033
External	4 482	5 388	5 672	6 377	7 453	8 343
Local authorities	80	87	93	121	123	. .
Social insurance organisations	369	221	242	191	193	. .
Other	96	96	93	90	76	. .
Inter-governmental debt	1 031	2 091	2 608	3 123	3 366	. .
General government debt	23 592	26 223	29 601	33 323	35 863	37 500
Domestic	19 100	20 825	23 919	26 945	28 410	29 157
External	4 492	5 398	5 682	6 377	7 453	8 343
Public enterprises	2 021	2 491	2 325	2 542	2 807	. .
Public sector debt[2]	25 613	28 714	31 926	35 864	38 670	. .
Memorandum items:						
Debt as a per cent of GDP						
Central government	113.9	116.4	118.2	121.4	118.6	115.5
General government	111.6	109.3	110.1	112.2	109.5	105.5
Public sector[3]	121.2	119.7	118.8	120.8	118.1	. .

1. Official estimate included in 1999 Budget.
2. Public enterprises inter-governmental debt has not been excluded from total debt.
3. Includes public enterprises.

Source: Ministry of Finance.

following the rapid reduction in domestic interest rates during the first three quarters of 1997. In late 1998, the Government issued Dr 250 billion of tax-free bonds (if held to maturity) and Dr 340 billion of privatisation certificates exchangeable into shares of public enterprises.[34]

To improve financial asset management, the Government created a new public enterprise, Public Enterprise for Liquid Assets (DEKA). It will receive equity of other public enterprises held by the Government and will benefit either from dividends from these stocks or from the sale of these assets. It can use these

revenues, *inter alia,* to provide transfers to other public enterprises or dividends to the government budget. Its original endowment comprised the National Bank of Greece's (NBG) convertible bonds and 3½ per cent of the equity in the national telephone company (OTE). The sale of OTE shares and about 30 per cent of the NBG asset garnered about Dr 200 billion for DEKA. It is also expected to receive the proceeds from the eventual sale of the duty free shops.

The 1998 Budget: meeting the Maastricht criterion and containing primary expenditures

The 1998 Budget aimed to reduce the general government deficit from 4 per cent of GDP in 1997 to 2.4 per cent of GDP in 1998, and thus meet the Maastricht Treaty's deficit criterion. The adjustment strategy relied on: *i)* broadening further the tax base; *ii)* continuing to make the tax system more equitable through the taxation of previously low-taxed assets; and *iii)* taking strong action to contain current primary expenditures. Most importantly, incomes policy for civil servants set cost of living increases at 2.5 per cent, following the large real wage increases obtained over the last few years. As (primary) pensions for the whole economy increase in line with civil-service cost-of-living increases, civil service wage restraint has an important additional budgetary effect.[35] Employment will be restricted by the continuation of the 1997 recruitment policy. Additional savings will be achieved through measures to cut expenditures on pharmaceuticals by social security funds (see Chapter III) and by the continued implementation of the structural measures begun during 1997 to improve expenditure efficiency.

Revenue enhancement is based on the introduction of over 10 new measures, together with continued gains from the changes to the tax system to fight evasion.[36] The strong rise in direct taxes arises mostly from the advancing of payment schedules for both personal and corporate taxes. In addition, a 0.3 per cent fee was introduced on stock market transactions. Regarding indirect taxes, a further reduction of tax evasion is expected to produce a high elasticity with respect to private consumption, despite the fact that excise taxes have remained unchanged in nominal terms, except for the tobacco excise tax (up Dr 50 per packet).[37] Public investment and capital transfers were projected to rise by 21 per cent with an increase in capital transfers from the EU (by the equivalent of ½ per cent of GDP to 2½ per cent of GDP) and equity acquisitions (by the equivalent of

Box 2. **Structural reform initiatives of the 1997 and 1998 Budgets**

Curbing expenditure

i) Reduction in the large number of public entities (some 1 750) by 170 which no longer serve useful purposes. Process began in 1997 and continues in 1998. Affected personnel of about 5 500 is transferred to other parts of the civil service, local authorities, prefectures, or hospitals.

ii) Requirement for all public entities receiving transfers from the budget, or a credit guarantee, to submit business and restructuring plans. Implementation during 1998.

iii) Introduction of a System of Logistics Inspectors which will assist public entities which receive transfers from the budget to formulate and implement their budgets.

iv) Reduction in automatic transfer of fees collected on behalf of public entities by 25 per cent (with the exception of social security funds and transfers related to wages and salaries). Started in 1997.

v) Introduction of a process of clearing inter-government agency and enterprise arrears. Process started in 1997, but slow progress in dealing with Dr 450 billion in arrears.

vi) Limitation of new government guarantees to 3 per cent of ordinary budget expenditure (with the exception of guarantees required for international institutions).

vii) Introduction of a new, more transparent, salary scale for civil servants which eliminates the plethora of *ad hoc* bonuses. Started in 1997 and completed in 1998.

viii) Reform of the costly health care system of civil servants. To begin in 1998.

ix) Legislation to rationalise local government (the Capodistrias Law, named after the Greek leader of the 1830s). The number of local authorities will be reduced from 6 000 to 1 300 in 1998. Agreement on sharing of central budget revenues (mainly a share of income taxes, car registration fees and taxes on interest from bank deposits).

x) Separation of the budget of the 13 regions (second level of regional government) from central budget. Following the shift of public hospital wage bill expenditure to the central budget in 1999, regional government budget expenditure is equivalent to about 1½ per cent of GDP.

Revenue collection

i) Tax police to randomly check establishments' tax reporting. Began in mid-1997, and comprises 1 300 officers.

ii) ''TAXIS'' computerised tax information system to permit the cross checking of tax returns between taxpayers. Four pilot programmes were in place since mid-1998, 120 will function by end-1998, and the remainder by end-1999. The electronic linkage between the tax offices will commence in 1999.

(continued on next page)

(continued)

iii) ESCORT computer system identifies potential VAT shortfalls. Introduced in 1997, and will be expanded in 1998.
iv) Implementation of US Internal Revenue Service recommendations for improving tax collection and tax auditing procedures. In 1998.
v) System of national tax audit centres in conjunction with regional audit centres. Commenced operations in 1998.
vi) Computerisation of customs offices. In the process of implementation.
vii) Real estate tax introduced in 1997. Creation of a land registry to support collection of this tax. Land registry in the process of being established, in part using special 1997 tax declarations.
viii) Introduction of tax identity numbers and a pension registry.
ix) Introduce into the budget the plethora of off-budget accounts.

0.3 per cent of GDP to 1.7 per cent of GDP) dampening the full impact on the budget. Overall, these measures were budgeted to produce a primary surplus adjustment of about 1½ per cent of GDP, of which one-third would be offset by a larger net investment budget, thus taking the recorded primary surplus to 6½ per cent of GDP. The far-ranging programme of structural reforms underpinning a prolonged effort to improve the efficiency of budgetary operations (on both the expenditure and revenue side of the budget) is described in Box 2.

At the time of the March 1998 devaluation of the drachma, the Government announced a package of measures to support the new exchange rate, including budgetary measures. To offset an estimated budget deterioration of the order of 1 per cent of GDP resulting from the impact of the devaluation on the budget and much higher than envisioned interest rates:[38] *i)* primary current expenditures would be cut mainly through reduced transfers to public enterprises and entities (0.3 per cent of GDP); *ii)* social security revenues would rise due to the increase in contributions from a programme to register illegal immigrants (0.2 per cent of GDP); and *iii)* the investment budget would be reduced through a cut in nationally-financed investments (0.3 per cent of GDP). At the same time, the EU's share of the co-financed investments would be increased from about 65 per cent to 70 per cent (0.1 per cent of GDP), as the EU agreed that the higher post-devaluation drachma value of the EU structural funds could supplement national resources to finance investments. Based on the revised official projections, the

cyclically-adjusted primary deficit would be contractionary, but by less than ½ per cent of GDP after adjustments for the increase in equity acquisitions and the additional financing of the investment budget by EU funds (Table 10). The Government also introduced a timetable for structural reform for the next 18 months leading up to Greece's hoped-for qualification for entry into the euro area. The structural reform programme includes:

- the sale of a third tranche of equity in the national telephone company, the sale of about 20 per cent of the public petroleum company, the privatisation of three small banks and 10 small public enterprises, resulting in revenues of almost 2 per cent of GDP;
- the implementation of restructuring programmes in loss-making public enterprises (see Chapter IV); and
- the introduction of a labour market reform and a small social security reform in 1998. The timetable for a major social security reform will be announced in late 1998 (see Chapter III).

Though an evaluation of the budgetary performance is difficult – especially in the case of Greece where a large share of expenditures are made in the last quarter – data for the first 10 months of the year (administrative basis) indicate that the growth of both revenues and expenditures at the central government level exceed that implied by their original budget targets. Revenue growth was 15 per cent compared with a budget target of 11 per cent, though this pace is expected to slow due to late-year cuts in indirect taxes. VAT revenues, which comprise a third of tax revenue, are exceeding their ambitious budget target. The strong growth of revenues also reflects the change in the payment schedules on direct taxes, and higher dividends from state-owned enterprises and the central bank. On the expenditure side, the main sources of slippage are the wage bill and interest payments. Overall, the deficit had declined by less than budgeted during this 10-month period. However, the Government announced that it was committed to take additional measures to contain primary expenditure during the remainder of the year and that it expects the general government deficit to have fallen to 2.2 per cent of GDP, overperforming relative to the budget target of 2.4 per cent of GDP. The OECD projections were based on announced measures as of end-October and, thus, embody a 1998 general government deficit of 2.7 per cent of GDP (Tables 7 and 10). The Government has accelerated the pace of debt reduction by applying privatisation revenues to reduce the debt. Excluding the

Table 10. **General government accounts**[1]

Per cent of GDP

	1993	1994	1995	1996	1997	1998 Official estimates	1999 Official projections
Current revenue	**35.0**	**36.6**	**37.7**	**38.0**	**38.9**	**39.3**	**39.1**
Direct taxes	5.8	6.9	7.3	7.1	7.5	8.0	7.8
Households	3.3	4.1	4.4	4.2	4.5	n.a.	n.a.
Business	2.5	2.7	2.9	2.8	3.0	n.a.	n.a.
Indirect taxes	14.0	13.6	13.5	13.7	14.2	14.3	14.3
Social security contributions	12.1	12.3	12.6	12.7	12.6	12.7	12.9
Other	3.1	3.8	4.3	4.5	4.6	4.3	4.1
Current expenditure	**43.0**	**43.7**	**44.9**	**43.0**	**40.6**	**39.8**	**39.0**
(Primary current expenditure)	30.3	29.6	31.9	31.0	31.0	30.6	30.3
Consumption	14.5	13.9	15.5	14.6	14.8	14.4	14.0
Wages	11.1	10.8	11.4	10.8	11.0	10.7	10.5
Non-wages	3.4	3.2	4.1	3.8	3.7	3.7	3.6
Interest payments	12.8	14.1	12.9	12.0	9.6	9.1	8.7
Transfers and subsidies	15.8	15.7	16.4	16.4	16.2	16.3	16.3
Business	0.5	0.4	0.4	0.5	0.5	0.4	0.3
Households	15.6	15.7	16.1	16.0	15.7	15.8	15.8
Rest of the world	–0.4	–0.4	–0.1	–0.0	–0.0	0.2	0.2
Net saving	**–8.0**	**–7.2**	**–7.2**	**–5.0**	**–1.7**	**–0.5**	**0.0**
Net capital transfers[2]	2.7	–0.2	0.3	–0.8	–1.1	–2.0	–2.0
Fixed investment	3.1	3.0	3.2	3.3	3.3	3.7	4.0
Net lending	**–13.8**	**–10.0**	**–10.6**	**–7.5**	**–4.0**	**–2.2**	**–1.9**
Primary balance	**–1.0**	**4.1**	**2.3**[3]	**4.5**	**5.7**	**6.9**	**6.8**
(Primary current balance)	4.7	6.9	5.8	7.0	7.9	8.6	8.7
Memorandum items:							
Net lending of:							
Central government	–15.3	–12.9	–12.4	–9.9	–6.1	–4.9	n.a.
Social security and hospitals	1.6	2.8	2.0	2.4	2.0	2.3	n.a.
Regional government	0.0	0.0	0.1	0.0	0.0	0.0	n.a.
Payments on called government guarantees	2.4	0.1	0.0	0.0	0.4	0.0	0.0
Equity participations and payments allocated to past budgets	0.0	0.0	–0.7	0.9	1.5	1.7	1.6
Net lending reduced by amount of equity participations and payments allocated to past budgets and increased by payments on called government guarantees	–11.4	–9.9	–9.9	–8.4	–5.2	–3.8	–3.6

1. On a national accounts basis.
2. Includes transfers from EU to support investment projects.
3. The primary surplus was 4 per cent of GDP prior to revision of the accounts to include payments for judges' back-pay that will be paid during 1996-2000 and other payments shifted to the 1995 Budget.

Source: Ministry of National Economy.

privatisation certificates, about Dr 0.6 trillion was applied to debt reduction (about 1.7 per cent of GDP).[39] The debt to GDP ratio was also favourable affected by lower-than-budgeted equity acquisition in 1998 (1.5 per cent of GDP compared with a budgeted 1.7 per cent of GDP).

The 1998-2001 convergence programme: the road map for EMU entry

The Government's new convergence programme outlines policy initiatives geared towards meeting the Maastricht criteria by early 2000.[40] EMU could then be joined at the start of 2001. The policies underpinning the programme are the structural reforms announced at the time of the March 1998 devaluation and a fiscal adjustment based on containing current primary expenditures. Expenditure restraint will focus on public consumption. Non-wage consumption is projected to decline by 0.3 per cent per year in real terms, and employment will fall through the continuation of the current recruitment policy. Additional gains will come from the continued implementation of the measures contained in the 1998 Budget to improve administrative efficiency and revenue collection (described in Box 2). Despite these efforts, the primary surplus is projected to remain basically unchanged between 1998 and 2001, reflecting the desire to fully utilise the EU structural funds. Specifically, higher capital spending is projected to broadly offset a 1½ per cent of GDP reduction in current primary expenditures over the period. The general government deficit could, nevertheless, decline to 0.8 per cent of GDP in 2001, owing to a convergence of interest rates toward the euro-area level. The decline in the debt to GDP ratio would only amount to 7 percentage points since equity acquisitions and other asset transactions are envisaged to add annually approximately 2 per cent of GDP to the public debt and the current privatisation programme was not extended (Table 11).

The convergence programme is based on a strong growth cum low inflation scenario. An increase in public investment (including public enterprises) from 6.7 per cent of GDP to 9.7 per cent of GDP is expected to be the driving force behind an output growth that rises from 3.5 per cent in 1998 to 4.5 per cent in 2001 (the respective ratios for general government investment are 3.8 per cent and 5.2 per cent). In the official projections, inflation reduction is supported by: *i)* sharp productivity gains, which dampen unit labour costs; *ii)* low import prices; and *iii)* diminished capacity constraints. Over the four year period, consumer price inflation declines from 4.5 per cent in 1998 to 1.7 per cent in 2001.

Table 11. **The 1998-2001 Convergence Programme**

	1997[1]	1998	1999	2000	2001
	Percentage changes				
Output					
Real GDP growth	3.5	3.5	3.7	3.9	4.5
Private consumption	2.5	1.8	1.9	2.1	2.2
Gross fixed capital formation	10.9	10.7	12.2	12.1	13.0
Public	18.2	18.5	19.1	17.2	17.4
Private	8.3	7.7	9.2	9.6	10.8
External sector[2]	–0.9	0.3	–0.5	–0.5	–0.4
Exports	5.2	8.4	7.4	7.8	8.6
Imports	5.9	4.2	5.9	6.2	6.6
Labour market					
Employment	–0.5	1.0	1.1	1.2	1.7
Unemployment rate	10.3	9.8	9.3	8.6	7.5
Labour force	–0.6	0.5	0.5	0.5	0.5
Prices and wages					
Private consumption deflator	5.5	4.5	2.5	1.9	1.7
Wages[3]	10.7	6.0	4.1	3.9	3.8
Unit labour costs	6.5	3.5	1.5	1.2	1.1
	As per cent of GDP				
Fiscal account					
General Government deficit	4.0	2.4	2.1	1.7	0.8
Total current revenue	38.2	38.8	38.9	38.8	38.6
of which: Non-tax	4.4	4.3	4.2	4.1	4.0
Total expenditures	42.2	41.1	41.0	40.7	39.3
Current primary expenditures	31.0	30.4	29.9	29.5	28.8
Wage	10.9	10.6	10.4	10.1	9.8
Non-wage	3.8	3.7	3.7	3.6	3.6
Transfers	16.3	16.0	15.8	15.7	15.5
Interest payments	9.5	9.1	9.0	8.6	7.8
Net capital expenditure	1.7	1.6	2.1	2.6	2.7
of which: Gross investment	3.3	3.8	4.3	4.8	5.2
Primary balance	5.5	6.7	6.9	6.8	6.9
Debt of General Government	108.6	107.8	105.8	102.5	99.8
Current account deficit[4]	2.4	2.7	2.7	2.6	2.5
(EU current transfers)	2.2	2.3	2.2	2.2	2.2
Memorandum items:					
Interest rate on 12-month T-bills	10.3	9.5	7.5	6.5	5.5
EU import growth	7.7	8.1	7.0	7.0	7.0
Eastern and Central Europe import growth	11.5	11.4	10.0	10.0	10.0
Import elasticity	1.4	1.1	1.4	1.4	1.3

1. National accounts subsequently revised. Output growth was revised down to 3.2 per cent.
2. Contribution to GDP growth.
3. Compensation per employee, total economy.
4. National accounts basis. Excluding capital transfers.

Source: Ministry of National Economy.

The current account remains broadly unchanged as the import elasticity remains below 1½ and exports are buoyed by strong growth in Greece's main markets.

The 1999 Budget: the budget for EMU entry required a tightening to dampen inflation

In view of the priority placed on inflation reduction and the need for a cushion in view of the more uncertain international environment, the 1999 Budget was tightened relative to the convergence programme, so as to more than fully offset the prospectively higher interest payments. The deficit target is 1.9 per cent of GDP rather than the programme's 2.1 per cent of GDP and thus will be about a ¼ percentage point more restrictive compared with the 1998 official estimate (2.2 per cent). However, the fiscal stance will be broadly neutral if judged by an unchanged primary surplus. The adjustment effort will focus primarily on cuts in current expenditure. In addition to the measures contained in the convergence programme, public sector incomes policy will remain tight, with the wage bill increasing by 4 per cent, as cost of living increases will be 2 per cent for civil servants (in line with the Bank of Greece's end of period inflation target).[41] Within the austere primary current expenditure target (4¼ per cent growth compared with 1998), and in line with Government objectives, the education, health and agriculture sectors and regional governments, will receive relatively favourable spending limits (Table 12). No major new revenue initiatives are envisaged, though tax brackets will not be indexed for inflation, presumptive taxation criteria for the self employed will be increased, and a programme of discounts for the closure of pending tax cases has been re-introduced from late 1998. The budget will have to make up the revenue slowdown from an increase in withholding taxes during 1998 (Dr 130 billion), and cope with diminishing yields from past efforts to clear tax arrears, the payments on maturing zero-coupon bonds, and a continued fast pace of public investment. In addition, the Government has reduced indirect taxes on cars, fuels, and electricity (and plans to maintain administrative prices broadly unchanged) with a view to dampen inflation. To this end, gasoline prices were reduced by 5 per cent in late September 1998, the excise tax on heating oil was reduced by 30 per cent in mid-October, the special excise tax on cars was reduced by 20-25 per cent on November 16 and the VAT on electricity was reduced from 18 per cent to 8 per cent effective from January 1, 1999. The total cost to the 1999 Budget is

Table 12. **Budget plans and outcomes**

	1996	1997		1998		1999
	Outcome	Budget	Outcome	Budget	Official estimate	Budget
	Administrative basis					
Central government						
Current expenditure	9 747	10 225	10 064	10 424	10 570	11 050
(per cent of GDP)	32.8	31.3	30.7	29.4	29.7	29.1
of which:						
Salaries and pensions	3 055	3 377	3 513	3 652	3 770	3 914
Goods and services	468	490	588	568	616	620
Grants and subsidies	1 491	1 511	1 495	1 543	1 540	1 619
Interest	3 501	3 468	3 216	3 220	3 270	3 350
Transfers to the EU	355	397	377	438	430	435
Current revenue	7 384	8 715	8 467	9 376	9 450	10 030
(per cent of GDP)	24.9	26.7	25.8	26.4	26.6	26.5
Direct taxes	2 316	2 790	2 767	3 111	3 260	3 474
Indirect taxes	4 300	5 014	4 834	5 397	5 331	5 615
Non-tax revenue	768	911	866	868	860	942
Transfers from the EU	88	67	45	65	37	60
Investment budget balance	–528	–849	–908	–1 115	–965	–1 135
Transfers from the EU	552	700	698	840	890	1 030
Primary expenditure (administrative basis)	7 341	8 423	8 474	9 209	9 175	9 895
(per cent of GDP)	24.7	25.8	25.9	26.0	25.8	26.1
Budget deficit (national account basis)[1]	2 890	2 010	2 002	1 560	1 498	1 535
(per cent of GDP)	9.7	6.2	6.1	4.4	4.2	4.0
Primary surplus (national account basis)[1]	611	1 458	1 131	1 660	1 710	1 815
(per cent of GDP)	2.1	4.5	3.5	4.7	4.8	4.8
Public entities deficit[2]	50	272	101	200	207	80
(per cent of GDP)	0.2	0.8	0.3	0.6	0.6	0.2
Excluding transfers, and EU and special funds	898	1 121	1 038	1 166	1 158	1 088
(per cent of GDP)	3.0	3.4	3.2	3.3	3.3	2.8
Public enterprises deficit	–122	75	36	–33	215	528
(per cent of GDP)	–0.4	0.2	0.1	–0.1	0.6	1.4
Excluding transfers, and EU and special funds	387	854	1 079	1 105	1 366	1 601
(per cent of GDP)	1.3	2.6	3.3	3.1	3.8	4.2
Memorandum items:						
Total additions to debt	944		1 085		1 075	1 074
(per cent of GDP)	3.2		3.3		3.0	2.8
Debt take-overs	368		432		165	181
Capitalised interest	179		33		27	. .
Equity purchases	200		420		525	620
Military debt amortisation	197		200		358	273
	Budget balances, cash basis					
Central Government[3]	–3 870		–2 620			
(per cent of GDP)	–13.0		–8.0			
Public entities	647		345			
Public enterprises	–83		133			
Public sector deficit[3]	–3 306		–2 142			
(per cent of GDP)	–11.1		6.5			

1. Including adjustments applied to arrive at national account basis; equity acquisitions and payments applied to past budgets amounting to: Dr 503 billion in 1997 (1.5 per cent of GDP), Dr 587 billion in 1998 (1.7 per cent of GDP) and Dr 620 billion in 1999 (1.6 per cent of GDP).
2. Only the 7 most important entities.
3. Includes capitalised and accrued interest amounting to Dr 179 billion in 1996 and Dr 33 billion in 1997.

Source: Ministry of Finance.

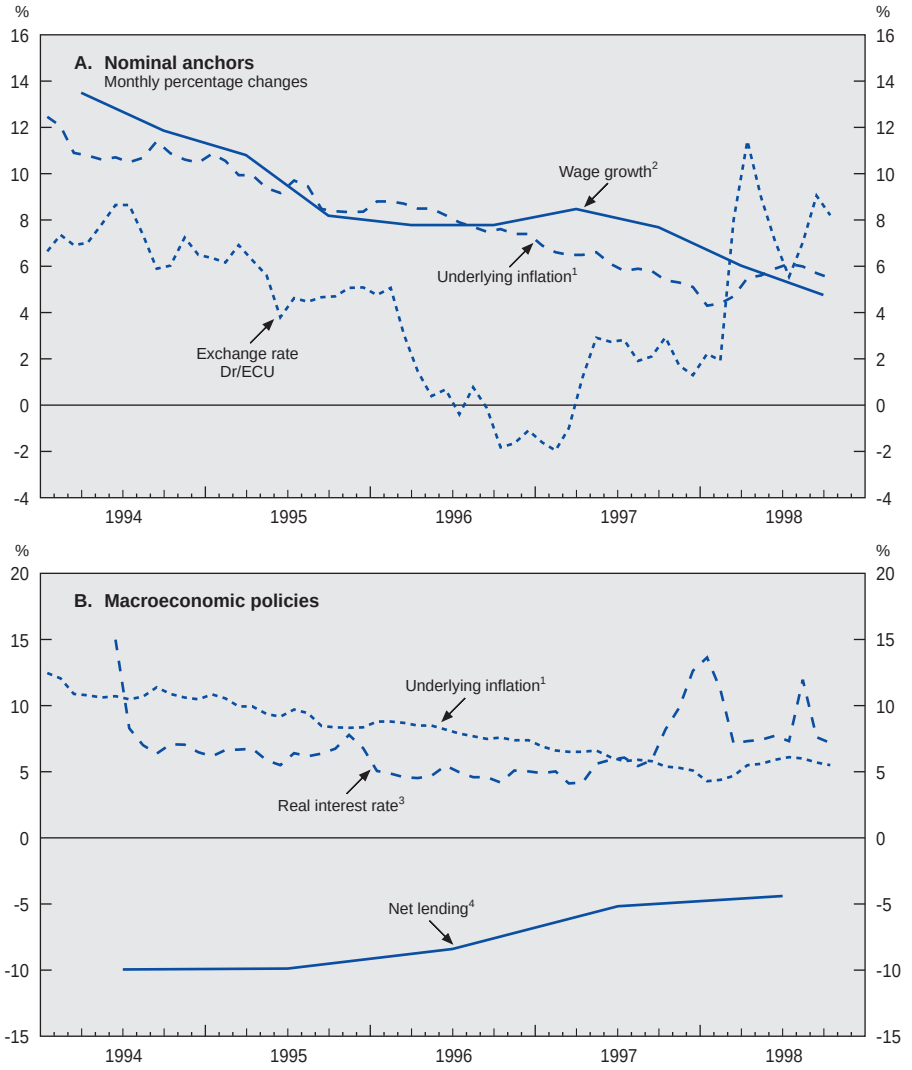
estimated by the authorities to be Dr 175 billion (about ½ a per cent of GDP) and the reduction in the CPI is expected to be nearly 1 per cent in 1999.

An evaluation of the anti-inflationary policy options

An even tighter macroeconomic policy stance may be required if inflation pressures remain high. Private sector credit growth is already rapid and, once financial markets sense that entry into the euro area is within reach, long-term interest rates will converge rapidly to a lower level and the exchange rate could move back towards the central parity – weakening control over monetary policy and concomitantly loosening it.[42] The transmission mechanism through the credit channel, however, may have become less effective than in other countries, because the prolonged existence of high real interest rates has crowded out private sector credit (and the credit to GDP ratio is among the lowest in the OECD). Inflation reduction might thus be achieved more rapidly through a more active exchange rate policy. With the new monetary policy framework, the central bank could raise interest rates further and allow the exchange rate to appreciate significantly above the central rate (without attracting large capital inflows). Inflation reduction would work mostly through the impact of the exchange rate (and the interest rate) as signals for price expectations – a policy which has been successful in the past (Figure 8). However, such a policy would risk reversing the recent gains in competitiveness, and hurt the external sector. In the period to 2000, moreover, a deterioration in the current account could provide a negative signal regarding the sustainability of the central rate.

In view of the constraints on monetary policy, an anti-inflationary policy relying more on a tighter fiscal policy appears advisable. It should have a larger impact on domestic demand, especially as it should have a greater contractionary effect on the more inflation-prone, sheltered sectors of the economy. Tightening fiscal policy would thus help contain wage drift as the current wage agreement will certainly not add to inflation if productivity growth remains high, but nonetheless wages are set to increase appreciably in real terms in 1998 and 1999. The source of financing for the deficit is also important to inflation reduction, and the largest possible share of the gross financing needs of the budget should be met by domestic non-bank financing, especially since an increasing share of the budget's financing is from EU capital transfers, which boost domestic demand.

Figure 8. **Fighting inflation: nominal anchors and macroeconomic policies**



1. Consumer price index excluding fresh food and energy products, 12-month per cent change.
2. Minimum wage growth agreed in national collective agreement.
3. Interbank rate, 1 month, deflated by the consumer price index.
4. General government deficit, as a per cent of GDP, increased by the amount of equity participations and payments allocated to past budgets and reduced by payments on called government guarantees. OECD estimates for the general government deficit in 1998.

Source: Bank of Greece and OECD Secretariat.

Pursuing a significantly tighter fiscal policy now, which would bring the budget quickly towards broad balance, would also provide a useful signal to the market that the central parity of the exchange rate can be maintained. Developing the capacity for a more active budgetary policy, in parallel with structural policies to address the source of inflation pressures, will also be important following an eventual entry into the single currency area, as price competition will be more critical for output and employment growth if recourse to exchange rate adjustments is to be forgone.

The medium-term fiscal strategy: an assessment

Between the beginning of the decade and 1998, Greece has corrected a major fiscal imbalance, with the general government deficit dropping impressively from 16 per cent of GDP in 1990 to an estimated deficit below 3 per cent of GDP in 1998.[43] However, the deficit has only recently fallen sufficiently to induce a gradual decline in the debt to GDP ratio, which exceeds 100 per cent of GDP. OECD simulations on Greece's debt dynamics suggest that a period of 14 years would be required to reduce the debt to GDP ratio to 60 per cent if: *i)* the Government's net primary borrowing requirement (primary balance minus asset transactions which add to the debt) is sustained at the 1998 level of 5 per cent; *ii)* real long-term interest rates are 5 per cent (a level near that in the euro area); and *iii)* real output growth is 4 per cent each year. Under less favourable assumptions for the primary balance (4 per cent), output growth (3 per cent), and real interest rates (6 per cent) the required period would be prolonged to 43 years (Table 13).

Maintaining a sizeable primary surplus for even a decade and a half will be difficult. In addition to adjustment fatigue, there exist many factors which will add pressure on the primary balance. On the expenditure side, social security expenditures on pensions and health care are set to rise due to the ageing of the population and the generosity of the pensions system, and the inefficiencies of the health system (see Chapter III). For example, the OECD in the 1997 *Economic Survey* on Greece estimated that in the absence of a new reform effort the financing needs of the pension system would increase by 5 per cent of GDP by 2015. Regarding public health care, the elasticity of health care expenditure with respect to per capita income is 1½ for OECD countries. Even if the system

Table 13. **Sensitivity analysis on debt dynamics**

Number of years from 1997 for the debt to GDP ratio to fall below 60 per cent[1]

	Real interest rate		
	4	5	6
Primary balance = 4 per cent of GDP			
Real GDP growth (in per cent)			
2	26	43	. .[2]
3	19	26	43
4	15	19	26
Primary balance = 5 per cent of GDP			
Real GDP growth (in per cent)			
2	18	25	40
3	14	18	25
4	12	14	18
Primary balance = 6 per cent of GDP			
Real GDP growth (in per cent)			
2	14	17	23
3	12	14	17
4	10	12	14

1. The calculation is based on the following formula. $\Delta d = d (r - y) - p$, where Δ denotes change and the variables are d: general government debt to GDP ratio; p: primary balance as a percentage of GDP; r: real interest rate; and y: real GDP growth.
2. Debt to GDP ratio increases over time.

Source: OECD Secretariat.

is brought towards the efficiency levels of other OECD countries, fast health care expenditure growth is likely. Another source of increased expenditure pressure may come from the Government's decision to expand the role of local and prefecture governments. Though recent legislation should reduce the number of local governments, experience of other OECD countries suggests that devolution of competencies to lower tiers of Government is often not accompanied by cutbacks at the central level. Finally, though one-off expenditure, Greece's commitment to a new large defence weapon procurement programme and the hosting of the 2004 Olympic games will also boost outlays.

Commitments on the revenue side may also place pressure on the primary balance. On direct taxes, the Government has promised to abolish the system of presumptive taxation on the self-employed professionals and small retail shop owners (as soon as 2000). It is also planning to introduce a new, lower and more progressive, personal income tax system.[44] Finally, tax competition may require

reductions in corporate tax rates and taxes on financial capital; at 40 per cent, the first is relatively high for banks and non-listed incorporated companies, and taxes on assets, such as government securities, bank deposits, and the stock exchange will be increasingly bypassed (*e.g.* through coupon washing) or will deter financial market development. A reduction in payroll taxes would be warranted but for the cost to the budget. They are also among the highest in the OECD area, and are necessary to finance the large social security expenditure. However, they are recognised as a deterrent to employment. Finally, the budget depends relatively heavily on non-tax revenue whose yield should decline in the future; non-tax revenues are equivalent to 4½ per cent of GDP and derive partly from dividends from the few profitable state monopolies which face increasing competition (essentially the national telephone company).

The Government's programme to obtain savings from the improved efficiency of the public administration and its continued fight against tax evasion will undoubtedly offset some of the above described pressures on the primary balance. A broad-based tax reform including, for instance, the further reduction in the plethora of tax expenditures that remain would also help in this respect.[45] However, the objective of such a reform should be to reduce distortions created by the tax system. In view of the need to reduce the deficit, a reduction in average rates should be delayed. The Government's intention to restructure loss-making state-controlled banks and public enterprises will provide significant budgetary relief by reducing capital outlays to finance their investment and outlays on ''below the line'' items, which reflect quasi-fiscal deficits, and have plagued budget policy in the past (see Chapters III and IV). Moreover, additional privatisation receipts would certainly improve the debt dynamics, as would a faster reduction in interest rates towards the level of other EU countries, though the latter would require the sustained implementation of a broad reform effort. If needed for budget consolidation purposes, recourse to cuts in investment expenditure would be difficult to contemplate by the Greek authorities in view of the need for infrastructure investment and the opportunity provided by the EU structural funds. However, high rates of EU fund absorption should not be achieved at the expense of sound investment projects, and in view of the priority placed on inflation reduction, delays in lower priority investments (mostly pertaining to the investment financed solely by national resources) may provide one way to cut the fiscal expenditure in the short run. Alternatively, shifting responsibility for

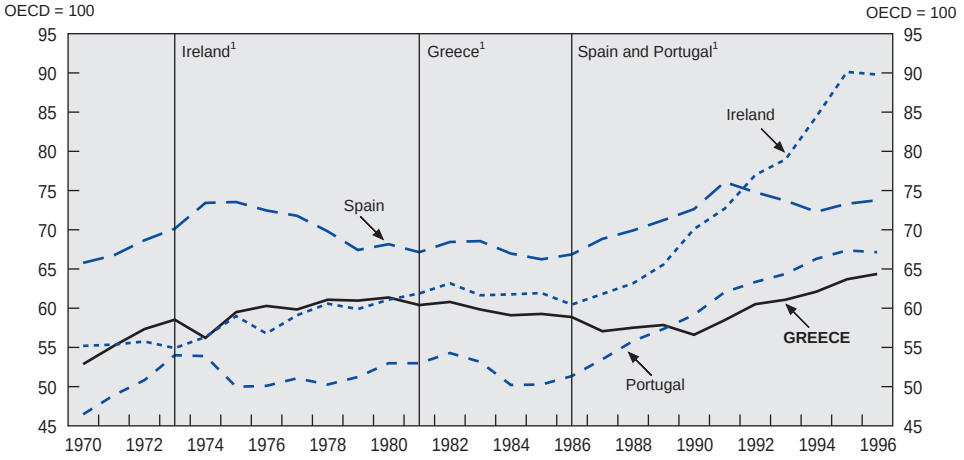
investment projects, including their concomitant co-financing, to the private sector would provide significant budget savings and might improve the efficiency of investment compared with it being undertaken by public entities or enterprises.

It is clear that further budget adjustment remains necessary if the debt to GDP ratio is to be reduced at a sufficiently fast pace. A level for the primary balance surplus of about 5 per cent of GDP (adjusted for equity acquisitions) will not be enough to reduce the debt to GDP ratio very quickly nor provide a buffer in the event of a cyclical slowdown in output. And spending pressures will probably reduce the primary budget surplus relative to the level targeted at the end of the convergence programme. Any strategy for further adjustment must inescapably focus on the social security system and personnel costs which together comprise 80 per cent of primary current expenditure. To this end, a thorough reform of the social security system is a necessary but difficult task, however it is a policy to which the Government is committed. Not only will ageing put upward pressure on these expenditures, but Greece already spends a high share of social expenditure on old-age pensions compared to other OECD countries to the detriment of other policies (*e.g.* active labour market policies). The OECD has analysed both the health system and the public pensions system in detail in the past and the options for reform are reviewed in Chapter III, along with the authorities' recent policy changes. Regarding personnel costs, the current recruitment policy has too many exceptions (a replacement policy of 1:1 for about one-half of total civil servants and 1:5 for the remainder). After two years of this policy, public employment remains broadly unchanged. Moreover, the expanding role of the lower levels of government and the need to transfer excess personnel from public enterprises and state-owned banks are likely to add to the total. Only a far more ambitious policy can reduce the size of the civil service (especially as there is a concomitant need to raise their very generous retirement age). The remuneration of civil servants has experienced rapid across-the-board increases during 1996-97 which are difficult to justify based on productivity gains, though they may have been the cost for introducing a more transparent wage scale. Looking forward, future wage bill savings can be achieved through the introduction of labour regulations which would permit more geographic and functional mobility and more flexible working time. Another option is an extensive use of merit based pay to replace time-in-grade increases, which would help retain and/or attract high quality staff.

III. Implementing structural reform: a review of progress

A rapid catch up of per-capita income to the level of other EU countries is a primary objective of the Government, and the current large difference in income levels suggests an important growth potential yet to be realised. However, per-capita income only showed signs of convergence towards to the OECD average from the beginning of the 1990s, despite Greece's 1981 entry into the EU – an event which has served as a catalyst for faster growth in Ireland, Portugal, and Spain (Figure 9). Critical for the turnaround in growth performance have been

Figure 9. **GDP per capita developments in Greece, Ireland, Portugal and Spain**
Indices using current PPPs (OECD = 100)



1. Line marks date of entry into the European Union.
Source: OECD, *National Accounts.*

macroeconomic policies which have corrected a large fiscal disequilibrium and concomitantly reduced inflation sharply. During 1997-98, output has accelerated and growth has surpassed the OECD average while inflation, though declining, remains significantly above average. Structural reforms have probably played a less important role so far, though most have proved to be very effective when implemented. Prime examples are reforms to the tax system and financial markets, and first steps towards reforming the pension system, the labour market, and public enterprises, which were taken in the early 1990s. A broad-based growth strategy would help Greece to continue to make up ground more rapidly. Its centrepiece should be a wide-ranging structural reform programme, covering labour, product and financial markets and the fast-growing public investment programme directed at improving the functioning of the economy. At the same time, such a strategy would facilitate meeting the Maastricht criterion for inflation in 2000 by providing strong signals regarding the prospects for the economy's performance once within the single currency area. Eventual participation in the European single currency area puts a premium on raising the capacity to absorb shocks.

The structural reform programme on the Government's agenda at the time of the last OECD *Economic Survey* of the Greek economy has been broadly completed[46] and the Government has commenced implementation of a new set of structural reforms in support of the 1998-2001 convergence programme (see Chapter II). The financial sector continues to be an area of rapid progress. In addition to the previously mentioned central bank independence, the restructuring of state-owned and state-controlled banks continues and their privatisation went faster than was initially expected: the privatisation of all small state-controlled banks was achieved and, in the face of strong union opposition, the privatisation of the mid-sized Ionian Bank is set to occur at end-1998 or in early 1999. In the areas of the labour market and the social security system, the outcome has been mixed. On the one hand, the envisaged social dialogue on labour market reform has been completed, and the Government has promised to implement a two-stage reform of the social security system, with the first stage to occur in early 1999. On the other hand, the labour market legislation that ensued from the social dialogue was rather below expectations, as it failed to embody a comprehensive reform. Moreover, the emphasis placed on it by the Government delayed other initiatives, most notably social security reform, as the 1999 changes will be

mostly of an organisational nature. In addition, progress has been slow in implementing the wide-ranging 1997 health care legislation. However, advances have been made in implementing the modernisation programme for public enterprises, which is reviewed in this year's special structural theme in Chapter IV.

Overall, the Government has identified the critical areas where structural reform is needed – the labour market, the social security system, and public enterprises and state-controlled banks. However, only a short period of time remains for Greece to attain the ambitious target of entry in the EMU, and the Government recognises that the current pace of structural reform will need to be accelerated, especially in view of the more uncertain international environment. To assess policy requirements within this context, this chapter surveys structural policies in the labour market, the public pension and health systems, and the financial system (the structural topics of the last three *OECD Economic Surveys*).

Debate on broad-based labour market reform

The direction of the Government's medium-term strategy for reforming the labour market was established by an agreement between social partners in November 1997. The resulting ''Pact of confidence between the Government and social partners on the threshold of the year 2000'' represents an important psychological step, as the Pact acknowledges that job creation requires broad-based structural reform to improve the competitiveness of the Greek economy. In this regard, the Pact's philosophy is often, though not always, in line with the OECD Jobs Strategy recommendations and also forms the basis for the National Action Plan for Employment presented to the EU in March 1998. The main areas covered by the Pact are: *i)* wage formation; *ii)* labour market flexibility; *iii)* active labour market policies; *iv)* taxation; and *v)* product markets. Subsequent legislative initiatives aimed to turn the Pact's guiding principles into practice, but have produced reforms which, in significant respects, have fallen short of the Pact's initial objective.

The first visible impact of the agreement between the social partners was on the negotiations of the 1998 national collective agreement described in detail in Chapter I. The agreement emphasised that real incomes should be protected and gradually increased, but that increases should also take into consideration the

growth of national productivity and the need to support employment. The important role of profits for investment and, thus, for output growth was also recognised. Although the Pact set the tone for a conciliatory accord, the emphasis placed by the Government on this aspect may have been at the cost of achieving deeper structural reform.

The Pact has also resulted in the formulation of a new labour law, passed by Parliament in late-August 1998, which aims to increase the flexibility of the labour market. The four main innovations that will be introduced are: *i)* a calculation of working time over a longer period so as to reduce overtime costs and the need to hire seasonal workers; *ii)* the introduction of employment contracts based on the national minimum wage in pre-determined geographical areas of high unemployment (*i.e.* opting out of the sectoral and enterprise level collective agreement); *iii)* the promotion of part-time employment by eliminating barriers to its use; and *iv)* the authorisation of private job placement agencies (with any fees to be paid by the employer).[47] Greater working time flexibility will permit work up to 10 hours per day (but with a limit of 48 hours per week) at normal pay for 6 months, if hours are reduced in the following 6 months to fully compensate for overtime in the previous 6 months. Under the previous legislation, employers had the unilateral right to request nine hours work per day but had to pay an overtime wage premium of 25 per cent for the extra hours.[48] The new law provides less flexibility to small enterprises which account for the bulk of Greek firms. The greater flexibility in working time has to be based on an enterprise level agreement (normally a collective agreement), and under current legislation small firms are not required, and usually do not have, organised representation. Thus under the new legislation, in enterprises with up to 20 workers, the employers can agree with groups of workers (at least 5) to increase working time to 9 hours per day for 2 months at normal pay, if these hours are made up in the following two months.[49]

The regional employment contracts will permit collective agreements for a specified task and duration to be based on the minimum wage level agreed by the national collective agreement. Such contracts will be permitted in pre-determined geographical regions of high unemployment, and will allow wage agreements to exclude the cascading wage effect of sectoral and enterprise level negotiations. In the same regions, firms will be able to hire (apart from regional employment pacts) new entrants into the labour force and long-term unemployed at the

national minimum wage (until 2001). No official estimates exist as to the potential wage reduction these measures might entail, but an indication can be drawn from the differential between the average earnings in manufacturing (for a blue collar worker) and retail trade compared with the national minimum wage (entry level for an unmarried worker). The respective ratios were for both 1.7 in 1997. However, the potential for labour cost cuts is lower since the minimum wage increases with marital status and years of work experience. For example, a married blue collar worker with 10 years' work experience in a specific profession is eligible for a minimum wage that is about 25 per cent higher than that at the entry level.[50]

The reform of part-time work regulations aims to eliminate barriers to its use. Greece has a very low share of part-time contracts (among the lowest in the OECD at 4 per cent of total employment). Though there are constitutional constraints to providing part-time contracts to civil servants, the draft law will introduce part-time work to the broader public service (public entities and public enterprises). Such part-time hires will not need to undergo the examination required of all full-time public sector employees, which were introduced in 1994 to stop placement through patronage. For the private sector, part-time contracts will become more flexible since they will be based on monthly rather than daily work. For example, the broader definition will permit part-time contracts which entail work for the first 15 days of the month and other forms of intermittent part-time work, rather than just part-day work.

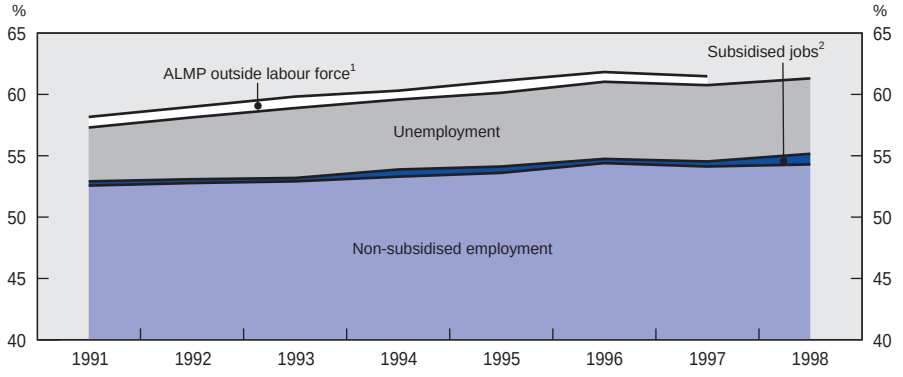
In the Pact, the social partners recognised that the effectiveness of existing active labour market programmes could be increased through their restructuring. *Inter alia*, agreement was reached on an improved targeting of employment subsidy and vocational training programmes, with a view to linking the former to social groups facing special problems (especially those of youth and long-term unemployed) and orienting the latter toward the needs of the labour market. A new employment subsidy programme (The Young in Active Life), provides young unemployed (age 20-27) the opportunity to gain work experience.[51] Such benefits will be provided to an estimated 50 000 young unemployed in 1998 and 200 000 over the three year period to 2 000, and will cost an estimated Dr 90 billion in 1998 and a cumulative Dr 290 billion over the three year period (almost 1 per cent of 1997 GDP). The second new employment subsidy programme will provide similar-sized subsidies for 8 to 22 months to about 11 000 long-term

unemployed in 1998 and 63 000 over the three year period, with the cost for 1998 being Dr 22 billion.[52] These programmes will be run by the Organisation for the Employment of the Labour Force (OAED) through an upgraded support system comprising (eventually) 28 new Centres for the Promotion of Employment. These new programmes will supplement the current employment subsidy programmes run by OAED covering some 50 000 individuals annually at a cost of about Dr 50 billion (Figure 10). Regarding training, the newly established National Centre for Certification (EKEPIS) aims to increase the quality of the plethora of vocational programmes provided by training centres (KEK) through the introduction of a new certification process. In the past, the KEK has often provided training of poor quality, indicating an inefficient used of public and EU funds.[53]

In the area of product market reform, the Pact acknowledged the important role of competition for development, and includes the objective of strengthening the role of the competition committee, which currently only approves mergers

Figure 10. **Active labour market policies and the labour force**

Per cent of population 15-64 years



1. OAED training programmes (accelerated vocational training, apprenticeship programme, intra-enterprise training programme). OECD estimates in 1997.
2. OAED employment subsidy program and lump-sum program for self-employment, and from 1998, new programmes for young and long-term unemployed.

Source: Ministry of National Economy and Ministry of Labour and Social Security.

and does not perform the role of a competition watchdog. Potentially more important, it also set the stage for a reform of the public sector by acknowledging the need to correct the many shortcomings and poor performance of large parts of the narrow and broader public sector. There was recognition that it is too centralised and bureaucratic, includes elements of favouritism and clientelism, and lacks modern infrastructure – and that these factors do not permit the public sector to play its purportedly linchpin role in the country's development. In this regard, it was agreed that special emphasis should be placed on the need for public enterprises to increase their efficiency and reduce the strain they place on the budget.

The Pact reflects an understanding by the social partners of many of the major problems facing the Greek labour market.[54] The implementation of the new legislation will certainly be a step forward in the liberalisation of the labour market, and it goes much further than the 1996 labour market legislation.[55] Nevertheless, the Pact side-stepped discussion of several major rigidities. (A synopsis of labour market reform and an assessment of what remains to be done is provided in Table 14.) For example, strong job protection for the head of household is clearly evident from the large deviation between the unemployment rate for prime-age males and the unemployment rate of women and youth, which are both high compared to other OECD countries. The low turn-over of workers is also reflected in the low unemployment inflow and outflow rates and the small number of laid-off workers, both indicators being among the lowest in the OECD. The difficulty for certain outsider groups to find a job is reflected in the large number of unemployed that have never been employed and the high number of long-term unemployed, both of which are among the highest in the OECD (representing, respectively, nearly 60 and over 50 per cent of total unemployment, with the two sets obviously overlapping). Unsurprisingly, the long-term unemployed comprise mostly women and the young (67.2 and 34.3 per cent, respectively, of total long-term unemployed), and many of them are being discouraged from job search because of their poor employment prospects.

A high level of job protection is provided by the restrictive lay-off procedures and high lay-off costs. The former restrict lay-offs to 2 per cent of a firm's labour force per month (for firms with more than 50 workers) but repeated recourse to monthly firings is not considered possible by employers.[56] Lay-off costs for white collar workers are above the EU average, and they can reach

Table 14. **Implementing the *OECD Jobs Strategy* – an overview**

Proposal	Corrective action since Jobs Strategy Review	OECD assessment/recommendation
I. Increase wage and labour cost flexibility – Lower minimum wages for vulnerable workers (low-skilled, youth) – Simplify and decentralise the wage bargaining process – Eliminate the explicit inflation catch-up clause – Reduce the regressivity of the payroll tax system	– Small reduction in non-wage labour costs for youth hirings – Introduction of an opt out clause from sectoral collective agreements in high unemployment areas	– Make-up the commensurate resources to the social security system – Timely implementation. Reduce further the role of the middle tier (sectoral) of collective agreements
II. Reform employment security provisions – Abolish administrative limits on dismissals – Reduce mandated severance payments for white collar workers		– Implement initial recommendation regarding employment security provisions
III. Increase working time flexibility – Permit limit to, and payment premiums for, overtime hours to be determined in individual collective agreement – Promote the transferability of pensions	– Introduction of 10 hour day for large firms over a 1 year period – Introduction of more flexible part-time work, including in the public sector (excluding civil servants) – Reduction of the segmentation of pension funds	– Timely implementation and enlargement – Tighten the links between earning history and pension entitlements to strengthen work incentives
IV. Reform unemployment and related benefit systems – Rationalise unemployment benefits		
V. Expand and enhance active labour market policies – Develop programmes for the long-term unemployed in lieu of employment subsidy – Allow for the creation of for-profit job placement agencies	– Consolidation of the system of employment incentives and unemployment benefits through the introduction of multi-use vouchers (for training, wage subsidy or unemployment benefits) – Allowed for-profit job placement agencies – Introduced subsidy programme for long-term and youth unemployed	– More emphasis of ALMP on training, rather than on employment subsidies. Strengthen PES
VI. Improve labour force skills and competencies – Expand and consolidate public vocational training programmes – Tighten gradually time limits on students earning a university degree – Improve evaluation at pre-university levels		

Table 14. **Implementing the *OECD Jobs Strategy* – an overview** *(cont.)*

Proposal	Corrective action since Jobs Strategy Review	OECD assessment/recommendation
VII. Technology and innovations – Strengthen the links between universities and enterprises	– Increase in the participation of enterprises in the vocational training system	
VIII. Enhance product market competition – Foster liberalisation in sectors dominated by the public sector – Continue the privatisation process	– Progress in clearing non-performing loan portfolio in major public banks – Continued introduction of management contracts for public enterprises – Equity sales of the national telephone company and the public petroleum company	– Accelerate the process of deregulation and privatisation

Source: OECD Secretariat.

24 months' salary for 24 years of employment. In the event, according to the Federation of Greek Industry (SEB) layoff restrictions have forced employers to provide severance payments far above the legal limits to attain the job cuts they require. Thus, employment flexibility would benefit greatly if effective layoff costs were reduced through a reform of the current restrictive firing procedures.

While some steps have been taken to raise wage flexibility, major rigidities remain. Less experienced outsiders, for instance, face low employment prospects because of the relatively high entry level wages (especially for less skilled blue-collar workers) compared with other OECD countries. This is partly due to the squeeze in wage differentials inherent in the wage indexation that existed during the 1980s and to the influence of the generous wage awards provided in the public sector on the private sector.[57] Evidence that the minimum wage is a barrier to employment for some groups of workers is provided by the high percentage of individuals paid at such a wage, especially in the case of outsiders, such as women. Its effect is also supported by the employment of an estimated 500 000 to 700 000 illegal immigrants (nearly 20 per cent of employment) at quite low wages. Their existence has prompted labour unions to call for their legalisation and 1998 legislation has resulted in the registration of approximately 375 000 immigrants. Though minimum wages are aimed at important social objectives, greater flexibility could be introduced without causing hardship: for example, lower minimum wages for youth who still live at home should raise their chance to find a job.[58] Furthermore, evidence from a number of OECD countries suggests that poverty is predominantly a phenomenon linked to non-employment rather than low wage employment. Therefore, a reduction in the minimum wage, especially for population groups which are manifestly having difficulty finding jobs, should spur employment creation and investment in human capital.

Employment has also suffered from the increase in social security contributions to above the EU average (35 per cent of compensation, excluding several sizeable *ad hoc* add-ons). In view of the financing problems of the pension system it is unlikely that basic contributions can be reduced without a reform of the benefit system. However, eliminating the *ad hoc* elements, *inter alia*, linked to financing housing subsidies, could provide a measurable reduction in costs and thus encourage hiring. Similarly, a reduction or elimination of the mandatory

hiring of the disabled and socially under-privileged (*e.g.* 8 per cent of the labour force for large manufacturing firms) would also reduce labour costs (and new legislation on the employment of handicapped persons is being discussed in Parliament). In fact, the Pact recognised the dampening effect non-wage labour costs have on job creation and thus there was the commitment to reduce them for the young unemployed in order to facilitate their job prospects. In another move to create new jobs and simultaneously reduce pension expenditure, the Government will introduce disincentives for retirees to work by reducing their pension. However, such schemes have not proven effective in other countries.

The concentration of the unemployed in certain groups and their resulting isolation as long-term unemployed may also partly reflect the inadequacy of active labour market policies (ALMPs) in Greece. Despite relatively low unemployment benefits – which have nevertheless increased substantially over the past few years to one-half the entry level minimum wage – the ratio of active to passive labour market expenditure is among the lowest in the EU and the EC noted a deficiency in the strategy for ALMPs in the recently submitted National Action Plan. International experience suggests that the priority should be the creation of a well-functioning public employment service (PES), whose role will be, most importantly, to act as a job placement agency (including counselling and job search assistance), but also to administrator efficiently the various job promotion programmes. The envisaged upgrading of the Greek PES (OAED) is a step in the right direction. However, the increasing emphasis placed on employment subsidy programmes suggests that care should be taken in their design. The experience of other countries shows that such subsidy programmes usually have very small employment effects. Nevertheless, for such subsidy programmes to have a chance to be effective, they need to be small, well targeted, and of short duration – and the new subsidy programmes in Greece contain some such elements. Moreover, programme evaluation is important for improving the programmes' design. Vocational training programmes, especially if targeted to younger individuals, and an improved education system in general, may be more effective than employment subsidies in creating employment. Regarding training, however, it must be concentrated on skills that are in demand by employers, and the further development of programmes combining work experience and training in private enterprises envisaged by the National Action Plan would be a move in the right direction.

A small reform of the social security system

At the time of the March exchange rate adjustment, the Government promised to introduce a two-stage pension reform. The first stage will introduce minor organisational changes during 1998 and reflects agreement between social partners in the social dialogue on pension reform. The second stage will aim at a major overhaul of the public pay-as-you-go pension system. The timing of the implementation of the second stage will be announced by end-1998 but is expected to coincide with the next legislature which will convene no later than October 2000. Delaying the introduction of significant changes to the pension system was partly due to the Government's desire to focus on labour market reform. The Government also did not implement cost saving measures that had been planned by the 1990-92 pension reforms to start in 1998.

The first stage of pension reform aims to improve the organisation of the pension system by reducing the large number of funds, which results in a complex and non-transparent structure.[59] By reducing their number, the reform also aims to eliminate certain deficitary funds through mergers with larger, financially healthier ones. First, a new fund will replace the three large primary funds for the self-employed. Second, the 12 supplementary funds for civil servants and the 48 funds for rural lawyers, respectively, will be unified. Third, the 7 supplementary funds experiencing financial difficulties will be merged with the large supplementary fund for private-sector employees. The creation in the near future of a unified pension fund for employees of state-owned banks is also envisaged. These operations will increase overall pension expenditure since no individual's pension will be reduced while in several cases – most notably in the case of the new fund for the self-employed and the supplementary funds – the regulations will allow a gradual convergence of pension benefits to the more generous level of the dominant fund in the new groupings.[60]

The reform also aims to generate some financial savings for the social security system. First, efforts will be made to reduce contribution evasion, including through the introduction of labour inspectors. Second, starting in 2001 pensioners who work and are below the age of 55 will not receive their pension, while those who are above the age of 55 will receive only 30 per cent of their pension for the amount of pension exceeding Dr 200 000 per month (which effectively excludes over 80 per cent of all private sector pensioners from any cut in pension from this measure). Third, in response to a constitutional court ruling

on the equality between the sexes, survivors' pensions will henceforth be provided to both sexes – males had previously not been eligible. However, the survivors pension will be provided for only three years unless the survivor is without work or pension (in which case after the third year the survivor would receive 50 per cent of the survivor's pension if above the age of 65 and nothing if below the age of 65). This measure will not be applied retroactively to women but men can apply for survivors pension if they qualify.

The Government did not implement certain planned aspects of pension reform, and also took decisions which are not commensurate with the final reform objectives. Most importantly, the 1990-92 reforms of the pension system envisaged restricting all supplementary pensions to at most 20 per cent of the pension base, raising the minimum retirement age for receiving a pension early on account of work in arduous and unhealthy environments, and raising the number of years of work required for eligibility for a full pension in public enterprises and state-owned banks. Though these reforms were legislated to commence in January 1998, the Government has decided to postpone them indefinitely. A less costly measure, but one which constituted an important change in philosophy, concerned the broadened access by low income pensioners to the means-tested pension supplement. This supplement was introduced in 1997 for pensioners above the age of 65 and was a first step at providing benefits by the social security system which did not include incentives for early retirement. The supplement will henceforth be made available to pensioners over the age of 60, and also apply to certain categories without age limit (*e.g.* survivors with under-age children). The supplement was increased by about 50 per cent in 1999, raising the budget cost to Dr 60 billion. In a similar move, the Government did not implement an envisaged reform of the expensive ''lifetime'' pension for mothers in families with more than two children, which would have made it means-tested and provided it only to women above the age of 65.[61] Finally, the Government raised significantly the base used for calculating invalidity pensions for individuals who entered the workforce after 1993 (the date of the last major reform). This had previously been based on a share of 1992 per capita income (increased by cost of living adjustments). Though several of these measures provided necessary support to low-income pensioners, they may be exacerbating incentives to retire early as they are introduced outside a comprehensive reform package.

The overall deficit of the social security system will be contained by the additional contributions from the new primary pension for farmers and the registration of illegal immigrants. The effect on overall government finances will be much smaller, however, since the Government pays two-thirds of the farmers' contributions, for which eligibility requirements are lax, and many immigrants are likely to contribute for the minimum period necessary to attain eligibility for health coverage and will receive health care throughout the year. Transfers from the central government budget to social security funds are expected to increase from Dr 840 billion in 1997 to an estimated Dr 900 billion in 1999 (equivalent to 2½ per cent of GDP).

Delays in the implementation of pension reform will make the eventual overhaul of the system more difficult. The Greek state-run pay-as-you-go pension system has remained one of, if not the most generous and inequitable in the OECD, despite the 1990-92 reforms. Current policy changes are insufficient to bring the system anywhere close to viability. Unreformed, the pension system will face rising and eventually unmanageable financing needs during the next two to three decades as the population ages and the system matures. The experience of other countries has shown that reform is usually introduced gradually, and takes decades to be fully implemented while complacency will lead to an even less palatable reform later on. Irrespective of the form chosen for the eventual pension system – maintaining a system with pay-as-you-go elements or moving to one with elements of capitalisation – the design of the long transition process will be very difficult as it requires a strategy to reduce the generosity of the benefits expected by the current generation of workers and pensioners.

The transition will not be easy as the unsustainability of the current system is not understood by the population, in general. Most pensions are low compared to wages but are nevertheless generous compared to contributions, as contribution periods are short or contribution levels low. Moreover, this generosity is quite inequitable between different employment groups, especially between private and public sector employees. With non-wage labour costs already identified as problematic for the labour market, revenue enhancement will need to focus on reducing contribution evasion and avoidance (the former often encouraged by a high minimum pension and the latter encouraged by distortions in the system such as the various clauses permitting early retirement at near full pensions). The most essential component of the reform process will need to be the gradual

raising of the average retirement age from between 55 and 60 years, and from an even lower age in the public sector, to at least 65. To this end, the retirement age of women (age 60)[62] and women with under-age children (age 55) will need to be reconsidered (especially because Greek women have among the longest life spans in the OECD). A further increase in the retirement age would come from a reform of the eligibility conditions for an early pension due to arduous and unhealthy work; currently 40 per cent of private sector employees ''qualify''. An implementation of means-tested survivors' pensions and a strict review of invalidity pensions would also work to this end. Some of these tougher eligibility standards already apply to entrants into the labour force after 1992, and bringing them forward could be justified by equity considerations.

Raising the retirement age will need to be accompanied by measures to reduce statutory replacement rates, which in Greece are the highest in the OECD, and often exceed 100 per cent of the final salary, though in a few cases (mostly for farmers) they are quite low.[63] These could be reduced by setting the pension base on a longer earnings period, covering most if not the whole of the working life income, compared with the current regulation which uses the last five years for private-sector employees and the last month's salary for public-sector employees. The average accumulation rate should also be lowered, while a lower accumulation rate for the early part of a career would provide an additional incentive to delay retirement. Less generous state pensions may also result in additional national savings by encouraging private pensions and may form part of a strategy of shifting to a partly capitalised system. An increase in savings would be welcome to offset the additional financing needs of the public pension system as well as the increased borrowing needs of enterprises as Greek firms turn more to capital markets to finance their investments. In view of the complexity of the changes that need to be undertaken, the introduction of a reform even at the time of the next legislature will require the technical work to commence as soon as possible.

Health reform: progress has been slow in introducing the 1997 health care bill

During the past 18 months, health reform has made headway, most visibly through the introduction of measures to reduce pharmaceutical expenditures.

However, progress in the introduction of the wide-ranging 1997 health reform legislation has been slow. Looking forward, the prospects for further advances have improved in view of a potential break with past reform strategies. Specifically, measures are now being considered by the Government which would reform the demand side of the system, while previous reforms, such as that of 1997, had focused almost exclusively on ameliorating the supply of health services.

The 1997 legislation should provide a basis for positive change in the health care system if implemented quickly and effectively, as it attempts to address the main weaknesses of the Greek health system – the dearth of primary care, the inefficient operation of public hospitals, and the excessive expenditure on pharmaceuticals. Regarding the most critical component of the reform, four pilot programmes have been established as precursors to a country-wide network of primary care providers. However, the establishment of a well-functioning network of primary care doctors will likely depend on the introduction of appropriate financial incentives to attract general-care physicians and supporting staff to primary health care centres. These are greatly understaffed in rural areas, and often staffed only by medical graduates performing their obligatory one year service. In urban areas, primary care is not well organised as health centres have not yet been established. The existing social security funds' polyclinics (mainly of IKA) are dysfunctional and often resort to private sector services or, in the case of other social security funds which operate through contracts with private sector doctors, low remuneration has resulted in inadequate services. This situation has led to referrals by doctors to their own private practices or private diagnostic centres, round-tripping in the case of fee-for-service contracts (especially in the case of public servants), and excess prescriptions, either as a substitute for primary care or due to (illegal) arrangements with pharmaceutical firms. Only if these distortions in the current primary care system are addressed will doctors have the incentive to move from specialised practices at public hospitals to primary care positions where they could play the critical gate keeper role. Doctors' pay incentives would be improved through the use of remuneration techniques successfully followed by other countries (capitation in combination with fee-for-service pay), along with measures which break the civil service mentality of health personnel (*e.g.* merit based pay and less job security).

Regarding public hospitals, the Government is still searching to appoint professional managers, following which another critical aspect of the 1997 reform should be introduced – the use of global budgets. In the very near future, the Government also expects to proceed with its plan to introduce diagnostic related groups payments in public hospitals (a measure not envisaged in the 1997 legislation). Fixed daily hospital rates would then differentiate between surgical and pathological services (with a 30 per cent differential between the two rates), and include separate rates for specific high cost services. In an effort to contain cost overruns, the daily hospital rates would no longer be increased for ''extra'' services. At the same time, daily hospital rates would be raised for the first time since 1992; the basic rate is envisaged to be increased by 20 per cent (12.5 per cent if one takes account of the ''extras'' that were usually added to the basic rate under the previous system). After this increase, the daily charges would cover about 40 per cent of the cost of hospital services, and it is the government's intention to gradually raise these daily rates to cover variable costs fully. The introduction of global budgets, diagnostic related group payments and professional managers are necessary steps to improve hospitals' efficiency but will be insufficient in the absence of hard budget constraints. These are currently lacking as public hospitals run substantial deficits (over and above the large transfers they receive from the budget). The need for hard budget constraints is more general. If these are in place for all service providers (hospitals and primary health centres as well as health funds, as third party payers), and are combined with improved administrative capabilities of all players, excess demand for health services, including pharmaceutical prescriptions, should be contained.

A significant reduction in pharmaceutical expenditures will be achieved through measures that reduced prices and restricted demand. Price reductions were achieved in 1998 through a large number of measures that were introduced in late 1997. First, the price of imported products must be below the lowest corresponding price pertaining in any European country.[64] Also domestically-produced products must be cheaper than the corresponding lowest price in Europe, and any increase from current levels (which are usually below the corresponding ones in Europe) must be based on documented cost accounting. Second, several large charges placed on pharmaceuticals to finance the National Drug Institute and the pensions of medical personnel were reduced (though the loss of financing to the pension funds was compensated by transfers from the

budget).[65] Third, regulated profit margins of pharmacists and wholesalers, as well as producers and importers were also reduced.[66] Overall, pharmaceutical prices fell by 21 per cent (corresponding to a 0.2 per cent reduction in the CPI index), though cost saving during the first three months of the year were substantially less as the volume of prescriptions increased by over 10 per cent during this period. Pharmaceutical demand was constrained through the introduction of a positive list of medicines in April 1998 (including a defined daily dosage requirement) which limits public drug reimbursements to some 3 000 drugs. It is too early to gauge the full impact on pharmaceutical expenditures from the introduction of the positive list, though partial evidence based on IKA data suggests declines in prescriptions during the first months of the list's existence (about 7 per cent). For all of 1998, the Government estimates a reduction in expenditure from both sets of measures of Dr 120 billion, equivalent to 20 per cent of total pharmaceutical expenditure (and 0.3 per cent of GDP). Looking forward, the authorities are considering measures to contain excessive prescriptions, *inter alia* through the creation of computer records of individual doctors' prescription profiles. Though not currently contemplated by the Government, further competition would benefit the pharmaceutical sector and help reduce pharmaceutical prices. To this end, the regulation of profit margins in this sector could be eliminated.

A successful health reform will need to raise the quality of care while containing its cost. Thus, reforms on the demand side are complementary to the reforms of the supply of health care services, which the 1997 legislation promotes. In the case of Greece, such reform work require systemic changes to the financial structure and organisation of the health care system. Specifically, price reform is needed, as many years of price containment have resulted in the inefficient use of services and other distortions. Raising daily hospital charges would thus be an important step. With prices closer to reflecting costs, the creation of an internal market would be feasible, with independent and efficient third-party payers which would serve to check wasteful practices. Of course, for this to occur, the demand and supply of services would also need to be separated (*i.e.* patients, or third party providers such as health funds, would need to purchase services from competing hospitals, clinics, and doctors). The separation of health funds from their parent pension funds and the unification of all existing health funds into a single fund (which should include public servants), as the

authorities contemplate to do in the next large social security reform, would be an important move in this direction. As a first step, the 1998 pension reform creates a service responsible for the control of health expenditures of all health funds. Such oversight of the health expenditures of all funds should facilitate cost savings by permitting co-ordinated purchases on better terms.

Financial sector reform: restructuring and consolidation propel the sector

The financial sector has been the area which has experienced the deepest structural reform. State-owned banks, with one notable example, have nearly completed a process of cleaning up their portfolios from non-performing loans, and are implementing programmes to improve their operating efficiency. Most of the critical players, both privately and publicly owned banks, have enhanced their capital base, and a significant market consolidation is occurring as the Government sold all the small state-controlled banks and now turns to sell the medium-sized banks. Nevertheless, the state-controlled banks' efficiency continues to fall far short of the performance of banks in other OECD countries as well as that of the Greek privately owned banks, as their return on assets is low despite maintaining one of the highest loan-deposit rate spreads in the OECD countries. Another impetus for change is coming from enterprises' tendency, especially the larger ones, to turn to foreign banks or to the rapidly developing stock market to meet their financing needs.

Spurred by aggressive competition from private-sector banks and the professionalism of new managers, state-controlled banks have continued to make progress in escaping from a *dirigiste* past which had left their asset portfolios in poor shape with a large number of non-performing loans and equity participations in loss-making non-financial enterprises. The dominant bank in the Greek market, the National Bank of Greece (NBG) changed particularly impressively. It strengthened its capital base by Dr 140 billion (*i.e.* an increase by 50 per cent) in 1997 through a rights issue and the issue of subordinated debt, thus raising its capital adequacy ratio to 11.9 per cent at end-1997.[67] At the same time, the NBG reported that it has completed the cleaning up of its balance sheet through write-offs of Dr 190 billion in non-performing loans and equity participations (and has brought in international auditors to produce semi-annual accounts), and has

raised provisions for doubtful debts to Dr 150 billion at end-1997. NBG's operating efficiency has also improved through a nearly 5 per cent reduction in staff during the two year period to end-1997 (to the still high level of 14 475) and the introduction of modern systems to contain the growth of other operating expenditure. Finally, the operations of the NBG Group should become more efficient through the merger of NBG with the Group's two recently restructured mortgage banks (the National Mortgage Bank of Greece and the National Housing Bank) which increased the size of the NBG by 25 per cent. In absorbing Greece's two main mortgage institutions, the NBG will have to address the issue of: *i)* finding the additional resources to meet the high required reserve ratio from which mortgage banks were previously exempt; and *ii)* improving the poor repayment rate on the subsidised housing loans provided by the Organisation for Workers Housing (OEK) through these two mortgage banks.[68]

Significant progress has been made in cleaning up the accounts of the Hellenic Bank for Industrial Development (ETBA), a perennial loss-maker. ETBA is reaching the end of a five year restructuring programme supported by a Dr 492 billion recapitalisation, distributed in annual tranches until 1999. The programme comprises the write-off of a substantial portion of the bank's assets including approximately Dr 350 billion of non-performing loans and some Dr 100 billion in mostly non-performing equity participations. The loan write-offs are to be completed in 1998 and, as a result of these operations, ETBA's loan portfolio is expected to be reduced by more than two-thirds, from over Dr 500 billion at end-1994 to Dr 140 billion at end-1998 (excluding some Dr 200 billion in loans to the armed forces).[69] In 1997, ETBA met international capital adequacy requirements for the first time in many years and made operating profits (prior to write-offs and provisions); in 1998 operating profits are expected to be Dr 7 billion.[70] However, clearing the balance sheet of equity participations in non-financial enterprises is proving to be a slow process. Some 20 enterprises have been sold since 1996, but ETBA continues to have large participations in 25 problematic enterprises (including large shares in Hellenic Steel, Hellenic Aluminium, and Hellenic Shipyards), with little prospect of attracting buyers for its shares.

In contrast, little progress had been made so far in restructuring the largest specialised bank, the Greek Agricultural Bank (ATE), which is the second largest bank in terms of deposits and the largest in terms of credits. However, a new

management was introduced in mid-1997, and subsequently a team of outside auditors is completing the process of estimating the size of ATE's non-performing loan portfolio (with preliminary estimates for loans with payment delays exceeding three months being about Dr 400 billion, equivalent to 17 per cent of outstanding loans). Moreover, a three-year restructuring programme is in the process of being prepared for the government's approval. ATE's poor financial performance has been the product of a long-standing government policy of lending to farmers or agricultural cooperatives on non-commercial terms (often without collateral requirements). Several loan rescheduling cum arrears amnesty programmes have enhanced poor repayment incentives and have led to repeated loan write-offs (especially to farm collectives).[71] As a result, ATE's operating profits were nil in 1997 (and in fact are likely to be negative due to the use of an accounting methodology which has delayed the recognition of certain payment arrears). Its capital adequacy ratio falls far below the 8 per cent minimum requirement, for which it has an exemption until 1999, and it holds hardly any provisions. The restructuring programme will likely be more difficult to implement than for ETBA in view of the large credits outstanding (Dr 2.3 trillion) predominantly to the declining agricultural sector, and ATE's poor net interest earnings. The latter arise from the relatively low (post subsidy) contractual interest on credits,[72] but could deteriorate further as it currently has a high dependence on relatively low yielding sight deposits for its financing.

A more consolidated market structure is emerging as the Government has completed the planned sale of the small state-controlled banks and is planning to sell Greece's fourth largest commercial bank, the Ionian Bank by end-1998 or in early 1999, if financial market conditions permit. The sale of Ionian Bank will represent a significant policy signal of the Government's commitment to structural change and, specifically, to a reduced role for State ownership. It will have overcome union opposition and given up control of a large public entity (comprising about 5 per cent of total bank deposits).[73] The sale is important for the restructuring of the banking system as it will provide much needed funds to recapitalise its parent, the Commercial Bank of Greece, and it will place the poorly performing Ionian Bank into private hands for its eventual restructuring and/or consolidation. The sale will also underline the Government's intentions for the future structure of the banking system, whereby eventually only the NBG and the ATE are likely remain in public hands. Regarding the other sales,

controlling shares in four small state-controlled banks, comprising about 5 per cent of the deposit base in total, were sold during 1998.[74] Following these transactions and the consolidation of the mortgage banks with the NBG, the NBG and the Commercial Bank of Greece are the only two (albeit very large) commercial banks that remain under the state's control.[75] The three main private banks (Alpha Credit Bank, EuroBank, and to a lesser extent the Piraeus Bank) are jockeying for position to determine which ones will survive in a more competitive market to challenge the NBG, and most have raised their capital base significantly in 1998. EuroBank has been the most aggressive to date, offering a surprisingly high bid for the purchase of the Bank of Crete (which the Government had unsuccessfully tried to sell twice before for less than half the final sale value), the purchase of a controlling share in the privately-owned Ergo Bank as well as the smaller Bank of Athens.

Despite these transformations and the large inroads made by private banks during the past years into the domain of the state-controlled banks, the banking system continues to be dominated by the state-controlled banks which account for over 60 per cent of deposits and credits, especially the NBG with control of over one quarter of the market. The oligopolistic structure, combined with the inefficient operations of the state-controlled banks and the imposition of various constraints on the banks, results in a spread between short-term working capital lending rates and deposit rates of about 7½ to 8 percentage points which is about 2 to 4 times that existing in other OECD countries. Policy measures push up the spread by about 1-1½ percentage points. First, all loans are subject to a 1.2 per cent fee which is passed on to subsidise agricultural loans. Second, required reserve requirements are 12 per cent and only one-half is remunerated at a fixed interest rate of 11 per cent.[76] Most of the remaining spread is accounted for by high operating costs and the need to provision for non-performing assets in the case of the public banks, and by high profits in the case of the private ones.[77] For example, the return on equity is about 8-10 per cent for the large state-owned commercial banks and 30 to 40 per cent for the private ones, compared to 30 per cent and 25 per cent, respectively, for UK and US banks (Table 15). Operating costs are dominated by personnel costs and if one measures productivity by the value of credits per employee, the NBG has about four times the personnel of Alpha Credit Bank and extends only about 1½ times the credit. Another factor explaining the low operating profits and the large intermediation spreads of

Table 15. **A cross-country comparison of commercial bank indicators**

1996

	Intermediation spread[1]	Required reserve ratio	Net interest	Staff costs	Net income	Loans		Non-bank deposits	Capital, reserves, and bonds	Net income	Net income
	Percentage points	Per cent	Per cent of total gross income			Per cent of assets	Per cent of GDP	Per cent of liabilities			Per cent of capital, reserves, and bonds
Greece	**7.9**	**12.0**	**44.5**	**47.0**	**31.9**	**31.5**	**21.0**	**75.2**	**4.6**	**1.3**	**27.4**
Spain	4.4	2.0	67.5	39.3	36.2	42.3	44.9	44.7	12.0	1.2	9.5
Portugal	4.3	2.0	68.9	35.0	35.7	32.7	73.3	49.0	9.8	1.1	10.1
Italy	5.6	15.0[2]	75.1	43.4	32.9	40.2	57.5	35.4	19.5	1.2	5.7
France	2.8	0.5-1.0	44.3	45.9	19.9	32.0	41.3	23.8	21.2	0.3	1.6
United Kingdom	4.5	0.35	61.5	34.0	37.7	55.3	75.7	54.0	17.1	1.4	7.4
United States	1.3	3.0[3]	62.9	26.1	37.5	63.8	38.0	68.5	9.4	2.2	22.8

1. The difference between the short-term bank lending rate and the 1-month time deposit rate. Period average for 1997.
2. The required reserve ratio is being drastically reduced towards the level of the euro-area and stood at 6 per cent in August 1998.
3. On deposits up to US$47.8 million.

Source: OECD, *Bank Profitability and Financial Statistics.*

public banks, and especially the NBG's, is the low share of assets accounted for by credits compared with private banks (this ratio is 30 per cent for NBG and about 50 per cent for Alpha Credit Bank). One explanation why public banks place a large share of assets into relatively low yielding government paper – for which the spread over time deposits is only a few percentage points – could be their low capital base (3 per cent of assets for NBG and over 8 per cent for Alpha Credit Bank). Other explanations could be a higher risk aversion or public bank management investing in government paper so as to dampen interest costs for the government (though public banks earn large operating revenues from fees related to the sale of Government paper). In the event, pressure to reduce operating costs will soon come from the advent of the single currency, which will decrease bank operating revenues, and a more unified European financial market, which should strengthen international competition.

Competition is being enhanced by the diminished direct and indirect role of the State, and the selling of the Ionian Bank and the Commercial Bank of Greece would be significant further steps. Thereafter, the Government will have to decide the State's role in Greece's largest bank, the NBG, as well as the role for specialised institutions such as a restructured ETBA and ATE, the Postal Bank and the Consignment and Loan Fund. Since the NBG is currently controlled by state-run pension funds, and is being turned into an aggressive commercial bank by its new professional management, there does not appear any rationale to keep it in State hands. However, the State will have to find an acceptable solution for the placement of surplus personnel. Giving control of these banks to the private sector would continue the necessary process of consolidation and mergers in the Greek banking sector. Private management and new foreign capital or strategic arrangements would help attract know-how and technology to the Greek financial markets. ETBA, and an eventually restructured ATE, will be much smaller institutions, and with industrial and agricultural policy no longer being channelled through these financial institutions, they too could be turned-over to the private sector. Finally, the Government will have to decide whether the Postal Bank and Consignment and Loan Fund are credit banks, and will therefore need to be subjugated to the standard rules (*e.g.* reserve requirement) or turned into special institutions, in which case they should refrain from extending credits to local governments and housing loans to public servants. To put Greek banks on a

level playing field with foreign competition, the two taxes on loan activity should be eliminated, as soon as budgetary considerations permit.

The dynamism of the banking sector in general and the Greek private banks in particular suggests the need to review the adequacy of the regulatory system. An indication of the potential growth of the banking system is the fact that the ratio of commercial bank credit to GDP is about one-half that in other OECD countries, and should rise considerably as real interest rates and intermediation spreads decline. In view of the prospects for expansion, the Bank of Greece took the appropriate step of imposing higher standards of internal risk management on the commercial banks. But perhaps additional steps are needed. For example, broadening the practice taken by the NBG to issue timely external auditors reports would raise confidence in a banking system with a history of state intervention and non-transparency. Better accounting practices would also facilitate the Bank of Greece's supervisory role. In addition to improving supervision, the elimination of obstacles which hamper banks in raising capital would promote better risk management. Specifically, the current unequal taxation of government bonds compared to corporate bonds has blunted the development of a domestic bond market and thus diminished a source for banks to raise capital. Combined with other factors, the underdeveloped nature of domestic bond market may form part of the explanation for why Greek banks have one of the highest ratios of deposits to total liabilities in all of the OECD. Regarding the other main source of funds, banks already account for an extremely large share of stock market valuation as well as turnover (about one-third in both cases) and the recent regulations to ease entry into the Athens Stock Exchange should broaden the relatively under developed stock market (whose market capitalisation is equivalent to only 45 per cent of GDP – following the recent surge in the stock market – compared to 75 per cent of GDP in the OECD area on average).[78]

A strategy for accelerating and improving the process of structural reform

Undoubtedly, 1997 and 1998 have witnessed a flurry of activity preparing for structural reform. As part of this process, the Government's attempt to obtain a social consensus so as to facilitate policy implementation has paid off in some respects. However, the nature of this process should be reconsidered as it has not

yet yielded as much as was expected concerning labour market or pension reforms, though some useful first steps were taken. In the future, a social dialogue would perhaps be more effective if its focus was on concrete proposals. In the event, reforms of this nature will entail winners and losers in the short run, so that reforms which aim to minimise the number of losers will fail to produce significant change. In the end, the Government must take the important decisions but should also be prepared to cushion the impact of change, especially on narrow but vocal interest groups. Although the difficulties in introducing structural reform must be acknowledged, Greece nevertheless requires a substantial acceleration in the pace of policy implementation in order to make up ground with respect to its EU partners, and subsequently attain the goal of performing satisfactorily within a single currency area. In this regard, it has generally been accepted that a strategy of reforming several areas concurrently will bring about the best results, as the inter-linkage of sectors make the reforms reinforcing.

IV. Reforming the public enterprise sector

For at least two decades, almost all public enterprises have been poorly managed and have often been used as vehicles for implementing broader policy objectives unrelated to their primary objective of providing goods and services. As a result, their operations are often inefficient, the quality of service generally unsatisfactory with repercussions on other sectors, and they are an extensive drain on the public purse. This negative assessment is generally acknowledged, including by the Government and social partners. In contrast to policy efforts in most other OECD countries, until recently progress towards restructuring public enterprises has been sporadic and overall very slow, while only a few legislative changes have diminished public enterprises' monopoly power. However, reform of public enterprises has gained prominence as a policy priority from 1996, – spurred by the need for change, including from the advent of globalisation and improvements in technology, as well as commitments to the EU. From a short-term macroeconomic perspective, an impetus to reform is the desire to meet the Maastricht Treaty criteria, and this objective would be facilitated by a significant reduction of the financial burden the public enterprises impose on the budget. Looking further ahead, EU membership necessitates an opening to competition and a levelling of the playing field in most of the markets in which public enterprises currently operate as monopolies. The survival of public enterprises, therefore, hinges on deep restructuring. Though this will be painful for the individual enterprises, the experience of other countries suggests that liberalisation cum restructuring policies will bring forth large efficiency gains for the economy as a whole. To meet the challenge, the Government has formulated a resolute reform plan aimed at improving the performance of these enterprises. A bold implementation will be required – and early indications of the Government's resolve are promising – as public enterprise reform in Greece lags

far behind that in other OECD countries and faces the opposition of strong domestic vested interest groups.

The objective of this chapter is to analyse the main issues concerning the public enterprise sector and assess the current policy framework, as well as planned changes to it. First, the key features of the public enterprise sector are outlined, including their impact on the public finances and product and labour markets. Second, recent changes in government policy are reviewed. Third, the economic conditions of the main sectors are examined (*e.g.* pricing policy, employment and compensation policies, investment programmes and technology lags, and the potential for enhancing competition). Fourth, an attempt is made to quantify the reduction in prices and the potential gains in output and employment from restructuring public enterprises and raising competition in their respective sectors. The chapter concludes with a set of recommendations based on OECD member country experience, consistent with those contained in OECD sectoral reviews such as the *OECD Report on Regulatory Reform*, and the International Energy Agency's *In-depth country reviews.*[79]

Key features of the public enterprise sector

The Greek public enterprise sector comprises nearly 50 enterprises and employs about 130 000 individuals, equivalent to about 3½ per cent of total employment but, due to the large number of self-employed in Greece, 6 per cent of wage earners. The first ratio is on the low side compared to most other OECD countries, while the second is high, portending an influential role for public sector employees in the collective bargaining process. This sector is dominated by about 10 firms, which include the largest enterprises in Greece and those that are usually monopolists (and in a few cases oligopolists) in sectors which provide critical inputs to the economy, such as communications (telephony and mail), energy (electricity, lignite mining, petroleum, and natural gas), and transportation (air, rail, and urban transport) (Table 16). There are few industrial firms in this group, except for those involved in the defence sector. Outside this group of 50-odd enterprises, which has traditionally been referred to as the public enterprise sector in Greece, there have been until recently several important industrial firms – most of whose financial viability is questionable – which were *de facto*

Table 16. **Main public enterprises by sector**

1997

	Employment (persons)	Revenue (billion Drs)	Compensation[1] (percentage)	Operating balance[2] (percentage)	Investment[3] (billion Drs)	Long term debt (billion Drs)
Transportation						
OASA (Athens urban transport)	9 586	38.1	74.6	–193.7	3.1	166.7
OSE (Rail)	11 758	25.4	68.9	–376.0	88.3	338.7
OA Group (Air transport)[4]	10 177	278.0	43.5	–2.5	19.9	7.7
Energy						
DEH (Electricity)	33 999	830.6	34.4	13.3	277.7	1 227.4
Hellenic Petroleum[5]	3 350	596.9	7.5	–4.2	93.4	8.0
DEPA (Natural gas)[6]	221	. .	. .	. .	113.8	75.6
Communication						
OTE (Telecommunications)	23 387	840.0	35.9	37.4	268.9	128.0
ELTA (Post)	11 581	83.2	83.6	–18.3	0.2	. .
ERT (Television and radio)	3 453	41.3	50.0	–15.7	2.3	29.0
Water and sewerage						
EYDAP (Athens water and sewerage)[7]	4 745	69.7	54.2	–3.2	13.4	12.4
Thessaloniki water and sewerage companies	635	17.0	47.2	37.6	5.2	12.1
Industry						
EAB (Military equipment)	2 886	33.1	67.4	–7.6	8.2	66.0
EBO (Military equipment)	1 731	15.1	47.6	–26.5	3.5	3.8
Subtotal	117 509	2 868.4	35.7	6.6	897.9	2 075.4
Per cent of total public enterprises	91.2	79.5			68.2	106.1
Total public enterprises[8]	128 797	3 609.8	35.4	18.1	1 316.7	1 956.9[9]
Memorandum item:						
Per cent of GDP, except employment and wage bill which are shares of corresponding economy-wide variable	3.3	11.0	9.3	2.0	4.0	6.0[9]

1. As a share of operating cost (excluding interest payments).
2. Operating revenue minus operating cost (excluding interest payments) as a share of operating revenue.
3. Administrative basis.
4. Comprises Olympic Airways, Olympic Catering, Olympic Aviation and Olympic Tourist Organisation.
5. Consolidated balance sheet excluding DEPA.
6. Until 1998, DEPA formed part of Hellenic Petroleum.
7. Excludes from revenue a subsidy from a fee on construction activity.
8. Comprising approximately 50 enterprises.
9. Excludes overdue liabilities to banks of Dr 705 billion at end-1997.

Source: Ministry of National Economy.

state controlled (*e.g.* shipyards, mining firms, textile plants, and until recently cement companies). These firms were either under the control of the Industrial Restructuring Organisation (IRO) – a public holding group for non-profitable firms – or are controlled by state-owned banks. In both cases, the control of private firms in dire financial straits passed into public hands when the State – rather than shut them down – converted their liabilities into equity. As a result of a nearly decade-old policy aimed at reducing the state's control of these industrial firms, there currently remain only a few, yet often important, such industrial firms still to be transferred to private-sector control. State-controlled banks are not reviewed in this chapter, but an assessment of their situation is contained in Chapter III.

How public enterprises burden the budget

The invariably poor financial performance by public enterprises has required financial assistance from the central government, equivalent to nearly half of Greece's large debt burden (with the latter amounting to nearly 110 per cent of GDP at end-1997). From 1984 to 1997, the annual gross financing needs of the nearly 50 public enterprises have averaged about 4 per cent of GDP per year, and current and capital transfers from the state budget averaged about 2 per cent of GDP annually (debt financing and depreciation account for the remainder of the financing requirement) (Table 17). Cumulated over this period, these (mostly capital) transfers are equivalent to about 30 per cent of GDP, but have been financed to a large extent by EU structural funds. Thus, the direct impact on the general government debt is much smaller than this number suggests, though it likely reflects the opportunity cost from the loss of EU funds to other sectors of the economy. In other countries which are recipients of EU structural funds, the amount of funds directed to public enterprises appears to be much smaller than is the case in Greece. In the event, this figure greatly understates total support to public enterprises as it does not include debt write-offs and equity injections. From 1990, the Government began to clear the back-log of arrears to state-owned banks which were covered by government guarantees. These ''called'' guarantees have cumulatively amounted to 7.5 per cent of GDP by end-1997, and the outstanding stock of such guarantees on loans to public enterprises stood at approximately Dr 1.5 trillion at end-1997 (equivalent to 4.5 per cent of GDP) (Table 18). Starting in 1996, the Government has supplemented direct financial

Table 17. **Public enterprise accounts**

Billion drachmas, 1984-99

	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998 estimates	1999 budget
Financing needs	196.8	283.0	295.2	329.3	357.3	517.8	565.3	567.0	656.5	838.8	1 018.2	1 018.0	1 032.0	1 548.2	1 921.8	2 176.8
(per cent of GDP)	4.3	5.1	4.4	4.3	3.9	4.8	4.3	3.5	3.5	4.0	4.2	3.8	3.5	4.7	5.5	5.8
Operating deficit [1]	37.1	71.9	76.8	107.3	112.9	168.6	209.4	153.6	48.1	51.8	298.1	162.3	–20.1	231.5	229.2	97.5
Investment	159.7	211.1	218.4	222.0	244.4	349.2	355.9	413.4	608.4	787.0	720.1	855.7	1 052.1	1 316.7	1 692.6	2 079.3
(per cent of GDP)	3.5	3.8	3.3	2.9	2.7	3.2	2.7	2.5	3.2	3.7	3.0	3.2	3.5	4.0	4.8	5.5
Sources of financing	141.1	190.2	208.4	272.9	348.5	421.3	466.5	510.9	608.2	767.2	1 003.4	914.9	1 153.8	1 512.1	1 707.2	1 648.4
(per cent of GDP)	3.1	3.4	3.1	3.6	3.8	3.9	3.6	3.1	3.2	3.6	4.2	3.4	3.9	4.6	4.8	4.4
Transfers	94.8	114.5	114.8	132.0	184.2	229.2	227.7	272.6	284.1	374.9	421.7	482.2	591.3	467.2	492.6	580.6
(per cent of GDP)	2.1	2.0	1.7	1.7	2.0	2.1	1.7	1.7	1.5	1.8	1.8	1.8	2.0	1.4	1.4	1.5
Operating budget	31.0	35.1	25.8	33.8	67.2	69.3	77.7	51.3	50.9	86.7	111.2	122.0	120.8	83.8	84.6	109.6
Capital budget	63.8	79.4	89.0	98.2	117.0	159.9	150.0	221.3	233.2	288.2	310.5	360.2	470.5	383.4	408.0	471.0
Depreciation	29.8	37.6	56.0	95.1	116.9	127.0	149.4	176.0	208.8	294.6	303.5	271.1	304.4	374.6	467.4	487.9
Other sources of financing	16.5	38.1	37.6	45.8	47.4	65.1	89.4	62.3	115.3	97.7	278.2	161.6	258.1	670.3	747.2	579.9
of which: Equity acquisitions	. .	. .	. .	. .	. .	. .	. .	. .	. .	. .	. .	. .	200.0	420.0	525.0	620.0
Net financing requirement (administrative basis) [2]	55.7	92.8	86.8	56.4	8.8	96.5	108.7	58.8	52.9	137.0	155.9	104.6	–37.2	83.9	389.0	553.5
(per cent of GDP)	1.2	1.7	1.3	0.7	0.1	0.9	0.8	0.4	0.3	0.6	0.7	0.4	–0.1	0.3	1.1	1.5
Net financing requirement (Bank of Greece, cash basis)	103.9	127.2	122.1	88.3	47.2	82.5	115.5	31.0	–35.1	–6.4	204.8	–0.7	83.5	–75.0	. .	. .

Table 17. **Public enterprise accounts** *(cont.)*

Billion drachmas, 1984-99

	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998 estimates	1999 budget
Financing needs of individual enterprises																
1. DEH (electricity)	69.9	94.6	65.9	70.8	80.7	155.5	119.1	100.1	124.6	209.9	182.2	153.2	230.7	284.2	295.1	287.0
2. Hellenic Petroleum[3]	–1.6	7.8	41.2	31.6	14.5	18.9	10.9	6.6	–0.7	–5.9	45.6	33.8	28.5	92.7	23.7	26.1
3. DEPA (natural gas)	0.0	0.0	0.0	0.0	0.0	–2.4	2.5	7.1	34.3	47.4	43.0	42.6	85.9	123.1	125.8	106.8
4. EAB and EBO (military equipment)	10.8	13.7	15.1	17.8	29.4	32.1	55.8	42.5	33.1	15.7	20.0	19.8	9.7	19.3	16.9	10.2
5. ELTA (post)	3.8	7.9	7.8	10.8	11.0	7.9	9.6	–5.8	18.1	11.9	18.0	45.7	59.2	80.3	73.0	88.8
6. ERT (television and radio)	0.4	–0.6	1.1	0.4	–1.3	11.7	13.3	8.0	7.7	22.2	15.0	4.9	12.7	6.8	8.4	4.8
7. EYDAP (Athens water and sewerage)	7.8	8.9	7.1	8.9	15.6	15.8	16.5	11.6	31.1	5.6	7.3	9.9	1.5	4.3	8.8	38.5
8. OASA (Athens urban transport)[4]	27.1	33.2	43.0	61.1	57.9	76.2	85.6	97.0	30.1	36.1	94.6	70.3	89.7	98.4	134.9	131.9
9. Olympic Airways[5]	16.7	30.1	30.1	14.9	22.5	48.8	67.3	35.4	45.5	44.3	64.7	–5.8	–8.1	23.1	125.3	71.4
10. OSE (rail)	14.7	21.9	27.1	41.1	43.7	57.2	75.2	77.2	88.2	138.5	125.0	184.7	193.2	241.2	353.6	401.5
11. OTE (telecommunication)	17.5	19.3	7.0	25.2	19.4	30.9	30.7	64.0	60.6	95.5	186.3	157.7	–7.3	152.1	270.4	428.4
Per cent of total	84.9	83.7	83.1	85.8	82.1	87.4	86.1	78.3	72.0	74.1	78.7	70.4	67.4	72.7	74.7	73.3

1. Including item ''other expenditure requirements''.
2. Including working capital borrowing from 1990 and excluding transfers to the budget.
3. Previously called DEP, it comprises several subsidiaries of which the main ones are two retail supply subsidiaries (ELDA and EKO) and an exploration subsidiary (DEP-EKY).
4. The company is an umbrella for 3 subsidiaries; buses (EThEL), trolleys (HLPAP) and the old Athens subway (HSAP).
5. The Group comprises Olympic Airways, Olympic Aviation, Olympic catering and Olympic Tourism.

Source: Ministry of Finance.

Table 18. **Debt assumptions from public enterprises**

Billion drachmas

	1990	1991	1992	1993	1994	1995	1996	1997	Outstanding stock of government guaranteed debt, end-1997
Total	385.4	255.9	303.8	157.4	120.3	120.5	101.0	147.1	1 574.9
(per cent of GDP)	(2.9)	(1.6)	(1.6)	(0.7)	(0.5)	(0.4)	(0.3)	(0.4)	(4.8)
Restructuring bonds	347.5	222.4	272.1	94.8	86.0	83.8	60.2	40.7	. .
Cash payments	37.9	33.4	31.6	62.6	34.3	36.7	40.8	57.3	. .
Transportation	126.7	121.9	62.8	55.9	50.4	66.3	59.1	65.4	564.3
OA	35.6	43.8	19.3	17.4	21.9	22.6	17.7	11.0	88.5
OSE	19.4	15.8	6.5	3.6	9.2	29.8	22.0	39.1	277.7
OASA	71.7	62.3	37.0	35.0	19.4	13.9	19.4	15.3	198.1
IRO	85.0	15.3	132.0	8.5	32.1	24.5	8.4	12.5	37.2
Military companies	82.7	58.7	34.0	63.4	28.0	15.2	24.3	9.2	81.5
EAB	69.7	52.3	16.2	57.4	18.5	10.6	7.4	5.8	73.3
EBO	0.0	5.6	16.2	6.1	0.4	0.2	6.7	0.0	2.2
Pirkal	13.0	0.8	1.6	0.0	9.1	4.5	10.2	3.5	6.0
EYDAP	69.7	8.2	6.6	4.5	0.0	0.0	0.0	0.0	14.9
ELTA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	41.0	. .
ERT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	8.0	. .
Other	21.3	51.7	68.3	11.9	9.4	14.4	9.3	11.0	877.0 [1]
Memorandum items:									
Capitalisation of State-controlled banks									
ETBA	0.0	0.0	50.0	255.0	0.0	152.0	110.0	85.0	
Agricultural Bank	0.0	0.0	110.5	537.8	7.5	0.0	0.0	150.0	
National Bank	0.0	114.0	0.0	0.0	0.0	0.0	14.0 [2]	0.0	
Bank of Crete	0.0	0.0	0.0	0.0	0.0	3.0	38.0	0.0	
National Mortgage Bank	0.0	0.0	0.0	30.3	15.7	0.0	0.0	0.0	
Total	0.0	114.0	160.5	823.1	23.2	155.0	162.0	235.0	
(per cent of GDP)	0.0	0.7	0.9	3.9	0.1	0.6	0.5	0.7	

1. Includes Dr 451 billion for DEH, Dr 75 billion for DEPA, Dr 66 billion for the Athens metro, Dr 28 billion for the Spata Airport and Dr 198 billion for the organisation for school building construction.
2. Re-issue of 1991 issue, augmented by capitalised interest.

Source: Ministry of Finance.

assistance to public enterprises with equity injections which have added an estimated 4 per cent of GDP to the stock of government gross debt over 1996 to 1998.

To the debt attributable to the public enterprises, one should add the debt taken over by the Government from the industrial firms that the Government

propped up during the 1980s for reasons of maintaining jobs and protecting firms of perceived importance to the economy. The operations of the IRO umbrella will have cost the Government about 1 trillion (equivalent to about 3 per cent of 1997 GDP); overall an expensive proposition because the IRO saved only a few firms from bankruptcy (in total 16 firms employing 9 000 are likely to be sold by IRO, prior to its planned closure at end-1998, and 63 firms will have been liquidated). The cost to the state-controlled banks, especially the Hellenic Industrial Development Bank (ETBA), from the firms that they took over (and which partly overlap with firms under IRO) is more difficult to determine. One indication is that these banks have written off several percentage points of GDP in non-performing loans during the past few years, of which a good part would be attributed to the loans they were encouraged to contract with some essentially bankrupt industrial giants.[80] The fact that the state has had to re-capitalise most of the main state-controlled banks can be partially attributed to lending policies to such industrial firms; however it should be noted that state-controlled banks also provided credits on non-commercial terms to the private sector during this period. The cost to the banks, and perhaps the budget, is not yet complete as the state-controlled banks still maintain control of some large non-financial concerns, most notably the large nickel producer, Larko, with debts of about Dr 60 billion, the remaining 51 per cent of Hellenic Shipyards, and Softex Paper Works (the largest industrial loss-maker in 1997).[81]

The total cost to the budget could be even greater if one adds several indirect costs. First, the poor management of public enterprises over most of their recent history has deprived the state of both tax revenue and the receipt of any dividend. It is striking that the National Telephone Organisation (OTE) provides virtually all of both dividend and corporate income tax revenue coming from a group of public enterprises with significant monopoly power in their respective sectors, with minor contributions from Hellenic Petroleum.[82] Second, public enterprises have very large outstanding arrears to banks as well as to each other, the tax authorities, and the public pension funds. The arrears to banks amounted to Dr 705 billion at end-1997 (equivalent to 2 per cent of GDP). The other arrears are often kept on the books as assets (though it is rarely the case that one finds the corresponding liabilities in the other entity's accounts). A snapshot of the extent of the problem at end-1994 presented inter-enterprise arrears and arrears to the state of about Dr 900 billion (equivalent to 3 per cent of GDP).[83] Some examples

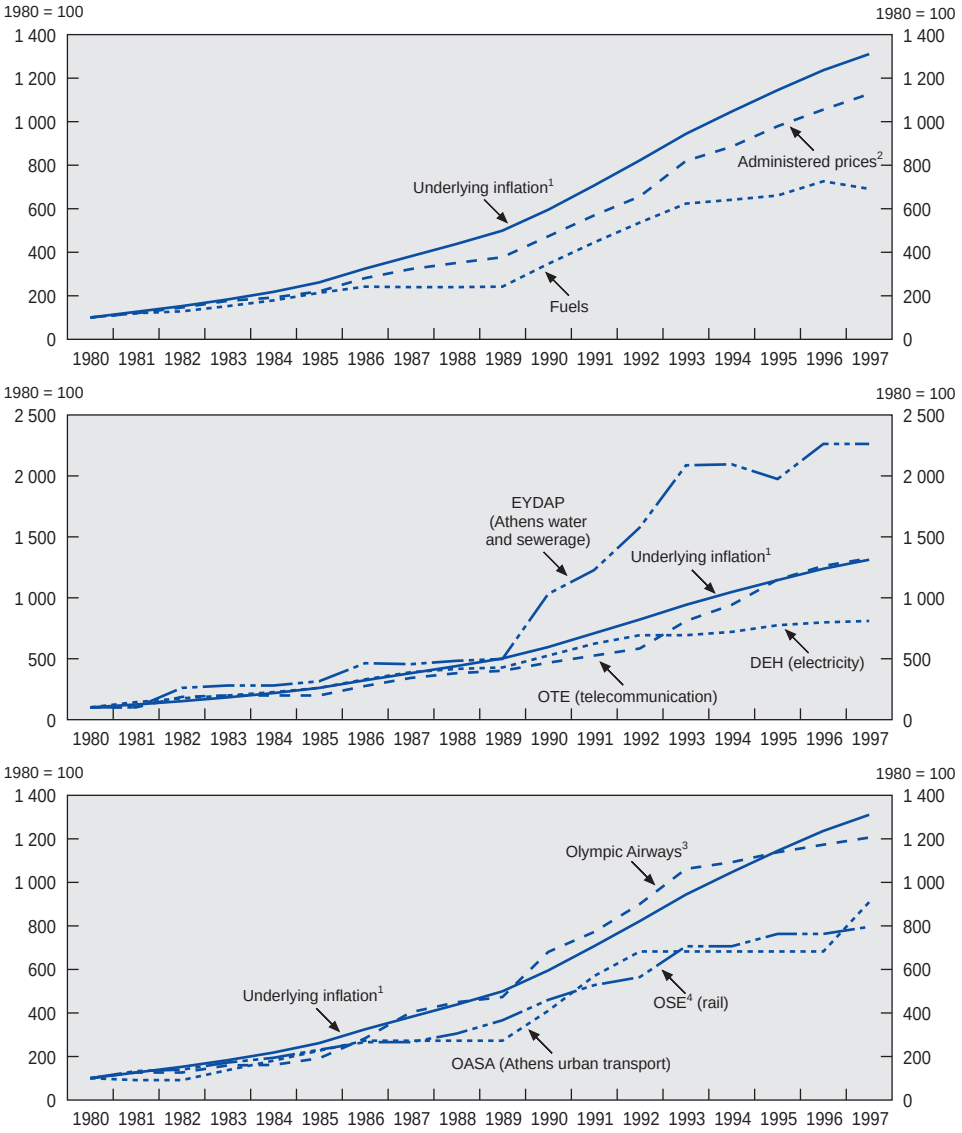
of inter-enterprise arrears include unpaid electricity bills to the Public Power Corporation (DEH) by the high energy consuming industrial firms (Larko Nikel Company, Chalkidas Cement Company, and Chalip Metal Works, which are controlled by the state-controlled banks and receive electricity at below market prices), unpaid water bills to the Water Authority of Greater Athens (EYDAP) by local authorities, and unpaid telecommunications bills by the National Radio and Television Company (ERT). Third, besides an outstanding public enterprise debt of Dr 2 trillion, all public enterprises have very large unfunded pension liabilities arising from the very generous pension benefits to their employees. The OECD in the *1997 Economic Survey for Greece* estimated these to be equivalent to about 7 per cent of GDP in 1994.

Main factors for the poor financial position of public enterprises

Pricing policy

Changes to the prices of goods and services produced by most public enterprises have lagged developments in underlying inflation, exacerbating their poor financial performance (Figure 11). Moderate price increases often have been related to the Government's social objectives, but also reflect attempts to diminish inflation expectations. Administrative prices currently comprise about 7 per cent of the CPI basket, and include most goods and services provided by public enterprises, with the exception of petroleum products, whose prices were liberalised in 1992 (Table 19). Administrative prices are often not adjusted fully for inflation for a period of several years, and then require large corrective action when the financial situation of the public enterprises have clearly been unsustainable. Prominent examples include Olympic Airways in the mid-1980s, most public enterprises in the early 1990's (though that of the water company was partly due to a drought) and subsequently for OTE in 1995 prior to the initial public offering, and the Athenian Urban Transport, OASA, in 1997 and 1998. Nevertheless, energy and telecommunication prices are high compared to prices in other countries, as there have been substantial productivity, and thus price, improvements in these sectors internationally, partly brought on by reforms and technological progress. On the other hand, prices for urban and rail transport and postal services are low compared to those elsewhere.

Figure 11. **Public enterprise prices and underlying inflation**
1980 = 100



1. Consumer price index, excluding energy and fresh food products.
2. Includes most public enterprises (see Table 19).
3. Domestic flights.
4. Passenger tariff.

Source: Bank of Greece and OECD Secretariat.

Table 19. **Weight of public enterprises in the CPI basket**

Sector	Main public enterprise	Weight in CPI basket (per 1 000)[1]
Energy		**93.82**
Electricity	DEH	22.96
Water and sewerage	Athens and Thessaloniki companies	7.04
Heating oil	Hellenic Petroleum	22.30[2]
Gasoline	Hellenic Petroleum	41.52
Transportation		**8.49**
Air travel	Olympic Airways	3.29
Urban transportation	Athens and Thessaloniki companies	4.91
Train transportation	OSE	0.29
Communications		**19.87**
Telecommunications	OTE	17.77
Postal services	ELTA	0.33
Television	ERT	1.77
Total		**122.18**
Memorandum item:		
Administered prices[2]	–	69.39

1. Based on the 1992 household survey.
2. The Bank of Greece index covering administratively controlled prices includes hospital services (1.84), road tolls (1.04), vehicle ''circulation'' fees (2.68) and local government fees (5.47), but not heating oil and gasoline as these latter prices are not administratively set.

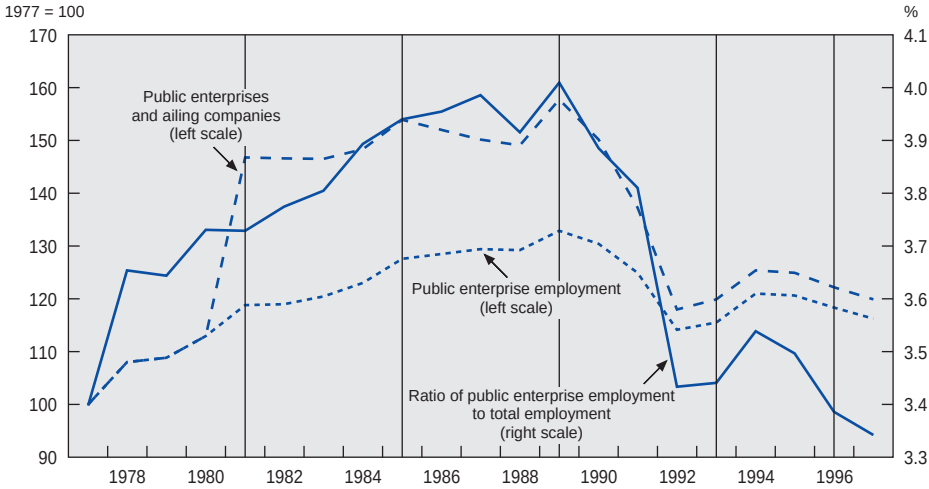
Source: OECD Secretariat.

Inflexible labour arrangements and high labour costs

- Employment policy: paying for past policies when public enterprise staffing was part of social policy

The excess employment burdening public enterprises, especially of low-skilled workers, is mostly due to their use as employers of last resort, as public enterprises have until the 1990s fallen victim to the political cycle (Figure 12).[84] The priority placed on employment is also reflected in the creation of the Industrial Restructuring Organisation (IRO) in 1982 and the use of capital injections by state-controlled banks to sustain – at a large eventual cost to the budget – effectively bankrupt firms. These policies resulted in a 33 per cent increase in employment in public enterprises between 1978 and 1989 (66 per cent if one includes the employment sustained in the firms under the IRO umbrella). Total employment underestimates the abuse of employment policy in public enterprises

Figure 12. **Public enterprise employment**[1]



1. The vertical lines represent election years.
Source: Ministry of National Economy.

since it does not capture: *i)* the poor skill mix resulting from indiscriminate hirings as well as outdated skills due to a rapidly changing technology; and *ii)* the fact that public sector employees have lower retirement ages (for most below the age of 55), with the result that the ratio of retirees to employees stands at 1:1.5 compared with a comparable ratio of 1:2.2 for the economy as a whole (excluding civil servants).[85]

During the 1990s there has been some effort to reverse the effects of these past policies, and employment in public enterprises was reduced by about 30 per cent and returned to the level existing of the late 1970s. In addition, most of the firms in the IRO were sold or liquidated in the early 1990s, and in 1994 a law was passed requiring public enterprise hirings to be based on merit (*i.e.* requiring public-sector wide entrance exams). Managers in public enterprises complained of the lack of hiring flexibility inherent in such a broad entrance exam, however, and a 1997 amendment to the law has permitted more targeted exams at the individual firm level. Moreover, managers in most firms are trying to reduce staff through attrition and in a few cases have introduced early retirement programmes

(*e.g.* Olympic Airways and OTE). Early experience with these programmes (especially Olympic's) has not been satisfactory – similar to other countries' experience – in that they have often had the undesirable result that highly qualified staff has departed rather than the low-skilled staff, and that the programmes have been very expensive. The reliance on strategies of attrition and early retirement to reduce staff has been due to the unacceptability of layoffs to governments. Public enterprise employees are *de facto* treated as if they were civil servants (whose jobs are protected by the constitution), though legally they are subject to private sector law which permits layoffs (subject to certain constraints).[86] Without more ambitious reductions in staffing, productivity indicators for individual enterprises have deteriorated substantially in relation to comparable companies internationally and are among the lowest in the OECD (see individual sector analysis presented below).

- Compensation policy: an extraction of monopoly-power rents by public enterprise unions

Though a judgement on the appropriate level of individual compensation in an individual enterprise is difficult, there exist several indications that it is inappropriately high for the average public enterprise employee. An economy-wide comparison indicates that compensation received by the average public sector employee is nearly twice as high as in the private sector and 29 per cent higher than in the manufacturing sector (Table 20). Moreover compensation in the labour intensive and loss-making sectors is as high as in the capital intensive ones. For example, average compensation for employees in Athens Urban Transport (OASA) is higher than that for the telephone company and the electricity company. The conclusion that compensation is excessive is also supported by an examination of wage developments. The growth of average wages in public enterprises has nearly always exceeded that of the private sector; and for the seven year period 1990-97 compensation in real terms has increased by nearly 9 per cent compared with 2 per cent and 8 per cent, respectively for blue collar and white collar workers in the manufacturing sector, despite the manufacturing sector having attained productivity gains of 26 per cent during this period. Wage increases in the public enterprise sector have been provided irrespective of the financial condition of the enterprise and the individual sectoral analysis indicates that real wages have generally exceeded productivity, notably during 1994-97. To contain such wage pressures, the Ministry of National Economy has

Table 20. **Average compensation per employee**

1997

	Monthly compensation (thousand Drs)	Ratio to compensation of private sector	Share of employees
Total economy[1]	379	122.9	100.0
Public enterprises	573	185.7	6.1
DEH (electricity)	520	168.5	1.6
Hellenic Petroleum (fuels)	991	320.9	0.1
OTE (telecommunication)	576	186.6	1.1
OASA (urban transport)	622	201.5	0.5
Olympic Airways Group	870	281.7	0.5
OSE (rail)	506	163.9	0.6
EYDAP (water and sewerage)	587	190.1	0.2
ELTA (post)	508	164.4	0.5
Private sector[2]	309	100.0	68.6
Manufacturing[3]	444	143.0	26.5
General government	486[4]	157.0	23.2

1. Based on national accounts data.
2. Excluding public enterprises and state-controlled banks.
3. Average of blue and white collar workers in manufacturing.
4. Assumes employer social security contribution rates equal to that in the private sector (28 per cent).

Source: Ministry of National Economy.

repeatedly attempted, since 1992, to introduce limits on wage bill growth. However, without an enforcement mechanism these have not been effective. In part, wage slippages are due to management being captive to the notoriously strong and intransigent public enterprise trade unions, whose control of key services or goods enhances their bargaining power, and they have often used this advantage to hold the country hostage to their demands.

The strength of the unions has often resulted in the granting of many other non-wage advantages. Perhaps the most costly are the pension plans. Replacement rates for pensions provided to public enterprise employees are the highest in Greece (along with those in the banking sector and for doctors), with statutory replacement rates over 100 per cent of an employee's last monthly salary (excluding in many cases generous lump sum end-of-service pensions equivalent to two years' salary). Because of these high pensions, total (employer and employee) contribution payments are almost twice as high as those in the private sector (of the order of 48 per cent for both primary and supplementary pensions *versus* 26 per cent for the private sector).[87] Nevertheless, the implicit net rate of

return on pension contributions is quite high, and future pension payments will be a major financial headache for public enterprises, especially in cases where enterprises have used part of contribution revenues for other purposes.[88]

Public enterprise unions have also succeeded in negotiating restrictive labour practices, which add considerably to labour costs. Limitations on geographic and functional mobility, as well as working time, have forced additional hirings in periods of high or shifting demand and add greatly to total compensation in the form of overtime, nightshifts, paid rest time, days-off, and other supplementary payment benefits (*e.g.* for high season service and for the purchase of meals). In the case of Olympic Airways, for example, these ''extras'' add 30 per cent to the basic salary. Finally, most employees receive ''in kind'' benefits, usually in the form of goods or services from their respective enterprise; these add about 3 per cent to the total wage bill. Some of the most notably generous examples are: *i)* free telephone time for employees of OTE; *ii)* discount electricity prices of about 75 per cent for employees and pensioners of DEH; *iii)* free (reserve-seat) tickets for employees and pensioners of Olympic Airways, and their close relatives; and *iv)* and a 40 per cent discount on water consumption for employees of EYDAP.

Investment programme

Despite substantial investment expenditure, which has averaged annually about 3½ per cent of GDP since the early 1980s (on an administrative basis), most public enterprises have not succeeded in updating to modern technologies or maintaining properly their infrastructure and/or equipment. Partly due to these modernisation delays, Greece has received a derogation from most EC directives related to the liberalisation of key sectors such as telecommunications, electricity, natural gas, and air transportation. Since the beginning of the 1990s, a more significant effort has been made to narrow the existing large investment and technology gaps. To this end, EU funds, most importantly the CSF I, but especially the CSF II, have played important roles. Regarding the 1994-99 CSF II, the main public enterprises will receive about ECU 6 billion (equivalent to almost 6 per cent of GDP) out of a total programme worth ECU 29 billion, with over half the financing in the form of EU grants. Despite some progress, examples of critical investment lags that remain are the following: OTE has one of the lowest shares of digital lines in the OECD (though it is currently pursuing a rapid

digitalisation programme). DEH barely has a sufficient and secure supply of electricity generation, and has built many inefficient plants (especially hydroelectric plants with a view to assisting agricultural irrigation). The construction of the natural gas pipeline and the upgrading of the rail infrastructure have suffered innumerable delays and cost overruns. In all transportation sectors (air, rail, and urban transport), the fleets are old and have been purchased from diverse producers, raising maintenance costs and spare part stockage. These developments are partly to blame for public enterprises' unsatisfactory quality of service and the resulting reduced demand. Characteristic of inefficient procurement practices was the December 1997 rush – ahead of an EC directive deadline on open competitive bidding on large supply contracts (EC 9338/1994) – to distribute to domestic producers without competitive bidding contracts worth Dr 1.3 trillion (equivalent to 3½ per cent of GDP).[89] The EC has subsequently decided to review these contracts to see if they breached competition rules.

Accounting practices

Public enterprises' accounting practices obscure their financial position, and make an analysis of their performance difficult. Most importantly, most enterprises do not have separate accounts for distinct components of their operations, prohibiting an accurate cost appraisal of the different goods and services that they offer. Moreover, it is generally accepted that public enterprises' financial statements and balance sheets are non-transparent, in particular due to the above-described practice of including items presumably owed as revenues and excluding implicit liabilities. Moreover, accounts are usually not produced on a timely basis. It is, however, envisaged that separate accounts for individual operations will be established and internationally accepted accounting practices will be implemented. For example, under the new policy initiative by the Government, international auditing firms are being hired, and audits had already been completed for the OTE and Hellenic Petroleum prior to their respective initial public offerings.

Current policy objectives

The Government acknowledges that Greece's real convergence to the level of its EU partners requires a successful resolution of these problems and it has therefore placed a renewed emphasis on reforming the public enterprise sector.

Prior to 1996, policy moved back and forth on choosing a reform strategy – the need of which has never been in doubt by successive Governments – and on the appropriate role for private sector participation, on which Governments have often held an ambiguous view. The first round of reform began in the early 1990s, with a considerable delay compared with most other OECD countries. Restructuring programmes were introduced in many enterprises and their financial position recorded a notable improvement (albeit partly due to a severe but short-lived wage compression). In addition, the Government moved to close most of the non-viable firms under the IRO umbrella by introducing a special fast-track liquidation procedure. It also privatised Athens bus transport and was prepared to sell large stakes in OTE to a strategic investor (49 per cent) and in Hellenic Petroleum (then DEP). Competition improved markedly in the domestic air transport market and the petroleum sector following the moves to liberalise the respective markets. Similar benefits emerged in the telecommunication sector following the introduction of two private operators in mobile telephony and in the media sector following the liberalisation of television and radio licensing. During the same period, the Government initiated efforts to attract foreign capital through Build-Operate-Transfer (BOT) operations; notable results were the contracts for new Athens Airport at Spata and the Rio-Antirio Bridge. Despite these initiatives, public enterprises' financial performance remained far from satisfactory, and the Government policy was put on the defensive due to the public's perceived lack of transparency in these operations. The initial reform effort was largely abandoned when the bus privatisation was subsequently reversed in 1993, and the other moves delayed significantly before resurfacing on the policy agenda. The prospects for a new initiative have improved significantly since 1996 in view of an apparent shift in popular opinion favouring privatisations, in part due to the positive results from reforms experienced both in Greece and in other OECD countries.[90]

To prepare Greek public enterprises for the challenges posed by a competitive environment at the EU level, as well as to underpin the budget consolidation efforts, the Government has decided that it will need to improve the quality of service, raise productivity, and attract private participation to the financing of their development. The cornerstone of the Government's strategy is the 1996 legislation which aims to change public enterprise management. The aim of this legislation is to introduce a management style equivalent to that existing in the

private sector. To this end, public enterprises are being transformed into *société anonyme*. New managers are being hired through public and competitive tenders. And each enterprise is preparing strategic and business plans (often with the assistance of international consulting firms). Management independence is to be strengthened through management contracts based on performance criteria consistent with the approved business plans. Almost all public enterprises have been included in this procedure (about 40), with the notable exception of the electricity monopoly (DEH) which has been granted a grace period to complete its current restructuring and to implement the new national electricity law. The new management in all public enterprises will also have to distribute consumer rights information. By mid-1998, new managers were in place in many enterprises, and completed strategic and business plans had been submitted to the Government for approval by several of the new managers. In the period August to October 1998, management contracts were approved for Greek Post (ELTA), the National Television Station (ERT), Athens Urban Transport (OASA) and its three subsidiaries, and Greek Railroads (OSE).

The Government has different strategies for the loss making compared with the financially more viable public enterprises. Regarding the former, the Government aims to harden their budget constraint, including by tightening access to credits under government guarantee. To improve their operations, these enterprises' business plans will include reductions in surplus personnel. For the most part, this will occur through attrition. However, the Government is also considering the option to transfer personnel to other entities, the central administration, or local authorities. Great emphasis has been placed on improving the flexibility of employment and thus improving productivity and reducing labour costs. To this end, the Government passed legislation in February 1998 that requires public enterprise management and the respective unions in loss-making enterprises to agree to revise the existing collective agreements so as to achieve more flexible working conditions. If no agreement has been reached within six months, the Government would then impose new conditions on the enterprise. This occurred in the case of all management contracts signed so far. In some cases, especially those where performance does not improve, the Government is considering a transfer of management control to strategic investors (*e.g.* Olympic Airways).[91]

Regarding the financially more viable public enterprises, the Government intends to sell equity to private investors with the hope that their participation

will improve enterprises' operations and at the same time reduce the borrowing needs. The sale of enterprises has been a contentious issue and it has only recently become politically palatable to sell even minority shares. Thus, the main operations to date have occurred only since 1996 and comprise the sale of 34 per cent of the national telephone company, OTE and 23 per cent of Hellenic Petroleum (Table 21).[92] The total revenue raised from all privatisations (including banks) is equivalent to about 3½ per cent of GDP (with the bulk raised in 1998), compared with 20.6 per cent in Portugal, 4.0 per cent in Italy and 4.4 per cent in Spain, respectively, during the period 1990-97. Additional sales are expected to occur, since in mid-March 1998, the Government pledged to sell over the next two years, minority shares of another 10, albeit small, enterprises or to offer tenders for managing their operations.[93] As a signal of its commitment to this process, the Government has issued privatisation certificates which will have guaranteed access to 40 per cent of future privatisations, and will receive a price

Table 21. **Main privatisations of public enterprises**[1]

1990-98

Company	Date	Share of capital sold	Receipts (billion drachma)
AGET – Heracles Cement	1992	49	129
Fix (beer)	1993	100	2
Neorion Shipyards	1994	100	2
Peraïki – Patraïki (textiles)	1994-96	100	3
Cassandra Goldmines	1994	100	11[2]
Chalkis Cement Company	1995	100	36
Hellenic Shipyards	1995	49[3]	8
Skalistiris Mining	1995	100	2
OTE (telecommunication)	1996	8	128
Chalipis Metal Works	1996	100	4
OTE	1997	12	298
Eleusis Shipyards[4]	1997	100	30
Duty-free shops	1998	20	24
MEL (paper)	1998	100	2
Athens stock market	1998	35	22
OTE	1998	4	126
Hellenic Petroleum	1998	23	90
OTE	1998	10	302

1. This list contains sales that raised more than Dr 1.5 billion; there were many sales for smaller amounts, especially of companies under the IRO umbrella.
2. The company was sold along with gold valued at US$204 million.
3. The remaining 51 per cent is owned by the Greek Bank for Industrial Development (ETBA).
4. The company had been privatised in 1992 but reverted back to the IRO in August 1995 following bankruptcy.

Source: OECD Secretariat.

discount. Nevertheless, the Government currently has no intention to sell more than half of any large public utility, and this upper limit of 49 per cent is set by law.[94]

The Government intends to implement several other policy moves in order to improve financial discipline. First, it will introduce a dividend policy for public enterprises which are not listed on the stock exchange. Most enterprises will henceforth need to improve their operations so as to issue a dividend to the state. Second, a process for clearing inter-enterprise arrears will be concluded quickly. Third, with a view to improving competition, the Government is planning to bring Greek legislation quickly in line with EC directives, and to state clearly the liberalisation strategy and timetable. The first beneficiaries will be the telecommunication and electricity sectors, for which important new legislation is planned for 1998 or early 1999. Fourth, the Government will make greater use of build-operate-transfer type concession agreements, most notably for the construction of motorways. Specifically, 40 per cent of new motorway construction will be reserved for such concession agreements, and a procedure has already been initiated for the Athens ring road. In addition, all major projects for the 2004 Olympic games will be built through concession agreements with the private sector. Finally, the Government intends to strictly contain administrative price increases for public enterprises and there will be few nominal increases during 1998 and 1999. This policy, however, is mostly related to reducing inflation and satisfying the Maastricht criterion for inflation, rather than public enterprise reform, *per se.*

Reform of specific sectors

Network industries

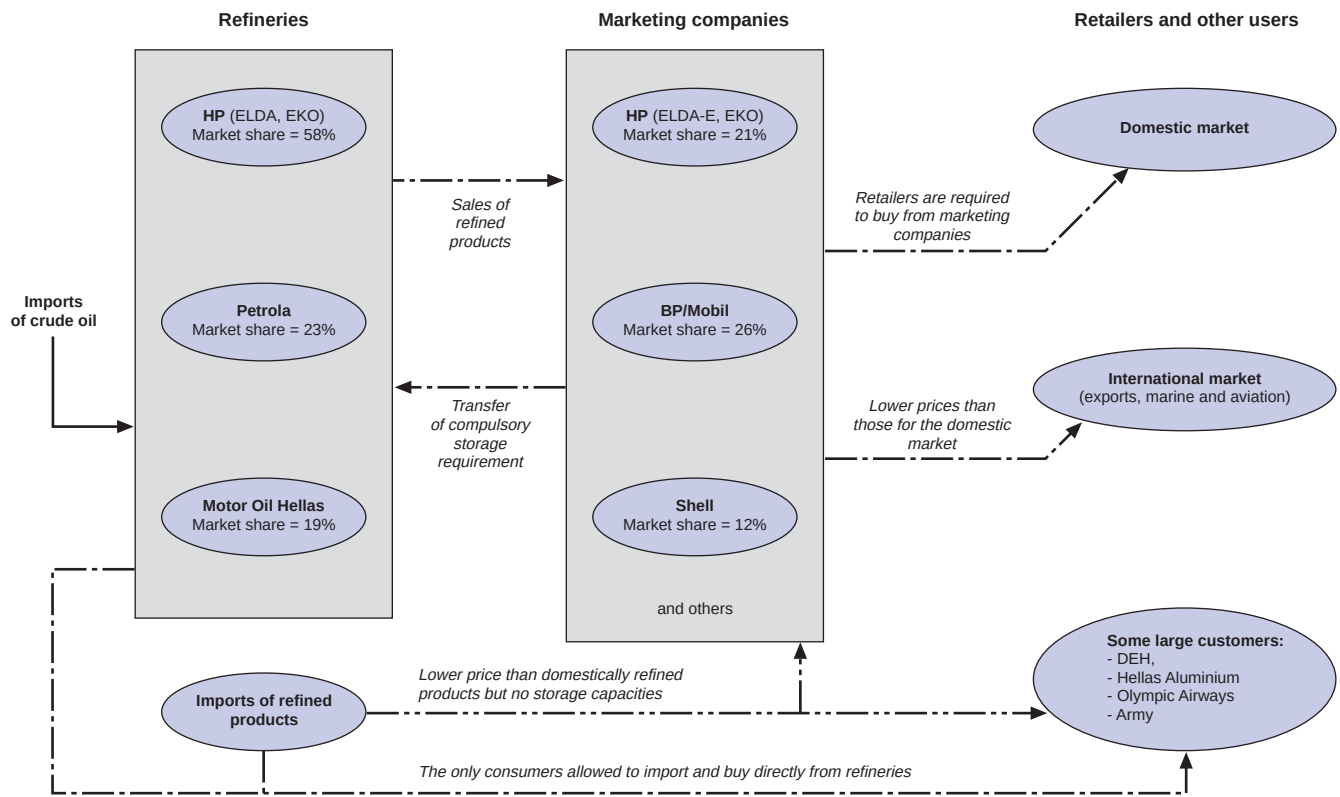
Oil refining and distribution

The oil sector plays a critical role in the Greek economy, as it accounts for more than 70 per cent of total energy demand, a share considerably higher than in most other OECD countries, and supply is mostly imported. Though a comparatively competitive market compared with the rest of the energy sector in Greece, the oil sector faces one of the most restrictive regulatory frameworks in the OECD, with a legislated segmentation of the upstream and downstream markets.

Specifically, refineries are not permitted to sell or distribute oil products directly to the inland market, but have to sell to marketing (wholesale distribution) companies. The latter compete to supply retail pump stations but, in turn, are prohibited from operating pump stations. Adding to the segmentation, pump stations cannot buy directly from refineries (Figure 13).[95] In an important step in liberalising this sector, oil prices were freed from administrative control in the early 1990s and the prohibition of refined fuel imports by marketing companies was lifted.[96] Subsequently, competition intensified in the downstream markets because of an increase in the number of marketing companies: the Ministry of Development licences for the inland market doubled to 35 between 1991 and 1997. The retail market is even more competitive, and is characterised by a high number of pump stations per vehicle by international standards and many independent owners. As a result, there exist significant variations in the prices offered by retail dealers, at least in the main urban areas.[97] Competition has helped contain distribution margins, after an initial increase following the 1992 liberalisation.

However, the upstream refining sector has essentially remained a duopoly dominated by the state-owned company, Hellenic Petroleum (HP) and the privately owned Hellenic Motor Oil/Aramco, which have a 60 per cent and 20 per cent market share, respectively. There is little competition from refined fuel imports, partly reflecting regulatory constraints; imports accounted for only 3 per cent of motor gasoline and fuel oil consumption in 1996. Though marketing companies have been free to import refined oil since 1992, legislation concerning the holding of compulsory stocks impedes imports by essentially forcing marketing companies to buy exclusively from domestic refineries as part of ''supply-for-storage'' contracts. The linkage of purchase agreements to storage contracts arises from the fact that Greek law, unlike that of other EU countries, does not permit compulsory stock obligations (equivalent to 90 days of sales) to be met by oil holdings in other EU countries. It does however permit marketing companies to transfer their storage obligation to Greek refineries which supplied them during the previous year and with which they have a supply contract for the current year.[98] At the same time, building new independent storage facilities in the key Athens area is uneconomical and, in any case, obtaining a permit is difficult due to environmental regulations. The EC has launched a proceeding against the Greek State in the European Court of Justice against what it considers

Figure 13. **Institutional structure of the oil sector**



Source: OECD Secretariat.

a violation of EU law regarding the free movement of goods. Regulations provide HP with another advantage over the other refineries as it is the sole supplier to several large consumers (Olympic Airways, the army, DEH, and Hellenic Aluminium).

The duopoly status combined with the restraint on imports appear to provide the refineries with the scope to raise their sales margins significantly in the domestic inland market. HP's ex-factory gross margin (*i.e.* before deducting refining costs) is about 50 per cent above the international market benchmark; specifically HP's ex-refinery average gross margin is estimated to be US$4.17 per barrel compared with the gross margin based on Platt's Mediterranean FOB spot quotation of US$2.64 per barrel. The bulk of this advantage is attained in the inland market where gross margins are three times that of the more competitive bunker market. Unsurprisingly, refining accounts for almost all of HP's profits. A competitive downstream market leaves pre-tax fuel prices near the OECD average according to International Energy Agency data.

Prior to a highly successful initial public offering in July 1998 for 23 per cent of its share capital, HP undertook a major investment programme and started a corporate re-organisation.[99] The re-organisation will focus HP's activities on the core refining and subsidiary distribution sectors, and employment will be gradually reduced by 20 per cent by 2002, especially in refining which accounts for more than 50 per cent of total HP's labour force. Additional productivity gains are envisaged from the implementation of a Dr 200 billion investment programme (of which Dr 95 billion on refinery operations) during 1998-2002. In this respect, a competitive advantage for HP is that it is expected to remain a beneficiary of public funds, while meeting environmental standards is expected to impose substantial financial cost on the other refineries in Greece.[100] Such funds have helped keep HP less-leveraged than its competitors, despite the recent completion of a large modernisation programme; its long-term debt was only Dr 8 billion at end-1997.

The oil market clearly suffers from a lack of competition at the refining stage. The main source for additional competition could be imports, but this requires legislative changes in line with EC directives on storage contracts. Though environmental concerns are constraining, a more lenient licensing procedure for the construction of storage facilities would also provide more room for competition from imported refined fuel products. Competition from domestic

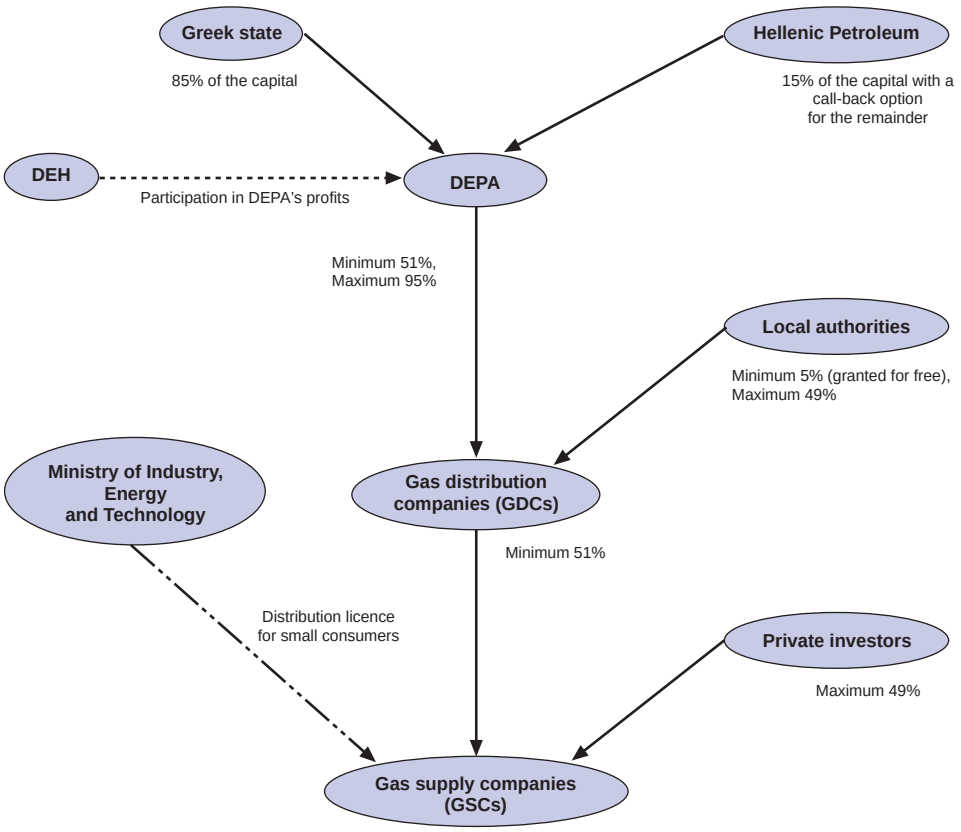
sources would be enhanced if retailers were allowed to buy fuel directly from a refinery (bypassing the marketing companies) as well as eliminating HP's exclusive right to supply large public consumers. The entry of more players in the refining stage would also provide more competition, and the Latsis Group's intention to upgrade the Petrola refinery would add a third major player.

Natural gas

Natural gas was introduced in Greece in 1997 through the recently built pipeline from Russia, and an additional source of supply is liquefied natural gas from Algeria.[101] Thus, demand for natural gas is in its infancy. To date, the public monopoly in natural gas, DEPA, has developed only a few sales contracts to some large industrial groups, outside a large contract with DEH. Demand from small users appears to be virtually non-existent in the near future, as the development of the low-pressure network is considerably behind schedule. The evolution of the natural gas market has been impeded by important infrastructure construction delays. Most of the high pressure pipelines have now been finally completed, but the cost overruns have been large; the total cost is currently estimated at US$2.6-3.0 billion compared with an original cost estimate of US$1½ billion. EU structural funds will bear about 40 per cent of the infrastructure costs.[102] The construction of the low-pressure system will only commence after the completion of a recent call for private sector tenders. Under the 1995 natural gas law, private sector entities are expected to contribute all of the construction cost of the low pressure system (net of EU contributions) in exchange for the right to be the exclusive suppliers for a defined region. Private sector participation is limited to 49 per cent ownership of the supply companies, which will be majority owned by DEPA's (distribution) subsidiaries (Figure 14).[103] After these projects reach maturity in the year 2020, 15 per cent of Greece's energy needs are projected to be met by natural gas.

Since natural gas will be distributed through a state-owned monopoly and there exist many barriers to entry, the sector is unlikely to operate in a competitive environment in the medium term.[104] Under the EC directive on natural gas, DEPA has no obligation to grant third party access to its network up to 2006, since the directive contains an exemption for states in which commercial gas supply was first introduced after 1988, and for which liberalisation would disrupt development of the sector. Without competition to ensure efficient pricing, the design of the price formulae for natural gas increases in importance, and in this

Figure 14. **Institutional structure of the natural gas sector**



DEPA's main rights	GSC's main rights
Imports Construction and operation of high-pressure networks Gas sales to GDCs/GSCs Gas sales to large consumers (annual consumption > 10 million m^3)	Gas sales to small consumers Construction and operation of low pressure networks under concession for 25-35 years *Note:* DEPA subsidiaries (GDCs) own the low-pressure networks

Source: OECD Secretariat.

regard the pricing policy announced to date risks being discriminatory and may contain some non-transparent elements. For industrial users, prices can be negotiated from a reference price based directly on the cost of alternative fuels (*e.g.*, post-tax heavy fuel prices) plus a fixed margin.[105] The reference price resulting from the formula appears to be in line with industrial prices of other EU countries, though it is biased upwards due to the relatively high tax on heavy fuels in Greece.[106] However, a few large public corporations, notably the public power corporation, DEH, have signed long-term take-or-pay contracts with prices determined on the basis of costs, *i.e.* import prices from Russia plus a large mark-up reflecting transport costs and DEPA's return on investments. The mark-up paid by DEH (US$7 per Gcal added to the border price of US$9 per Gcal) appears to be high compared with prices in the EU of about US$13 per Gcal. However, DEH has been granted compensation in the form of a profit-sharing agreement with DEPA.[107] The dual price system could hinder competition in the electricity sector, if the incumbent DEH and potential entrants in gas-fired generation are not treated on an equal footing.

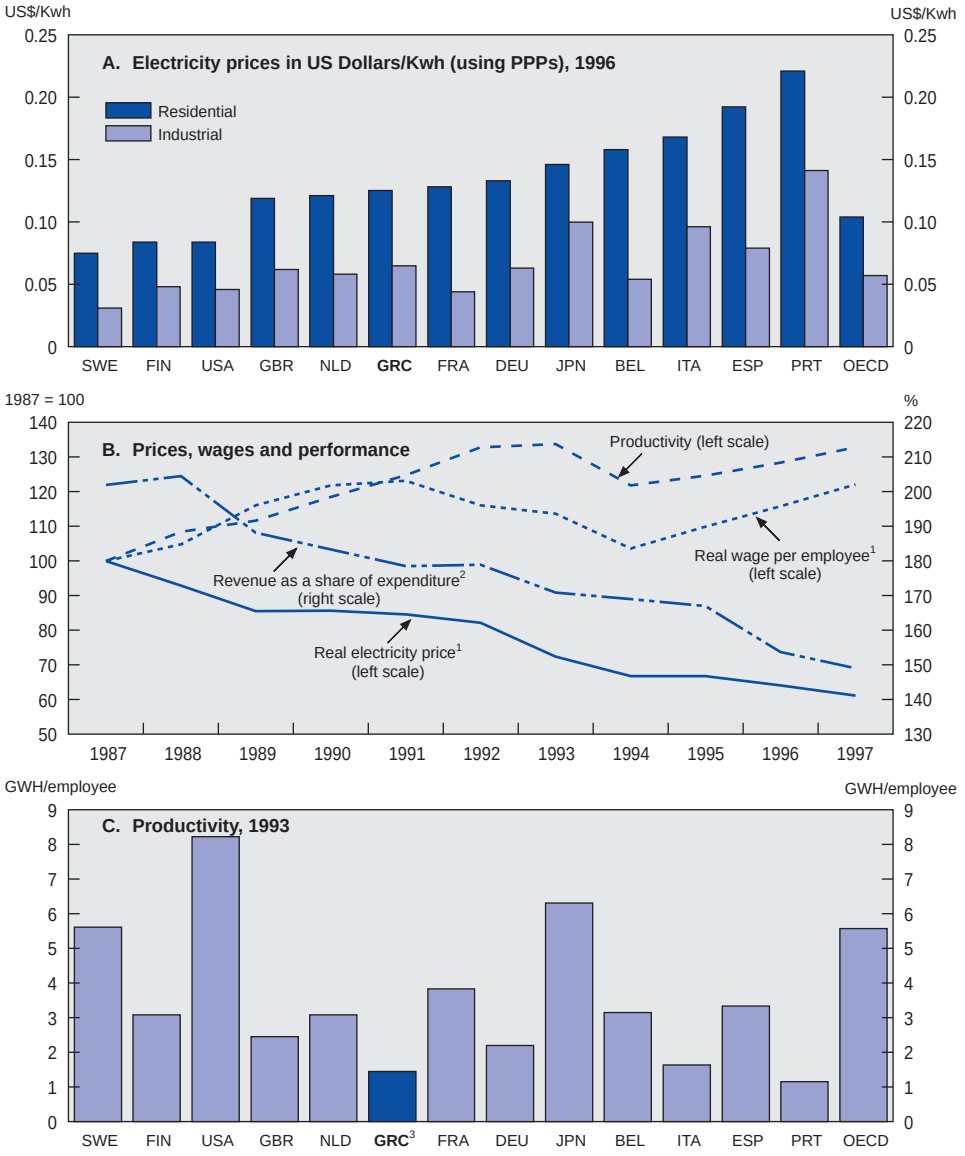
Finalising the legal and administrative framework for gas distribution is key to speeding up the introduction of natural gas. Delays in building the network could result in a significantly lower than projected growth in demand. To give a boost to the use of natural gas will require a rapid involvement of the private sector in distribution. Their interest will obviously depend on an appropriate design of the supply contracts. Regulatory arrangements will also be critical to attracting private sector participation to the sector, and these remain in their infancy in Greece. In other countries, competition has been enhanced by the introduction of a regulator with independent and strong enforcement capacities, as well as the right to mandate price setting regulations and access conditions.[108] Furthermore, the need to co-ordinate policies in the energy sectors whose products are close substitutes has led most countries to expand the coverage of the regulator to the electricity and oil sectors. The need for co-ordination is even stronger in Greece in view of the close inter-relationships between monopolies in the three sectors. DEPA and HP are the main suppliers of DEH, there exists a profit sharing agreement between DEH and DEPA, and DEPA's recommended gas prices are set on the basis of petroleum ex-refinery prices. Moreover, HP has an option to gradually re-purchase the 85 per cent of DEPA that belonged to it, prior to its recent restructuring.

Electricity sector

The electricity sector is dominated by a vertically-integrated public monopoly, the Public Power Corporation (DEH). The (high-voltage) transmission grid and the (low voltage) distribution network are owned by DEH, and there exist only a few limited exceptions to DEH's monopoly situation, as the legal framework that existed up to 1998 constrained the development of independent power producers.[109] Thus, DEH accounted for 98 per cent of both total supply and capacity in 1997. As the main source of electricity supply, DEH has struggled to meet a rapidly growing electricity demand. Though its generating capacity was increased by 25 per cent from 1985 to 1995, current total available capacity is barely sufficient to meet peak load demand, and as a result Greece suffers numerous brown outs, and there is little security back-up.[110] DEH's market power is further protected by the geographic position of Greece which has no interconnection links with other OECD countries, though an interconnection project between Italy and Greece is under construction, partly financed by the EU.[111] Net imports, mainly from Bulgaria and the FYROM, accounted only for about 2 per cent of Greece's total electricity supply over recent years.

Electricity prices are above the average of the OECD countries, according to the data of International Energy Agency (PPP adjusted) (Figure 15).[112] The standard international comparison of industrial prices is distorted, however, by the fact that DEH is required to provide large amounts of electricity at about half price to the aluminium and publicly-controlled nickel firms; adjusting for this effect, the industrial price appears to be significantly above the OECD average.[113] The relatively high electricity price in Greece is surprising as generation benefits from a strong reliance on lignite, the primary energy source for over two thirds of total generation. The dependence on lignite occurred as a result of the international oil crises in the 1970s, when Greece purposefully diversified its energy base. DEH owns and operates most of the lignite production, and exploitation costs are low since lignite is extracted from open cast mines, for which DEH does not pay depletion costs to the state. Moreover, electricity prices would be much higher had they not formed part of the anti-inflationary policy. They have been reduced by 35 per cent in real terms since 1987 – a slightly larger fall in electricity prices than has occurred in other OECD countries (Figure 15, Panel B).

Figure 15. **Electricity sector: prices and performance**



1. Deflated by the CPI.
2. Operating revenue as a share of operating expenditure, excluding interest payments and depreciation.
3. 1997 figure. Productivity calculation excludes employees in mining.

Source: DEH and *The OECD Report on Regulatory Reform.*

Government interference in several other aspects of electricity pricing and management policy are a financial burden on DEH. DEH has been required to comply with public service obligations without receiving commensurate compensation. For example, hydro-electric plants have been operated inefficiently in order to meet water supply and irrigation needs. Another important cross-subsidy arises from the fact that prices are kept uniform despite substantial geographic cost differentiation arising mainly from much higher supply costs in the islands not interconnected to the grid, which operate on more expensive fuel fired plants.[114]

DEH's financial performance is also affected by high labour costs, reflecting mainly low productivity by international standards; about one third of the OECD average after excluding employees in the lignite sector (Figure 15, Panel C). High labour costs also arise from a poor skill mix, including too many individuals in administrative positions. Operating costs are also inflated by the poor strategic choices regarding the construction of generation plants and procurement procedures. Examples of the latter are the entry into large non-competitive supply contracts. Examples of the former are hydro-electric plants with specifications partly aimed for water supply considerations and more recently the decision on a new lignite-fired plant at Florina, at a time when there is a general shift in emphasis to lower cost and more environmentally friendly gas-fired plants. Overall, with a large increase in volume offsetting the decline in electricity prices in real terms, DEH's operating income covers current expenses (including interest payments equivalent to nearly 15 per cent of operating revenues) but the rate of operating return on assets has been consistently low (close to 2 per cent). Thus, investment has been mostly financed through debt and, to a lesser extent, public and EU funds, leaving DEH highly leveraged (with a debt of Dr 1.2 trillion at end-1997, equivalent to 3½ per cent of GDP and 1½ times 1997 revenue).

Greece was granted a 2-year derogation for implementing the EC directive on the liberalisation of the electricity sector. Thus, the market for generation will begin opening up to competition by February 2001. To comply with the EC directive, the Greek government is preparing new legislation. The key features of are expected to be:

i) DEH will remain a public-controlled company;
ii) DEH will remain the owner of the transmission system, and both the high and low voltage distribution systems;

iii) DEH will remain vertically integrated but split into four business units with unbundled accounts: generation, transmission, distribution, and lignite mining;

iv) eligible (*i.e.* large) customers (as defined by the EC electricity directive) will be able to enter into direct contractual relationship with electricity supplier;[115]

v) a system operator will be responsible for dispatch and will assure the security of the network; and

vi) an independent sector regulator will be introduced.

Though the envisioned electricity law and the associated restructuring of DEH are a first step to open the electricity market, effective competition is likely to remain limited. The envisaged unbundling of DEH's accounts (especially for mining) will certainly improve transparency, but the plan leaves DEH with significant monopoly power. Given the proposed market structure, DEH will remain the exclusive supplier of the ''captive'' market of small consumers for some time to come. As this is the most profitable segment of the market, it could provide the potential advantage to the enterprise to cross subsidise operations in the competitive segment of the market from its monopoly position in the non-competitive market. Another potentially unfair advantage for the incumbent firm, DEH, appears to be the differentiated access conditions among generators to primary energy sources, which may deter potential competition, especially from the more efficient generators based on combined cycle gas turbine technology. Specifically, DEH will maintain its effectively exclusive right over low cost lignite, while the price of natural gas could be excessively high.[116]

A well functioning electricity market will become even more critical as there is an urgent need for increasing generation capacity, including through the improved functioning of existing plants; DEH's planned construction projects are barely sufficient to match a conservative projection for the increase in demand, and will in part replace inefficient plants (especially hydro-electric plants). Although, DEH intends to build 5 large new power plants – partly financed by public and EU funds – which would increase generation capacity by 15 per cent from its 1997 level, demand is projected to grow by at least 3 per cent annually. To attract more entrants will require fostering a more competitive environment and a level playing field. As a first step, the interest of new entrants and the

functioning of the new design for the electricity sector could be gauged by an early offer of licences to new entrants. Thus, new entrants who wish to enter can have their generation capacity in place by the time the market is opened up; otherwise competition to DEH will be delayed by 2 to 3 years. If there is little interest shown by new entrants, a more radical reform of the sector may be required. As in many other countries, a healthy competitive environment could be created by splitting DEH into several independent and competing distribution and generation companies. Regarding generation, somewhat smaller operating units would not be costly in terms of sector efficiency, as most economy of scale benefits can be captured by individual plants. These producers would compete to sell electricity to one or more distribution companies while the high-voltage distribution system is in the hands of a third party. However, this option has to overcome several difficulties, including a fair distribution of existing plants based on the efficiency of their generation.

In any event, the creation of a level playing field requires re-examining the obligations imposed on DEH and the liabilities it inherited from the past. These comprise: *i)* the large investment programmes since the 1980s, which have left DEH with a heavy debt and some inefficient plants; *ii)* the cost of providing cheap electricity to certain energy-intensive industrial plants; and *iii)* the implicit liabilities stemming from the firm's unfunded pension system.[117] To these, one could add the costs incurred in the construction of the natural gas pipeline, which should be borne equally by all gas consumers (*i.e.*, this is precluded by the profit sharing agreement between the two public enterprises DEH and DEPA). The Government would need to recognise these ''stranded costs'' and design a transparent and equitable funding mechanism for their recovery.[118] One strategy to cover these stranded costs, adopted in the United States and in Spain, has been to recover them through the electricity tariff. In these cases, the tariff has been set so as to recover costs but to provide a sufficient margin for innovation and entry. In view of uncertainties surrounding the accuracy of the cost estimates, the price cap has been set for a period of about 3 to 5 years and reviewed periodically. Here again, the sector regulator would need to play a critical role and the prospective legislation provides only advisory powers to the proposed new regulator. Moreover, its relationship with the Board of Energy Planning proposed in the 1995 natural gas legislation has to be clarified.

Telecommunications

The Greek telecommunications market is one of the economy's most dynamic sectors, and the publicly-owned monopoly provider of basic voice telephony services (OTE) is the most profitable enterprise in the country (in absolute terms). Nevertheless, the Greek authorities requested a 5 year derogation from the EC regarding the date of the opening of the basic telephony market. The request was based on the need to complete a large capital investment programme which would digitalise the network, and in general modernise OTE's infrastructure and commercial structure. In the event, Greece obtained a two-year derogation to 31 December 2000 in October 1997.[119] However, the EC did not delay liberalisation due to the existence of an underdeveloped network structure, but rather to provide OTE sufficient time to rebalance its tariff structure. In fact, network development is not a central issue. First, telephone penetration as measured by main lines per inhabitant is close to the EU average (though many are for second, summer homes). Second, the universal service obligation is not in fact rendered substantially more difficult in Greece by its more complex topography as this can be dealt with through modern technology (*e.g.* wireless in the local loop). Third, the degree of digitalisation (at close to 50 per cent at end-1997 compared with 35 per cent in 1995) was below the EU average, but not far below that of other EU countries that have already liberalised their markets (*e.g.* Germany and Spain) (Table 22). In the event, the EC froze financing to a large investment programme by OTE in September 1998 due to the slow process in reforming regulations to conform with EC directives.

Though OTE is the monopoly provider of basic telephony services, it faces stiff competition from the two private mobile telephone operators (Panafon which is partly owned by the British Vodafon, and TeleStet, which is partly owned by the Italian Stet). Greece was the last country in the EU to grant licences to mobile phone operators (1992 compared with the mid-1980s for most other EU countries), but mobile phones have already achieved an 8.5 per cent market penetration, compared to 14 per cent for OECD countries on average.[120] Both companies are achieving high profit rates (a return on total assets of the order of 50 per cent in 1997), and plan stock market flotations in 1998. These high profits in part reflect low interconnection costs but also mobile phone charges which are high compared to operators in other countries. The latter may reflect both a relatively inelastic demand at low market penetration as well as dissatisfaction with the

Table 22. **An international comparison of telecommunications sector indicators**

1995

	Mainlines per 100 inhabitants	Degree of digitalisation[1]	Revenue as a per cent of GDP	Wages as a per cent of operating expenses	Employees/ total employment	Taxes as a per cent of total revenues	Net interest paid as a per cent of operating expenses[2]	Investment/ revenue	Return on capital[3]
Belgium	45.6	66.0	1.6	36.7	0.7	2.9	6.2	20.7	4.9
Finland[4]	55.0	89.8	2.0	67.0	0.7	2.0	1.6	18.9	2.7
France	56.3	100.0	1.7	47.3	0.7	13.0	9.3	21.4	2.5
Germany	49.5	56.0	2.0	24.9	0.6	5.4	15.5	25.4	3.6
Greece	**49.4**	**35.3**[5]	**2.4**	**36.7**	**0.7**	**12.6**	**5.1**	**25.5**	**18.0**
Ireland	36.7	79.0	2.9	46.7	1.0	4.6	11.0	16.4	4.0
Italy[6]	43.4	75.6	1.8	24.8	0.5	11.1	5.0	18.7	3.5
Japan	48.8	90.4	2.1	31.3[7]	0.3	4.9[7]	3.9[7]	32.9	1.8
Mexico	9.7	87.2	2.6	26.8	0.2	2.0	0.0	19.9	9.2
Netherlands	51.8	100.0	2.1	46.2	0.5	8.8	0.0	19.3	7.2
New Zealand	46.4	97.0	3.5	29.6	0.6	11.8	6.9	14.0	15.6
Norway	55.8	81.8	2.2	28.3	0.9	3.0	2.8	21.6	5.4
Portugal	36.1	70.0	3.1	28.1	0.5	9.8	8.6	24.2	4.1
Spain	38.5	56.0	2.0	24.8	0.6	1.1	15.4	33.4	3.8
Sweden	68.1	91.0	2.6	28.7	0.9	2.3	5.7	27.5	3.3
Switzerland	62.3	66.2	2.7	19.1	0.5	0.0	0.0	24.7	9.4
Turkey	23.0	77.0	1.1	26.9	0.4	29.0	0.0	24.4	1.4
United Kingdom	50.2	87.7	2.6	37.3[8]	0.7	5.4[8]	1.7[8]	14.4	8.5
United States	62.7	89.8	2.6	27.9	0.7	7.0	5.6	17.1	5.0
OECD[9]	47.2	82.8	2.3	29.9	0.6	5.7	6.2	22.6	4.3

1. Per cent of digital mainlines on the fixed network.
2. Interest expenses include only public telephone operators with revenues higher than one billion dollars in each country.
3. Net income as a per cent of total assets.
4. For taxes, Telecom Finland only.
5. Raised to 47 per cent by 1997 and estimated to be close to 70 per cent by end-1998.
6. Taxes include income taxes and concession fees.
7. NTT only.
8. BT only.
9. Weighted average based on GDP.

Source: Ministry of Finance and OECD, *Communications Outlook.*

service provided by OTE. Mobile phone charges have been falling but their decline has been limited in the long-distance segment by a clause in the license agreement which requires mobile tariffs to be above the comparable OTE tariffs in this segment. A third operator, CosmOTE, which is a joint operation between OTE and the Norwegian Telenor, received a licence in 1995, but so far has had difficulty getting its operations off the ground.[121]

The short period prior to the advent of a fully liberalised market in basic voice telephony requires OTE to achieve a substantial restructuring and capital investment programme. OTE's profitable operations are due to an overall high but very unbalanced tariff structure, which leaves it vulnerable to new entrants, especially those aiming at the domestic long-distance market (Table 23). Fixed

Table 23. **A comparison of telephone charges**

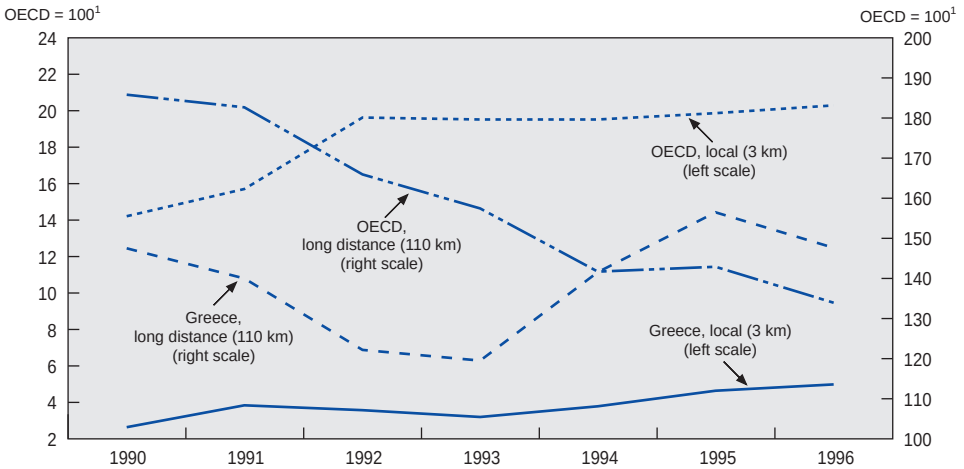
January 1996, expressed in PPPs

	Business charges[1]	Residential charges[2]	International business charges[3]	International residential charges[3]
Belgium	824.5	443.9	90.3	103.8
Finland (HTC)	391.4	308.3	52.8	63.7
France	747.8	345.3	86.1	93.5
Germany	862.6	385.4	56.3	62.2
Greece	**1 159.0**	**605.6**	**110.8**	**107.0**
Ireland	1 176.7	601.0	87.2	94.1
Italy	1 241.2	479.6	94.0	97.2
Japan (NTT)	958.7	377.5	83.7	77.3
Mexico	3 011.2	1 062.5	362.1	331.6
Netherlands	488.0	299.5	75.3	78.4
New Zealand (TCNZ)	924.0	395.3	117.0	103.9
Norway	412.3	287.9	49.8	58.0
Portugal	1 661.3	714.0	147.4	153.8
Spain	1 207.7	538.0	134.3	135.6
Sweden (Telia)	438.2	257.3	59.7	67.1
Switzerland	989.3	440.6	50.0	49.9
Turkey	952.2	374.6	167.0	164.4
United Kingdom (BT)	826.2	381.4	71.4	77.9
United States (Nynex)	1 026.3	404.7	110.5	91.2
OECD average	893.8	408.5	100.0	100.0

1. Average annual spending by a business user, based on a common basket of calls, in 1995 US dollars excluding tax.
2. Average annual spending by a residential user, based on a common basket of calls, in 1995 US dollars including tax.
3. Zone Distribution Method in US dollars. OECD countries are divided into three regions (Europe, North America, Pacific) and international call destinations are assumed to be distributed equally to each area. The output is expressed in the form of an index with the OECD average equal to 100.

Source: OECD, *Communications Outlook.*

Figure 16. **Tariff rebalancing in telecommunications**



1. Indices based on the average price for the OECD long and short distance telephone calls in 1990.
Source: OECD, *Communications Outlook 1997*.

fees and local calls are low, while long distance (especially domestic) are high compared to other OECD countries. During 1995-97, OTE has made some progress in rebalancing its tariffs and it aims to complete the process by 2000 (Figure 16). There exist three important obstacles to eliminating this cross-subsidisation. First, OTE's profitability is critical to the financing of the budget, both through its contribution of about 20 per cent of the annual corporate income tax revenue, and through dividend payments of a broadly equivalent size. Moreover, OTE's profitability enhances the amount of the potential privatisation receipts, which will help the budget indirectly, either by reducing the debt or by reducing the financing needs of other entities via transfers from the holding company, DEKA (see Chapter II). Second, the Government intends that public enterprise tariffs be used to reduce inflation expectations, thereby possibly delaying the needed rise in local call rates.[122] Third, OTE lacks an accounting system capable of costing the different calls, though a new accounting system is scheduled to be in place in 1999. Analytical cost accounting is also necessary to price interconnection charges, where there is much debate in the mobile phone market.[123]

A faster pace of digitalisation could offset some of the revenue loss, since it would permit time-based charging on local calls (which cannot be done on analogue lines and leads to revenue losses of about 40 per cent per line); the current low level of digitalisation partly explains OTE's very low revenue per main line compared with most other countries. OTE's process of digitalisation has suffered long delays. It was started only in the 1990s due to poor management decisions and a general situation of under-investment.[124] The factors behind the subsequent slow progress, despite the financial assistance from the EU's CRASH programme until 1993, are the decision not to outsource the installation process, legal disputes with suppliers of digital switches, and other managerial issues.[125] In the event, the new management has made speeding up this process a high priority and the current timetable is to digitalise 70 per cent of the network by end-1998, and complete the process by 2000. The cost will be partly financed by EU funds, and forms part of a Dr 800 billion investment and modernisation programme (equivalent to over 2 per cent of GDP) covering the period 1998-2002. The modernisation programme also aims to improve the quality of service to international standards by the year 2000.[126]

Besides revenue enhancement, OTE needs to reduce its operating expenditures. Following the pattern of all Greek public enterprises, OTE's personnel costs are too high. The wage bill represents nearly 40 per cent of total operating costs compared with an OECD average of 30 per cent, and the share of telecommunications employment in total employment is about 20 per cent higher than in other OECD countries. Moreover, about one-half the staff are employed in the construction of the network for whom there will soon be reduced demand. Nevertheless, OTE has reduced its labour force substantially over the past decade: a workforce of over 30 000 in 1988 has been reduced to 22 500 in 1997. A large part of the resulting productivity gains, however, was absorbed by substantial increases in compensation; compensation rose by 30 per cent in real terms during the past decade. Management intends to reduce staff by another 20 per cent by 2002, through the continued use of attrition and an early retirement programme.[127] The plan also calls for an improved skill mix, including hiring personnel trained in telecommunications technologies.

Since OTE is among the most profitable (based on a return on assets), and least indebted, telecommunications operators in the OECD, the necessity of rebalancing its tariffs at the same time as it finances a substantial modernisation

of its infrastructure does not seem problematic. However, it will need to avoid certain pitfalls. For instance, too ambitious an expansion into operations outside of its core business of providing adequate basic telephony services could distract management and require expending operating profits committed to the investment programme. Such problems could arise from OTE's investment in eastern European and central Asian countries.[128] In addition, OTE's entry into the mobile telephone market has so far resulted in losses, from both the initial investments and the offering of discount offers to attract clients. Similar problems could emerge from an aggressive entry into new generation services such as satellite TV and cable television.[129]

In view of OTE's current dominant position and its earning of sizeable monopoly rents, an earlier-than-required opening of competition to a second operator, similar to the strategy of other OECD countries, could result in substantial and rapid reductions in telephone tariffs and a higher demand for a multitude of related productivity-enhancing services. The floor on mobile phone tariffs should also be removed as it deters competition to OTE. In the event, with the advent of several operators in mobile and fixed telephony, the responsibilities of the regulator established in 1994 for this sector (the National Telecommunications Committee, EET) will be increasingly more important (*e.g.* transparent access and pricing to the network). To date, however, these responsibilities appear to be quite limited and the committee remains understaffed. Its role is solely to monitor the telecommunications market so as to assure competition, give an expert opinion to the Government on licence approvals, and propose measures to the Government. The 1998 telecommunications law plans to increase the EET's responsibilities to be able to award licences. A promising sign was EET's fining OTE Dr 80 million for non-competitive practices in early October 1998.

Water and sewerage

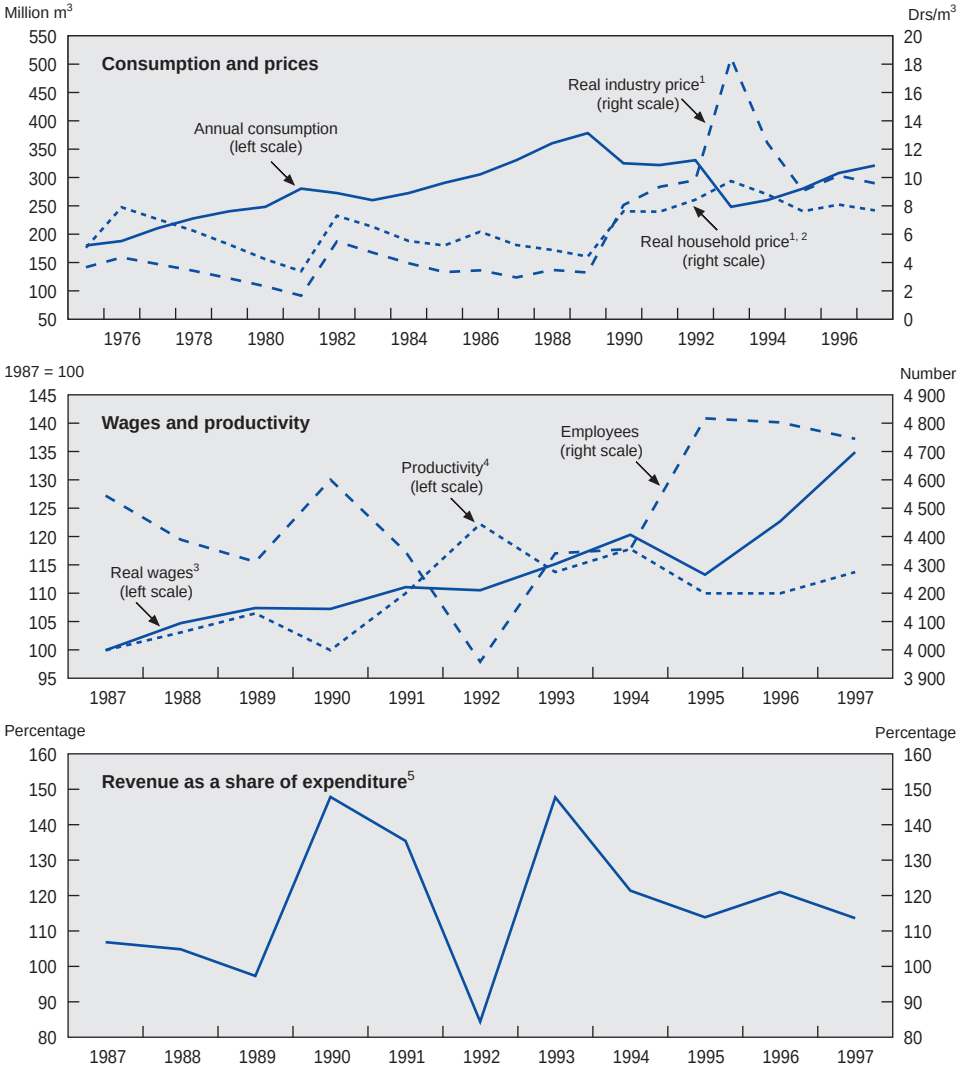
In view of the concentration of the population in two large urban centres and the natural catchment areas surrounding these cities, there is no national water grid and the bulk of water and sewerage services are provided by public enterprises in Athens and in Thessaloniki. In Athens, the Company for Water and Sewerage of the Capital (EYDAP) serves 40 per cent of the total Greek population. In Thessaloniki, there are two separate companies, the Water Organisation

of Thessaloniki (OYTh) and the Sewerage Organisation of Thessaloniki (OATh), which serve another 12 per cent of the population. Public entities run by local governments are responsible for these two services in their respective districts.

Regarding Athens, current water supply capacity is sufficient to meet demand until 2017, except under drought conditions, while distribution could become a constraint as early as 2001.[130] These projections envisage demand rising towards the level in other EU countries, but its growth can be slowed by price increases, as has occurred in the past. Specifically, water prices had been maintained very low until 1990, when a drought forced an increase in prices of the order of 200 per cent in real terms over the period 1990-92 and the introduction of a sharply more progressive pricing system to deter large consumers. These changes resulted in a 33 per cent contraction in demand (Figure 17, Panel A). Subsequently, however, prices have not been increased.[131] The price of residential water would now need to be raised by 23 per cent in real terms (for the critical 5-20 cubic meter monthly consumption) to cover a large part of long-run marginal cost, while industrial prices are considered to be at a broadly satisfactory level as they had initially been raised by more.[132] The price of water to local authorities would need to be raised much further. EYDAP has been obliged to provide water to local authorities in the greater Athens vicinity at highly subsidised rates equal to about one-third that charged to other consumers; and the water is supplied in bulk without an escalation in price corresponding to usage. The result is that local authorities are inefficient consumers – with water often used for agricultural purposes – and use water re-sales as a source of revenue.[133]

Despite the price increases attained at the beginning of the decade, EYDAP's financial position continues to be unsatisfactory. Though EYDAP does not have separate accounts for water and sewerage operations, the latter is considered to be the biggest loss maker. Sewerage fees are low in international comparison. Specifically, sewerage fees in Athens are equal to 40 per cent of the water bill for residential consumers and 28 per cent for industrial consumers, compared to an international standard of above 60 per cent for both types of consumers. Moreover, local authorities and the public sector receive large discounts. EYDAP's revenue also suffers from poorly functioning meters, which are estimated to lead to a 15 per cent loss in revenue. Finally, the financial impact of subsidies provided to local authorities is compounded by the fact that they are constantly in arrears.[134]

Figure 17. **Water: prices and performance**



1. The increase in the price of water in 1996 was due to the inclusion of the special tax to finance the construction of the Mornos aqueduct into the tariff.
2. Prices were not increased after 1992, however various charges and indirect taxes were included into the price.
3. Deflated by the CPI.
4. Number of connexions per employee.
5. Operating revenue as a share of operating expenditure, excluding interest payments and depreciation.

Source: EYDAP.

EYDAP's financial position is also burdened by excess labour costs. These represent over 60 per cent of operating costs, in what is considered to be a capital intensive sector. A comparison with an average of UK companies indicates that EYDAP's productivity in 1995 was about 25 per cent lower (3.2 employees per thousand connections *versus* 2.5 employees per thousand connections), in part due to a 20 per cent increase in employment during 1992-95 (Figure 17, Panel B). EYDAP's labour force also suffers from a poor skill mix, with one-half the employees in administration and finance. Wages per head also appear high as they are about 30 per cent higher than in the manufacturing sector. The low revenues and high labour costs have resulted in an operating deficit equal to 10 per cent of operating costs, and an outstanding debt of Dr 42 billion.[135] The financial situation is better in the Thessaloniki companies, partly because their labour costs are lower.

Looking forward, EYDAP and the water and sewerage companies of Thessaloniki have been included in the Government's list of enterprises to be partially privatised by end-1999.[136] The companies will be split into an operation management firm and an owner of the infrastructure, and the operations will be sold in the form of a concession agreement in the second half of 1999. For the sale to take place on schedule, the new management teams will need to complete, and then begin to implement, their respective business/restructuring plans. To generate sufficient profitability to attract private investors, these will certainly need to reduce labour costs and contain an agreement with the Government regarding the financing of the future infrastructure works. Water price increases would also need to form part of the reform package. Together with an indexation of future tariff increases to inflation, these have been projected to make EYDAP capable of self-financing its Dr 72 billion water investment programme albeit with a debt build-up over the medium term (replacement of one water treatment plant at Galatsi, improved security of conveyance, introduction of sludge treatment facilities).[137] However, the financing of flood control projects would not be covered and the Government would henceforth need to take over this public service obligation. A study similar to that undertaken for Athens water supply would need to be undertaken regarding the viability of the Athens sewerage operations, especially as the current infrastructure does not achieve the desirable waste treatment processing prior to its ejection into the depths of the Saronic Gulf (40 per cent compared to an end-target of 95 per cent). The upgrading of the

Psytalia treatment plant of Athens is estimated to cost around Dr 30 billion. In the event, sewerage charges would likely need to be raised towards international levels, similar to those of UK firms, which include substantial infrastructure amortisation charges. An extension of this strategy would be for EYDAP to be split into two companies – if the network permits – each providing water and sewerage services to approximately one-half the greater Athens area. Creating this possibility for benchmarking has enhanced competition in other countries (*e.g.* the city of Paris). Moreover, with three large players (two in Athens and one in Thessaloniki), there would also exist a sufficient mass of know-how and economies of scale which could provide services or operate the local water and sewerage entities, which are for the most part managed poorly.

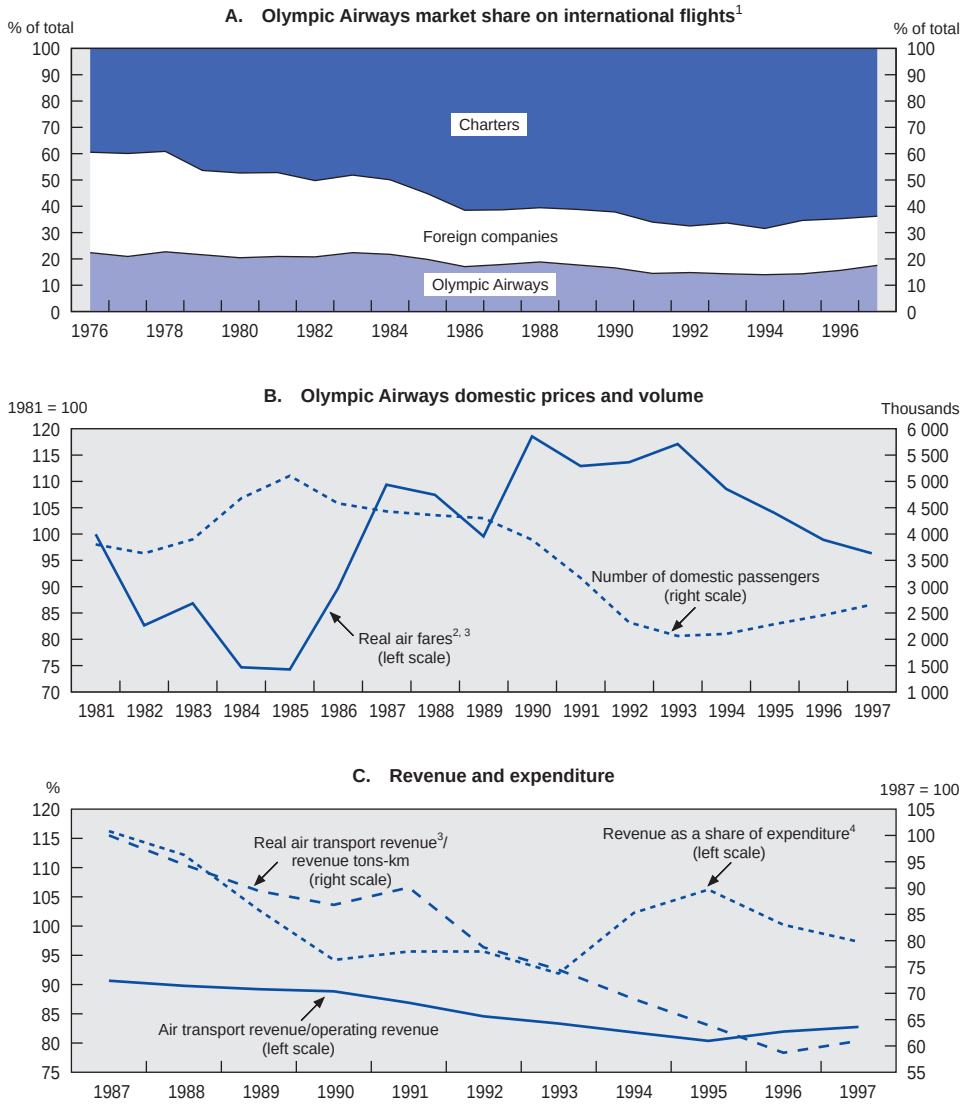
Potentially competitive sectors

Air transport and Olympic Airways

Similar to developments in many other EU countries, competition has intensified on international flights over the past 15 years, but it remains more limited in domestic air transport. In the context of a rapidly growing international market, foreign carriers with unscheduled flights (charters) have gained sizeable market shares. As a result, the publicly-owned Greek flag carrier, Olympic Airways Group (OA), accounted for about 16 per cent of international passengers to Greece in 1997, compared with over 20 per cent in the early 1980s (Figure 18, Panel A).[138] On domestic flights, several domestic private carriers have entered a few key routes following the market liberalisation in 1992, and their market share has increased notably to about 25 to 30 per cent of the domestic market.[139] On these routes, price competition has forced down OA's prices, with the fare on the high-volume Athens to Thessaloniki route declining by 25 per cent in real terms between 1992 and 1997 (Figure 18, Panel B), yet it remains about 30 per cent above that of its competitors. Largely as a result of price competition, which has also been extended to some key international flights, almost all of these new airlines initially faced serious financial difficulties, and remain heavily indebted.

Olympic Airways' monopoly control of handling services at all Greek airports has been a major hindrance to the development of competition in air transport. Ground handling costs represent a significant element of an airline's overall costs, and these are high at Greek airports.[140] Handling operations provide a critical source of revenue for OA, as they are the only major profit earner in the

Figure 18. **Air transport: prices and performance**



1. Share of total international passengers to and from Greece.
2. On the Athens-Thessaloniki route.
3. Deflated by the CPI.
4. Operating revenue as a share of operating expenditure, excluding interest payments.

Source: Olympic Airways.

company, bringing in about 15 per cent of total operating revenues (Figure 18, Panel C). Following an EC infringement procedure in 1994, the Greek authorities have improved the international terminal in Athens and OA also committed itself to establish a tariff structure which is better related to the actual costs of the services. Moreover, a licence for a second passenger handling operator (Goldair Handling) has been provided by tender for the international airport of Athens in February 1998. Following these actions, the EC closed the procedure. However, EU airlines have also complained about the preferential treatment of the incumbent carrier in the allocation of slots at Greek airports and opening hours of certain Greek airports geared exclusively to OA's needs.[141]

Following the approval of an EC-approved restructuring programme and a massive debt write-off in 1994 (following one in 1990), the airline turned only briefly into profit, and the programme is now far off-track. The restructuring plan included a reduction in labour costs (through a wage freeze and an early retirement programme),[142] the reformulation of the route network to reduce loss-making flights, and a further opening of the Greek air traffic market to competition, including the removal of any constraints on charter services (*e.g.* seat-only sales) and full access to the Greek islands from June 1998. The agreement also comprised a cancellation of Dr 500 billion of debts (equivalent to 2 per cent of 1995 GDP). Due to the slippages in the programme's implementation, the EU has delayed disbursing the second tranche of the agreed Dr 54 billion capital injection until August 1998, when it approved an additional Dr 14 billion in state aid.

The deterioration in OA's financial performance mainly reflects a dramatic slippage in wage and fringe benefits, which have put the company back into the red in 1997 despite the large relief from financial costs stemming from the reduction of OA's debt. Specifically, wages increased by 34 per cent between 1995 and 1997, and both off-day entitlements and overtime payments increased three-fold, so that total compensation rose by 50 per cent during 1996-97. The timing of these increases was especially unfortunate as OA was already in a poor position *vis-à-vis* its international competitors prior to these recent wage slippages; its labour costs accounted for 40 per cent of total operating costs, compared to an average of 26 per cent for companies reporting to IATA. Excess ground personnel and low productivity are the main culprit for OA's high labour costs (Table 24). However, social benefits are also generous, in

Table 24. **An international comparison of airline performance indicators**

1996

	Transported passenger-km/ employee	Revenue tonne-km/ employee	Flight and cabin crew/total personnel	Passenger load factor[1]	Total load factor[2]
Lufthansa	1 856	369	0.39	0.69	0.70
Air France	1 492	280	0.26	0.76	0.73
British Airways	1 869	245	0.30	0.73	0.69
Olympic Airways	**850**	**93**	**0.17**	**0.65**	**0.52**
Alitalia	2 083	295	0.33	0.69	0.69
TAP (Portugal)	1 157	135	0.21	0.67	0.57
Iberia	1 155	138	0.20	0.70	0.61
Delta Airlines	2 566	267	0.38	0.70	0.58

1. Passengers-km transported/passengers-km offered.
2. Total tonnes-km transported/total tonnes-km offered.

Source: Institut du Transport Aérien.

particular for pensions where eligibility conditions are looser than in most other EU carrier (*e.g.* stewardesses are entitled to receive a full pension after 17 years of work from the age of 42). A lack of technical and working time flexibility and an inadequate skill mix of OA's labour force requires excess staffing and overtime payments, and non-basic salary payments account for 30 per cent of total earnings. Regarding other operational costs, the age and disparity of the fleet and the stockage of diverse spare parts raise maintenance costs significantly.[143] An improvement of OA's cost control is hindered by the poor accounting practices, with little accounting separation of activities, thus making it difficult to identify inefficiencies or to spin-off and sell non-core services.

On the income side, OA's low revenue is due to weak demand, increased price competition on domestic and international flights, and a poor marketing strategy. Revenues are also hurt by the obligation to provide some services at prices which do not reflect cost, even though OA's monopoly in ground handling and maintenance were intended to compensate for the various public service obligations. These include the provision of a national service, in particular to the islands, links to the main Greek communities around the world, and free tickets for certain groups (public servants, students, armed forces personnel and the clergy). In addition, the airline is burdened by the requirement to fly the daily press delivery (at prices far below cost and with larger than otherwise-required

planes) and by arrears of political parties for services rendered. Revenue losses also arise from OA's commercial strategy which is considered to be poor, with insufficient efforts on marketing and a reliance on a simple strategy of commissions with ticket agents. Indicative are the expenses on marketing and distribution; they account only for 10 per cent of OA's operating costs compared with an average of 18 per cent for other airlines. These factors are reflected in OA's load factor, which is one of the lowest in the world.

To restore OA to financial viability, the Government hired a new board of directors and charged it with drawing up a new business plan. The plan that was finally approved by Parliament in early 1998 aims at achieving a one-off Dr 43 billion improvement in OA's results – albeit relative to a pessimistic baseline scenario – mainly by controlling labour cost developments and reducing maintenance and other costs.[144] The proposed strategy is a step in the right direction but is not sufficiently ambitious when account is taken of Olympic's starting position and the expected strengthening of competitive pressures in the near future. OA lost its monopoly in air services to the islands starting in July 1998 and ground handling, currently a major source of OA's income, will gradually be opened to competition in the larger airports starting from 1 January 1999.[145] Moreover, work stoppages in 1998 – in protest at the latest restructuring plan – have led to a drastic decline in passenger demand of about 12 per cent compared to a year earlier, and have thus put the integrity of the plan further into doubt.

To be able to resist tougher competitive pressures, and survive an eventual downturn in the world market, OA should follow some of the general trends in the OECD. First, it will need to focus more on its core activities (which entails terminating the loss-making long haul flights), and an outsourcing of peripheral activities (*e.g.* handling, which in any case will become less profitable, and catering) would allow OA to increase its cost efficiency. A substantial further reduction in labour costs is the other main area where OA has to align with other airlines' practices, in particular, to compete with new airlines where working conditions are more flexible and wages are significantly lower. In this context, if renegotiating the existing contracts with the employees cannot succeed, OA could examine the establishment of a two-tier scale system (as did Air France, Austrian Airlines, or Alitalia). Adapting the labour supply to the seasonality of traffic may require a more flexible assessment of working time and an increased

use of fixed-term contracts. In line with other carriers, social benefits should also be curtailed. In particular, for pensions, the retirement age in some airlines has been increased substantially. Forging strategic alliances with other international carriers (*e.g.* through code-sharing agreements, technical co-operation, providing feeder flights, and the pooling of aircraft) has been another general trend in the air transport service industry. Such a move would also permit OA to focus on regional middle distance routes.[146] In this regard, the brand new Spata airport could be in a position to act as the main hub for eastern Mediterranean countries.[147] With the view of identifying a strategic partner for Olympic Airways, the Government hired an international investment firm in October 1998. Regarding the air transport market in general, the more competitive domestic market needs a more active and independent role from the Civil Aviation Service, both in the area of regulation (licensing, equitable access to infrastructure and transparent pricing of support services) as well as airport operation, maintenance and construction.

Television and radio

The public operator, National Broadcasting Corporation (ERT), lost its monopoly when the market was liberalised in end-1989. Its market share fell rapidly, to below 50 per cent in 1990 and subsequently to 6 per cent in 1995 before rising to 10 per cent in 1997. Advertising revenue suffered more, as the increased competition from 9 other TV stations and a plethora of radio stations reduced the per unit price of advertising substantially in real terms. Though ERT's revenues are enhanced by a Dr 1 000 monthly charge on all electricity bills, it nevertheless remains a substantial loss maker, with an annual deficit of the order of Dr 10 billion on operating revenues of Dr 45 billion. As a result, ERT's debts currently stand at Dr 35 billion and it is Dr 15 billion in arrears to DEH and OTE. In contrast, the main private stations are earning profits. Part of ERT's financial difficulties arise from an inability to downsize in line with its decline in market share; *i.e.* it still operates three TV stations, and 5 radio stations. Moreover, despite a three year hiring freeze, total employment of 3 480 has not adjusted substantially from the level at the start of the decade. As a result, its productivity is estimated to be about a third of the private channels. The five-year restructuring programme that was agreed between management and the Government in September 1998 contains a 15 per cent reduction in the workforce through early retirements, a transfer of local stations to regional authorities, as

well as productivity gains of 40 per cent over the 5 years, so that profitability is restored by 2000.

Enterprises implementing public policy

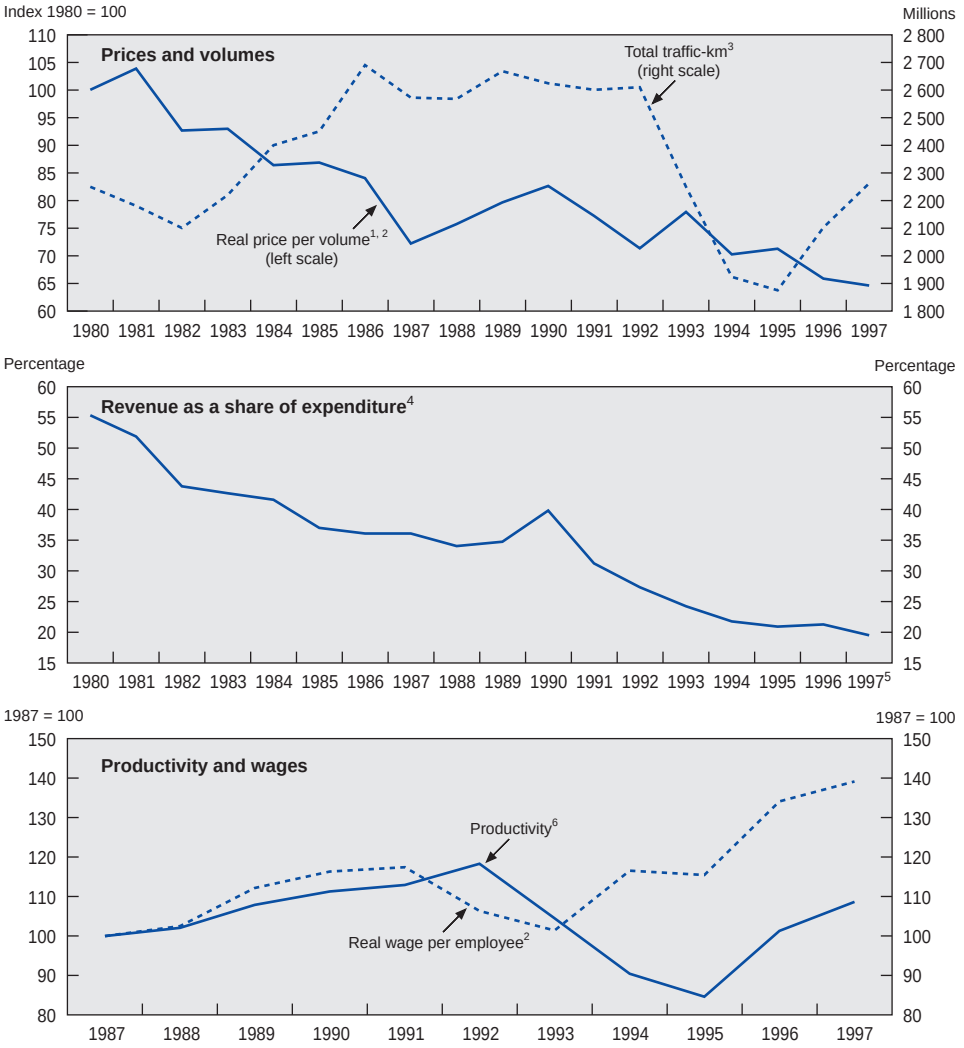
Rail transport

The lack of a clear strategy for rail services over many decades has reduced the national railroad company, OSE, to a lamentable state as regards both its financial position as well as its infrastructure development. With emphasis focused on developing the road network, investment in infrastructure had been minimal until the 1990s, when a large scale upgrading of the system began under the CSF I and was continued under CSF II. As a result, modernisation has commenced on the main route of Athens to Thessaloniki, and then on to the international network, with a view to the construction of a dual track network, and the introduction of electrification and automatic signalling. Travel time, however, remains disadvantageous for rail passengers compared to car or plane travel. The other parts of the network are in worse shape. There exist lines with different gauge track relative to the main line (requiring cargo transfers), the geometric characteristics of the rail layout and many level crossings slow train speeds, and wagons have not been modernised.

Pricing policy has partly offset the deterioration in the quality of service and the increased competition from substitute sources of transport. The Government – for social and anti-inflationary reasons – has increased passenger and cargo fares by far less than inflation. As a result, both prices fell by 25 per cent in real terms between 1980 and 1997. Despite the decrease in price, total (cargo and passenger) demand has fallen substantially during this period, though it has recovered somewhat in 1996 and 1997 (Figure 19, Panel A).[148] Nevertheless, the volume of demand relative to the size of the network remains far below comparable indicators of other EU countries, and revenue per volume (passenger and cargo kilometres) and the revenue per employee indicators are among the lowest in Europe (Table 25).[149]

By 1997, operating revenues covered only 20 per cent of operating costs compared with 55 per cent in 1980 (and compared with about 55 per cent on average currently in most EU countries) and the operating deficit was equal to Dr 120 billion (Figure 19, Panel B). Despite large operating and capital transfers

Figure 19. **Rail transport: prices and performance**



1. Weighted average of revenue per passenger-km and revenue per tonne-km. Weights are the average volume shares of the period 1987-97.
2. Deflated by the CPI.
3. Passengers-km + tons-km.
4. Operating revenue as a share of expenditure, excluding interest payments and depreciation.
5. OSE estimates. OECD estimates for maintenance cost borne by the government as per law 1300/1972.
6. (Passengers-km + tons-km)/employees.

Source: OSE and OECD Secretariat.

Table 25. **An international comparison of the rail transport sector**

1994

	Volume/Network[1] (millions)	Productivity[2] (thousands)	Revenue per employee (thousands ECU)	Revenue per volume (ECU/thousand)
Austria	4.1	355.9	20.2	55.3
Belgium	4.1	379.4	17.4	52.9
Denmark	2.9	380.5	35.3	104.0
Finland	2.2	752.2	29.5	40.2
France	3.2	578.5	33.7	60.7
Germany	3.1	403.0	37.8	95.1
Greece	**0.8**	**160.2**	**6.1**	**39.4**
Ireland	1.0	163.0	35.0	206.3
Italy	4.5	490.4	26.4	51.5
Luxembourg	1.9	296.4	36.5	226.9
Netherlands	6.2	648.0	37.4	58.2
Portugal	2.5	486.0	12.2	25.5
Spain	2.0	569.7	21.3	35.1
Sweden	2.6	1 143.6	43.9	38.8
United Kingdom	2.6	351.2	29.9	81.0

1. (Passengers-km plus tons-km)/length of the network.
2. Traffic unit per employee *i.e.* (passengers-km plus tons-km)/employees.

Source: Union Internationale des Chemins de Fer.

(about Dr 120 billion in 1997), the accumulated debt was Dr 650 billion at end-1997 (equivalent to 2 per cent of GDP).[150] The poor financial condition is also due to inflated operating costs. The ratio of personnel costs to operating costs is far above the comparable ratio in the EU, and nearly double that in the United States (75 per cent compared with about 55 per cent in the EU and about 40 per cent in private carriers in the United States). A comparison of international productivity indicators has OSE operating at one-half the efficiency of the EU (despite a nearly 20 per cent decrease in staff during the last decade). There are too many employees in maintenance compared to operations (accounting for nearly two-thirds of the wage bill), and too many employees in the underused southern part of the network in the Peloponnese (30 per cent of the employees *versus* 10 per cent of the demand). Compensation per employee also seems to be high, as wage gains in real terms have exceeded productivity gains, and the average wage exceeds that in manufacturing by 15 per cent (Figure 19, Panel C).[151] Finally, operating costs are inflated by a diverse rolling stock (the

new management plans to reduce the 18 types of engines to 4 and thus the requisite types of spare parts from 185 000 to 70 000).

The new OSE management put forth a five year restructuring plan in early 1998, which was approved in October 1998.[152] The plan will be supported by an ambitious 10-year investment programme of Dr 1.7 trillion equivalent to 5 per cent of 1997 GDP, whose main objective will be to: *i)* eventually improve the Athens – Thessaloniki main line so that the time of travel falls to 4 hours 20 minutes for the 520 kilometres; *ii)* complete the north-south link of the network; and *iii)* renew the rolling stock.[153] However, construction is behind schedule and is suffering from massive cost overruns (of the order of 100 per cent), both jeopardising the plan and implying the need to find additional sources of funding.[154] The plan focuses on reforms to cut costs and raise demand. Significant cost savings could come from the closure of non-viable lines and a focus on mainline operations and cargo transport from Greek ports towards the Balkans. However, the Government is not currently considering to downsize operations, and thus the plan expects OSE to operate these lines as a public service obligation and receive payments to cover the cost of these operations.[155] Labour costs are expected to fall due to a reduction in employment from 12 000 to 8 000 and the introduction of a more flexible labour agreement. Equipment standardisation and competitive bidding are expected to reduce maintenance costs, while private-sector participation in the construction of stations and the inner city rail network would provide needed capital for OSE to further develop its services. With a view to encouraging an increase in demand of 60 per cent, price increases will be contained over the five year programme; an increase of 2½ per cent per year in real terms for passengers and a decrease of 9 per cent per year in real terms for cargo. Nevertheless, operating subsidies (albeit reduced) will remain high: near Dr 90 billion per year, equivalent to 200 per cent of operating revenues (and amounting cumulatively to nearly Dr 500 billion, equivalent to 1.5 per cent of 1997 GDP). The plan acknowledges that due to large investment lags, only the Athens to Thessaloniki line will be profitable by 2000 and OSE will need until 2007 to cover its operating costs, excluding public service obligations. OSE's financial situation will obviously remain precarious even if the ambitious and expensive programme meets its objectives. Moreover, the Government will need to have a contingency plan for the eventuality that the

programme goes off-track, including options for more drastic cost cutting measures and a more significant downsizing.

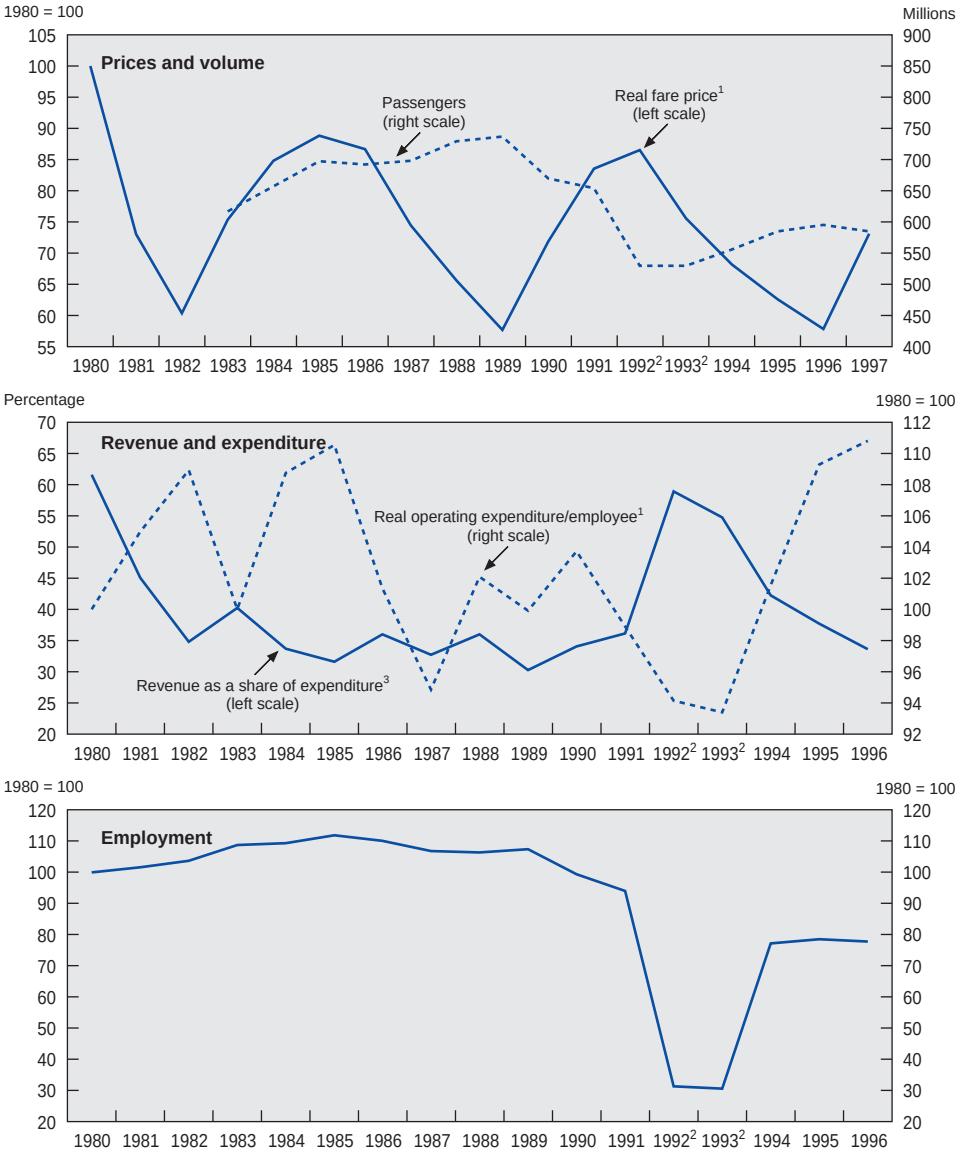
Urban transport

The urban transport sector is dominated by the large, publicly-owned and heavily loss-making Athens Urban Transport Authority (OASA), which provides bus, trolley and rail services to the nearly 4 million inhabitants of greater Athens.[156] The much smaller urban bus company of Thessaloniki is privately operated through a franchise agreement.[157] Demand for urban transport services in Athens has declined by over 20 per cent since the late 1980's (Figure 20, Panel A), and its market share of urban transport has fallen to 30 per cent – reflecting the increased usage of private vehicles. As a result of increased road congestion, and an ageing fleet (average age is 15 years), service quality has declined, which has reinforced the fall in demand.

The decline in demand is even more striking in view of the dramatic fall in the cost of a ticket adjusted for the effect of inflation. From 1980 to 1997, fares have fallen by over 40 per cent in real terms from an already subsidised price. As a result, fare revenues only cover 33 per cent of operating expenditures compared with 62 per cent in 1980 (Figure 20, Panel B), and OASA had a deficit of Dr 90 billion in 1996 and outstanding debt of nearly Dr 900 billion (equivalent to 2.5 per cent of GDP), the bulk of which was in arrears.[158] OASA's financing needs are met by subsidies from the central government budget (Dr 40 billion in 1996) and bank credits with a government guarantee (Dr 45 billion in 1996). Despite the fare increase from Dr 75 per ticket in 1997 to Dr 125 per ticket in 1998, OASA estimates that an additional increase to about Dr 250 would be needed to cover operating costs.[159]

The fare increase should be less severe since operating costs need to be brought down. International comparisons suggest that labour costs are high; they account for 82 per cent of operating costs. Despite labour shedding and a 20 per cent increase in productivity during the past 10 years (passengers per employee and vehicle kilometres per employee), the new management envisages a further improvement in productivity. For the most part, a reduction is required in the number of maintenance personnel.[160] While salaries have not experienced the sustained large gains in purchasing power observed in many of the other public enterprises, wage growth has been rapid during the last 4 years (about 3 per cent

Figure 20. **Urban transport: prices and performance**



1. Deflated by the CPI.
2. Figures for 1992 and 1993 have been affected by the privatisation of the Athens bus company.
3. Revenue as a share of operating expenditure, excluding interest payments.

Source: Athens Urban Transport Organisation.

per annum above inflation) and the average wage of an OASA employee is double the national average and significantly higher than the level in the manufacturing sector. Wage costs are burdened by large expenditure on overtime due to inflexibilities in labour regulation combined with the uncertainty of timing shifts in the face of unpredictable traffic congestion.

The new business programme was approved in October 1998 and includes both fare increases and the negotiation of more flexible labour regulations. The plan expects fares to remain below cost as part of environmental and social policies. This strategy would also serve to raise demand. Higher demand would also come from a better service quality. In this regard, the fleet will be upgraded through a Dr 130 billion investment plan which will provide for the purchase of about 1 000 new vehicles and 120 wagons, to be financed from public funds. Improving quality would also require co-ordination with traffic control to improve route speeds and reduce the uncertainty of travel time. The source of the residual financing requirement is under debate but a tax or charge on cars or fuel, though initially considered, was not finally accepted. The Government is not contemplating more radical changes in the management of OASA, either through a switch of control to the municipal government (the international norm) or by introducing a role for the private sector, as is the case for inter-city buses or in Thessaloniki, though these have been operating more or less satisfactorily.[161]

Postal service

The public monopoly, Greek Post (ELTA) is being forced by EC directives and competition from overnight mail and other new services to reform its operations so as to prevent a further deterioration in its already poor financial position. Currently, ELTA dominates the postal sector and faces competition only from international couriers, the private inter-city bus service (KTEL), and small (underground economy) inner-city operators. EC Directive 97/67 delays full liberalisation of the sector until 2003, when the currently protected market for items weighing less than 350 grammes will also be deregulated.

The sources of ELTA's current poor financial position are manifold. On the revenue side, postal tariffs have been adjusted by less than inflation – declining by over 10 per cent in real terms during 1987-97. Thus, the domestic tariff for the standard 20 grammes letter is the lowest in the EU (except for Spain), and 33 per cent below the EU average (Table 26). Despite these low prices, demand is

Table 26. **An international comparison of postal service indicators**

	Domestic prices[1]	International prices[1]	Number of postal objects/population	Population/number of postal offices	Percentage of packages within 3 days	Number of postal objects/employee
Austria	. .	175	321.6	3 128	64.8	75 374
Belgium	173	136	328.9	6 206	69.8	75 457
Denmark	171	180	317.1	4 218	83.3	65 467
Finland	147	160	379.3	. .	67.1	83 631
France	155	156	416.5	3 420	68.5	84 761
Germany	191	191	245.7	5 065	74.1	70 614
Greece	**100**	**140**	**44.4**	**8 173**	**11.9**	**42 936**
Ireland	149	140	149.7	1 832	56.1	65 360
Italy	143	144	. .	. .	14.0	. .
Luxembourg	135	128	314.5	3 962	71.8	77 797
Netherlands	123	. .	. .	. .	78.6	. .
Portugal	137	136	107.1	3 792	52.1	65 498
Spain	66	140	108.0	. .	48.1	65 369
Sweden	179	280	502.9	5 140	73.1	95 420
United Kingdom	132	140	312.3	3 040	68.2	86 853
GDP weighted EU average	149	157	240.4	3 107	60.7	62 367

1. Prices for a letter weighting 0-20 g in drachmas in April 1998.
Source: ELTA and Universal Postal Union.

exceedingly low compared with the rest of the EU, in part due to a service quality which is considered unsatisfactory. For example, the timeliness record of mail delivery and the population per postal office are the lowest in the EU. The poor ‘‘on time’’ record reflects both lack of automation in distribution as well as the more difficult geographic characteristics of Greece, with its many mountains and islands. Regarding postal offices, their distribution is also problematic, as far too many are in the rural areas compared to the densely populated urban centres, and are especially sparse in the newer urban suburbs (such as the north of Athens).

On the cost side, labour costs are high in a highly labour intensive sector. Productivity indicators are by far the lowest in the EU, and rigid labour regulations constrain an improved distribution of personnel and thus the closure of the many loss-making postal offices. Moreover, 75 per cent of rural outlets are offices owned by ELTA rather than agencies through which postal services are provided as a secondary operation (*e.g.* in small stores or gas stations). In other EU countries, the comparable average is 25 per cent. Wages also appear to be high, as gains in real terms have exceeded productivity for at least the last decade (Figure 21), and compensation in 1997 exceeded that in manufacturing by 15 per cent. Though ELTA has received resources from other sources, deficits equal to over 25 per cent of revenues resulted in a debt of Dr 70 billion by end-1997.[162]

The restructuring plan put forth by the new management and accepted in July 1998 by the Government, sets forth several important measures which would bring ELTA to profitability by 2000, excluding payment for a public service obligation.[163] Perhaps most importantly, it introduces more flexible labour regulations, which should improve the geographic and functional mobility of staff and permit more elastic timetables and the use of part-time work. However, it does not anticipate any sizeable reduction in its work force. It also envisages the redesign of the network to fit the changed distribution of the population; *i.e.* closure of loss-making offices, especially in rural areas and their replacement with agencies. Revenue enhancement is expected from a 10 per cent per year average nominal increase in fees during the five year period and an increase in demand of 3 per cent per year arising from an improved quality of service. The Dr 41 billion investment programme for the period 1998-2000 (of which 60 per cent will be EU financed) will focus on automation of the central offices. The programme also reduces operating costs significantly since it entails a Government take-over of ELTA’s liabilities and a capital injection (Dr 60-80 billion).[164] Nonetheless, if

Figure 21. **Postal services: prices and performance**

1. Deflated by the CPI.
2. Operating revenue as a share of operating expenditure, excluding interest payments and depreciation.

Source: ELTA.

ELTA fails to hold its own in a more competitive environment, management would need to consider concentrating on basic services and leaving the remainder of the market to private sector operators. A leaner operation would also require fewer personnel. In the event, increased private sector participation may be attracted by the implementation of currently existing draft legislation which sets out the institutional framework and creates a market regulator by expanding the role of the telecommunications regulator to cover the postal sector too.

Quantification of the effects of liberalisation

This section provides estimates of the potential long-run effects at the macroeconomic level of restructuring the main public enterprises and introducing competition into the sectors they currently dominate; the analysis covers eight sectors (petroleum, electricity, telecommunications, air transport, urban transport, rail transport, water services, and postal services). The analysis consists of three stages and is based on the methodology developed by the OECD for its regulatory reform project.[165] It is a static analysis, based on the situation existing in 1996 so the timing neither of the changes nor of their eventual impact form part of the analysis; one-off reforms are assumed for each sector.

In the first stage, an assessment is made of the potential reform's effects on prices. The quantification of these effects is partly based on benchmarking (*i.e.* the observed effects of the introduction of reforms in other countries) and partly on an assessment of the conditions currently existing in these sectors in Greece, as they have been described in the sections above. Reviewing the main components of price performance, the following observations can be made (Table 27). First, it is clear that large savings can be made from reduced labour costs – both from improved productivity and lower wages. The average reduction in labour costs is about 50 per cent – with wages assumed to fall to the average of the manufacturing sector. Second, the cost of intermediate inputs, which depend heavily on energy, and thus, reflect the effects of reforms, would provide a further downward impetus to prices. Third, the reforms would result in substantial changes in relative prices, with prices of some goods and services falling and others rising. Specifically, a downward stimulus to prices comes mainly from the telecommunications and petroleum sectors, while in the case of enterprises with public service obligations, even the assumption of negative profit margins of

Table 27. **Assumptions and effects of sectoral deregulation**

1996 base

	Petroleum	Electricity	Water	Train	Bus	Air transport	Post	Telecom-munications	Total
Assumptions (% change):									
Cost of intermediate inputs	0	–1	–3	–1	–2	–1	–7	–4	–0.1
Fuel inputs (petrol)	0	–1	–1	–1	–1	–1	–1	–1	0.0
Coal and mining inputs	0	0	0	0	0	0	0	0	0.0
Electricity	–14	0	–14	–14	–14	–14	–14	–14	–0.4
Water	–22	–22	0	–22	–22	–22	–22	–22	–1.0
Train	161	161	161	0	161	161	161	161	7.8
Bus	47	47	47	47	0	47	47	47	2.2
Air transport	–20	–20	–20	–20	–20	0	–20	–20	–0.9
Post	–13	–13	–13	–13	–13	–13	0	–13	–0.6
Telecommunications	–36	–36	–36	–36	–36	–36	–36	0	–1.1
Labour costs									
Labour productivity	20	50	25	35	25	25	25	30	1.7
Wages	–35	–15	–25	–20	–30	–30	–15	–15	–0.9
Capital costs	10	25	10	25	25	20	15	20	1.0
Price mark-up, current [1]	3	12	8	–98	–82	6	–16	51	1.2
Price mark-up, assumption [1]	5	5	5	–30	–30	5	5	5	0.2
Innovation: effect on output	0	20	0	0	0	0	0	30	0.9
Price elasticity of demand	–50	–50	–50	–25	–25	–150	–50	–50	–2.9
Sectoral effects (in per cent):									
Direct price effect	–1	–14	–22	161	47	–20	–13	–36	–1.0 [2]
Price-induced output effect	1	7	11	–40	–12	31	7	18	0.6
(including innovation effect)	1	28	11	–40	–12	31	7	53	1.7
Employment effect [3]	–16	–14	–11	–56	–29	4	–15	18	–0.3
(excluding innovation effect)	–16	–29	–11	–56	–29	4	–15	–9	–0.6
Memorandum items:									
Baseline									
Wage share [4]	6	27	46	303	221	43	96	24	30
Capital share [5]	3	38	26	–300	–179	9	–12	52	23
Intermediate inputs share [6]	91	35	28	96	59	49	16	24	48
Reform scenario									
Wage share [4]	3	18	35	69	84	30	75	24	21
Capital share [5]	5	42	30	–5	–23	9	8	39	24
Intermediate inputs share [6]	91	40	34	36	39	61	17	37	56
Sectoral value added as a share of total GDP [7]	0.2	1.8	0.2	0.0	0.1	0.5	0.2	1.9	4.8
Sectoral employment as a share of total employment	0.1	0.9	0.1	0.3	0.3	0.3	0.3	0.6	2.9

1. Price mark-up is defined as gross operating balance, minus depreciation and interest, divided by the sum of the cost of intermediate inputs, labour costs and depreciation.
2. The total is the sum of the sectors using their weights in the CPI.
3. Sectoral employment effects include impacts of productivity growth, price-induced output growth and the assumed effect of innovation-induced output growth, but ignores the possible impact of lower wage premia.
4. Labour costs as a percentage of the sum of labour costs, gross operating surplus and intermediate consumption.
5. Gross operating surplus as a percentage of the sum of labour costs, gross operating surplus and intermediate consumption.
6. Cost of intermediate inputs as a percentage of the sum of labour costs, gross operating surplus and intermediate consumption.
7. Value added as a percentage of total GDP.

Source: OECD calculations based on 1990 Input-Output Table of Greece.

30 per cent implies large price increases.[166] Fourth, the need by all public enterprises for extensive infrastructure programmes will raise their capital costs. Fifth, innovation in the most technologically dynamic sectors is assumed to provide further room for price reductions (mainly in telecommunications but also in the electricity sector due to the arrival of natural gas). The results of the analysis indicate that, on balance, a more efficient public enterprise performance could have a downward impact on the aggregate price level (1 percentage point off the CPI price level). Economy wide wages are expected to remain broadly unchanged in real terms, though they could be significantly lower for employees in public enterprises and 1 per cent higher in the rest of the economy.

In the second stage, demand elasticities are used to derive the increase in each sector's output arising from the potential price gains described above. These are aggregated across sectors to produce the direct effect of public enterprise reform on output and employment; a 1½ per cent increase in output and a ½ per cent decrease in employment. Overall, unit labour costs would fall by about 2 per cent. The impact on output could be underestimated and that on employment overestimated due to the fact that some of the more important sectors (energy and telecommunications) are still maturing in Greece and their weight in the economy is below that of other industrial countries.

This static approximation of the effects of reform, however, underestimates the full impact on the economy since it does not take into account the multiplier effects arising from the lower prices of these critical inputs on other sectors of the economy. In a third stage, these second-round effects are estimated using the OECD Secretariat's INTERLINK model, and result in a cumulative increase in total output (direct and second-round) of the order of 5-7 per cent of GDP; in simulations for other OECD countries which did not include the loss making public service enterprises (water, train, bus and post), output gains ranged from 5 to 6 per cent for France and Germany to 1 per cent for the United States.[167] This estimate understates the impact on Greece since it does not capture either the impact from the provision of higher quality products due to the large investment programme that will be undertaken, or the improvement in the budget balance. Rough estimates of these factors would push up the total income gain to at least 9-11 per cent. This potentially large increase in output could lead to substantial increases in employment if labour market flexibility permits the gains in productivity to be shared between real wages and employment.

Assessment

The Government has embarked on a programme to revitalise an inefficient public enterprise sector, which has been costly to sustain, has provided less than satisfactory services, and is now struggling to make up a large investment gap relative to comprable OECD enterprises. The cost to the economy consists not only of an annual drain on the budget of about 3½ per cent of GDP, but a sizeable brake on activity as the economy is often deprived of essential inputs at low cost. To a large extent, budget transfers to public enterprises are financed by EU funds but they nevertheless represent a diversion of these funds from other uses. If accumulated over the past decade and half, and added to other government liabilities due to public enterprises, they represent a burden equivalent to about 50 per cent of GDP. The force pushing for policy change is Greece's quest to satisfy the Maastricht criteria, and then to perform satisfactorily within EMU and achieve real economic convergence to the level of the EU countries. In order to catch up with other OECD countries, a reform strategy must be bold and decisive, since further reversals would risk leaving Greece at the economic periphery of the EU. But an advantage for Greece is that it can benefit from of the experience of other countries which have preceded it in their reform efforts. Their experience suggests that much emphasis must be placed on designing the pertinent regulatory framework. An appropriate design would generate sufficient competition so as to attain the full potential of the reform for the economy, which as noted above, is estimated by the OECD Secretariat's simulation to amount to an eventual increase of at least 10 per cent of GDP. In view of the large overmanning in public enterprises, appropriate social safety net procedures need to be developed in parallel to the reform effort, though the output gains should generate many new jobs for the economy.

The Government's strategy rightly places much emphasis on the need for improved management and operating efficiency in public enterprises. The decision to hire new experienced managers through competitive tenders should certainly benefit those enterprises for which the process is completed quickly, and in this regard the exclusion of the Public Power Corporation (DEH) from this process is a missed opportunity. However, new managers by themselves are not sufficient if their independence from the Government and the respective line ministries is not assured. In this regard, managers will need to be responsible for hiring and remuneration policy, since high labour costs and low productivity are

an overriding factor for the poor financial performance of public enterprises. Productivity indicators in almost all public enterprises are very low in international comparison. The issue of how to deal with current excess staffing and the inappropriate skill mix has to be placed high on the agenda, as a quick reduction of personnel is necessary to hasten the enterprises' resurgence. The adjustment process will otherwise be more costly, in terms both of resources as well as operational efficiency. In view of the obvious social issues at stake, the option under consideration by the Government to transfer a portion of surplus public employees to other public entities or local government could be a workable solution – though caution is required not to transfer employees where there is no need. Perhaps the development of the new regional governments is an opportunity in this regard. Concerns about moral hazard, nevertheless, dictate that public enterprise employees need to be treated *de facto,* as well as *de jure,* similarly to their brethren in the private sector and not feel the complacency which comes from a guaranteed job security. A normal degree of job security would also temper unreasonable demands for compensation and other benefits. Management independence is also needed for procurement decisions. Managers should be free to choose supplies based on criteria relevant to the well-functioning of the firm, without pressures based on other considerations, such as domestic production and employment. Prices should be set with a view to enhancing firms' efficiency while providing an adequate revenue performance, and not be based on short-term macroeconomic considerations. Therefore, price decisions should be best left to the discretion of an independent sector regulator responsible for setting price-cap regulation. Eventually, prices should be allowed to be set by market forces once sufficient competition has been generated in a sector.

International experience has been disappointing regarding the success of management contracts in providing adequate firewalls and preventing conflict of interests between management and short-term Government considerations. In many cases, government intervention led to breaches in initial financial targets, and left tax payers the bill, as these contracts are essentially unenforceable. In others, the lack of a credible hard budget constraint was the source of similar outcomes. In essence, corporate governance suffers because the shareholder function is not separated from the regulatory function within the state. One option for overcoming this shortcoming is the introduction of independent boards (including adequate remuneration of its members). Nevertheless, dissatisfaction with

this model has led many countries to use more competitive environments to ensure the efficient operation of public enterprises. This solution obviously requires more participants and thus Governments have liberalised the relevant markets so as to attract new participants and also turned to the private sector for the management and/or ownership of their public enterprises. This tendency has been facilitated by technological innovations which have reduced the sectors which previously were considered to be natural monopolies as well as by successful experiments in auctioning the rights to operate public services and to construct infrastructure. A necessary complement to this approach has been the introduction of independent and powerful sector regulators for the large sectors so as to ensure fair and competitive practices, which also serve to attract more participants. The creation of an overall competition watch-dog would ensure the well-functioning of individual markets as well as permit arbitration of cross-sector issues and provide a venue for the private sector to address its complaints regarding non-competitive behaviour of public enterprises. The Greek Government recognises the shortcomings inherent in management contracts and considers them to be a first step in the reform process. After undertaking the appropriate due diligence procedures, which clarify the financial position of the firm, and preliminary restructurings, the Government intends to divest from more public enterprises, especially as their markets are opened to competition.

In the case of Greece, the timetable for opening up markets to competition is in line with EC directives (and their derogations), but Greece could follow the example of some other countries and liberalise at a faster pace. Regarding auction tender techniques, these have already been used in Greece. The construction of the Athens metro is a prime example, and the agreements for the new Athens Airport and Rio-Antirio bridge combine both operations and construction in Build-Operate-Transfer (BOT) agreements. The same philosophy can be applied to the existing public enterprises in Greece and the Government's decision to proceed with partial sales in profitable enterprises (either equity or operations for pre-determined periods) is a step in this direction. They should provide the double benefit of injecting more exacting and dynamic minority owners to spur operational efficiency while attracting new capital to support needed investment. Of course, this form of corporate governance would be enhanced the larger the ownership share held by investors as well as the diversity of ownership. The Government's decision to seek strategic investors, which would also operate the

firm, as well as risk their own capital would also work towards this objective. However, leaving majority ownership in the hands of what the private sector could consider a whimsical public owner weakens investors interest. Larger privatisations would also increase revenues accruing to the Government. However, these should be a secondary objective and should not delay the timing of sales in view of the opportunity costs. On the contrary, the process should be accelerated. International experience suggests that the biggest gains come from higher future tax revenues rather than the upfront privatisation receipts. Moreover, clawback clauses in the sales contracts, based on an expected profit stream, can protect the Government from accusations of underpricing the sales.

Nevertheless, international experience indicates that it is not easy to introduce a suitably contestable environment, with easy entry and exit for potential competitors. Such pervasive institutional changes are difficult to design and even more difficult to implement. And if this aspect of the reform fails, private or public enterprises which retain monopoly power will have few incentives to operate efficiently. Most mistakes have been made in designing contestable markets in sectors containing elements of natural monopoly (*i.e.* electricity, natural gas, water, and to a lesser degree telecommunications), where Greece is facing important institutional changes.

The reform of the electricity sector is probably the most critical area of reform. It is an essential input for the rest of the economy, and the incumbent monopolist has not been successful in providing sufficient and secure supply, while its prices could be significantly lower in view of its free access to lignite and public resources to finance its investments. The general direction of the proposed reform is in line with the 1996 EC Directive for the sector. However, several factors make it apparent that only a bolder reform will result in a contestable market. Greece lacks potential competition from imports from other EU countries and its Balkan neighbours due to geographic and economic/infrastructure reasons, respectively. This isolation places a larger onus on stimulating the potential for domestic competition. However, non-transparent and distorted access and pricing for key inputs (natural gas and lignite) as well as recourse to public resources appear to be highly advantageous to the incumbent and thus provide barriers to entry. Competition between energy sources also seems to be jeopardised by cross-sector financial interests between the incumbent public sector enterprises; *e.g.* the public power corporation's (DEH) control of lignite

mines, Hellenic Petroleum's call option to re-purchase the natural gas company (DEPA), and the profit sharing agreement between DEH and DEPA on natural gas operations. Moreover, the long transition period contained in the EC directive permits the vertically integrated incumbent in electricity to price discriminate between the captive and competitive markets. Leaving all these impediments to competition at the initiation of the reform will likely leave the energy sector as an effective monopoly for the foreseeable future.

A bolder reform would sever the financial links between the natural gas, lignite, petroleum, and electricity sectors. Such separation could enhance competition between these sectors and especially within the downstream electricity sector which uses the others as key inputs. Fair access to inputs and customers would attract new private-sector players to generation. However, the long lags until the entry of full competition under the EC electricity directive and the above described advantages of the incumbent are not encouraging. If there is no immediate interest by private sector participants to the new legislation, two additional steps to encourage entry into generation would be to legally separate DEH's generation and distribution operations (including a shift of the high-voltage grid to an independent operator), which would permit competition for the supply of the cheapest source of electricity, and to separate the generation capacity of DEH into several competing firms. This second step would obviously require a fair distribution of DEH's stranded costs so as to ensure a level playing field; and the example of other countries can be followed to this end (*e.g.* cost compensation could be achieved through the price cap regulation in combination with a redistribution mechanism). This strategy would permit the electricity sector to follow the example of the petroleum sector. Though the latter is showing the benefits of previous liberalisation, eliminating barriers to imports arising from storage requirements would reduce high refining margins. Similarly, entry into the natural gas sector would be encouraged by setting clear and uniform rules on access prices and providing private participants a majority ownership in distribution companies. With the advent of new players and many complex issues likely to arise, an impartial referee in the form of a truly independent regulator with broad decision making powers will be needed. Its powers should extend to all three energy sectors.

In the sectors more susceptible to competition, a Government strategy of opening up the market to new entrants should provide sufficient impetus to the

creation of an environment which efficiently provides low-cost goods and services to the economy. Two prime examples in Greece are air transport and telecommunications. Soon after the respective partial liberalisation of these sectors (mobile telephones and domestic flights), new entrants have provided alternatives to the public monopolist and provided the impetus for improved services and/or forced a decline in prices for the respective public enterprises. With EC derogations soon ending in both sectors, the incumbent public enterprises, especially Olympic, will need to restructure quickly, and compete with dynamic new entrants or face bankruptcy. International experience indicates that there exist few viable options other than to focus on niche markets, rather than attempting to maintain (or attain) extended markets. Reaching out to larger markets through international alliances could be used to establish feeder links to other operators. The Government's intention to turn to a strategic investor with broad decision making power may improve operations and import know-how. If this fails, sale to another private carrier should also be considered. In the case of telephony, OTE's healthy financial position and rapidly changing technology provide the opportunity for the Government to move faster than required by the EC directive and issue a second license for basic telephony and additional licenses for mobile telephony (if legal obligations are not constraining the latter). In the case of Spain, a second licence issued to domestic interests for basic telephony during the derogation period has produced surprisingly quick results in the form of domestic price competition. Price competition would also be enhanced if the floor on mobile phone prices was eliminated. Such relative price changes would also assist in the attainment of the macroeconomic objective for inflation and offset necessary increases in the tariffs of other public enterprises (especially in urban and rail transport). In the event, the powers of the existing regulators in both these sectors – the National Telecommunications Committee and the Civil Aviation Service – appear to be grossly insufficient compared to other countries. In view of the prospective competition and the need for co-operation in the use of network infrastructure (basic telephony and airport infrastructure and services, respectively), here again, an impartial referee is needed to arbitrate between the public sector monopoly and new private sector entrants, as the government faces a conflict of interest.

Almost all public enterprises have been burdened with uncompensated and ill-designed public service obligations. These are partly to blame for the poor

financial situation of some of the largest loss makers; *e.g.* rail service, urban transport, air transport, water and sewerage services, and postal services. When restructuring these firms, the Government must decide the extent of the public service obligation and base this decision on a full cost evaluation of the provision of such services. In many cases, the current budgetary constraints (combined with commitments to other social objectives) or simple need-based targeting considerations could suggest a more limited public service obligation. For example, the highly under-used trains in the southern Peloponnese region appear to be a luxury (especially given the parallel development of road networks), as do the under-utilised postal offices in the rural areas, the long-distance flights to countries with Greek emigrants, and extensive public television which serves less than 10 per cent of the viewing audience. In all cases, significant downsizing appears necessary.

Regarding pricing, raising ticket prices for urban transport, rail or postal service from the current extremely low level should not be considered as regressive, even if externalities or social considerations suggest that a degree of subsidisation should be maintained. Use of urban transport can be increased through taxation of the competing, socially more expensive, car transport, as has been done in other countries. This tax could serve to raise the cost of operating cars towards their social cost to the economy. More generally, direct means-tested assistance from the budget would probably be a more efficient and cost effective method of providing subsidies to the truly needy. In the event, clear contractual agreements with the enterprise stating the service provided (quality, distribution, etc.) and the compensation to be received, must replace the current policies of residual deficit financing (*i.e.* soft budget constraints) and the other non-transparent arrangements. Once such clear agreements are reached, the rationale for public operation of the service lessens, and further efficiency gains have been achieved by countries experimenting with tenders to private operators for a predetermined duration period. The finite period of the arrangement provides the operator the incentive to perform well so that the contract will be renewed. In the case of Greece, water and sewerage service, urban transport, and rail service, and postal services could all be candidates for tenders, and the Government intends to proceed along these lines in the case of water and sewerage services. In several sectors, splitting the public monopoly into several competing firms provides the option for benchmarking, which would be another means for judging performance.

Though the infrastructure could probably remain in the hands of the public sector, over the longer run such a division of responsibilities skew the incentive for the operator regarding the maintenance of the infrastructure. If sales are linked to infrastructure ownership, then private sector capital could provide significant relief to the budget, and supplement EU funds. Moreover, competition by private sector applicants to use EU funds, and the infrastructure to which they apply, could improve the efficient utilisation of these resources, deter cost overruns and completion delays. Such incentives appear to have made the completion of the new Athens airport at Spata the only large project to be ahead of schedule. Similar structures are possible for the construction of the low pressure natural gas distribution system and can be extended to electricity distribution as well. Nevertheless, for existing public enterprises, raising additional resources from the private sector will require share capital increases (as equity sales accrue to the State), and for the new ownership to risk the additional investment will necessitate the implementation of ambitious restructuring plans for these enterprises.

Notes

1. For the first time since 1992, tax brackets were corrected for inflation in the 1998 Budget. However, a change in the schedule of personal income tax payments, the introduction of several new taxes, and measures implemented to enhance the fight against tax evasion have led to an increase in the 1998 effective tax burden for households (see Chapter II).
2. The data source is the Federation of Greek Industry's annual survey of a large group of enterprises.
3. The new investment law places more emphasis on capacity building in tradable sectors. It limits the eligible projects to the manufacturing and tourism sectors. Cash assistance is conditional upon the number of jobs created, with an upper limit of Dr 15 million per new job. In addition, the new law gives a preferential treatment for investment in the regions with high unemployment; these are the same regions where local employment pacts can be signed according to the new labour market law (see Chapter III). It also puts more emphasis on tax allowances rather than cash grants.
4. Viewing output growth from the supply side reveals that the main source for the 1997 acceleration in activity was the reversal in agriculture after two consecutive years of drops in production. The recovery in agriculture was also responsible for the sharp slowdown in food prices. For 1998, the contribution to output growth should be more balanced between agriculture, industry and services.
5. Apparent productivity gains could be overstated by an imperfect recording of a large number of immigrants by the labour force survey.
6. A more accurate picture of the employment situation cannot be provided because the labour market survey, the only reliable source of information on labour market developments, while conducted once a year during the second quarter, is only published during the first months of the following year.
7. In January 1998, the government adopted a new policy of offering work and residence permits to all immigrants. Requirements include finding employers willing to employ them legally, and paying 40 days of social contributions. Once registered, an immigrant is eligible for health insurance coverage upon payment of the standard payroll tax.
8. Inflation reduction in 1997 was also facilitated by several exogenous factors. First, the decline in energy prices was furthered by a sharp reduction in the excise tax on heating oil, which reduced CPI inflation by 0.3 percentage points, while the overall impact on the CPI inflation during 1997 from lower fuel prices was 0.6 percentage point. Second, an 18 per cent fall in olive oil prices (after an increase of 30 per cent in 1996) reduced CPI inflation by

an additional 0.6 percentage point. Third, the reform of pharmaceutical prices reduced CPI inflation by an additional 0.2 percentage point. On a period average basis, changes to the tax on heating oil reduced inflation by about ½ a percentage point.

9. Fiscal policy was less supportive as a lower borrowing requirement was met mostly through bank and external borrowing.

10. Wage awards in 1997 were substantial in public enterprises and state-controlled banks as well, at 11 and 10 per cent, respectively.

11. The inflation catch-up clause specifies that if inflation exceeds 3.5 per cent in December 1998, there will be corrective payments. One at the beginning of 1999 for the deviation in inflation between 3.5 and 4.5 per cent. If inflation exceeds 4.5 per cent, a second payment will be made at end-1999. For 1999, the agreement contains a 2.8 per cent minimum increase in wages, which is provided in two equal amounts at the beginning of January and July. A catch-up clause is triggered if inflation remains above 2 per cent at year-end, and the catch-up payments are awarded at the beginning of 2000.

12. The current account based on settlements data is considered to under-record exports. In an attempt to provide a more representative picture of external developments, the OECD has constructed a proxy for the balance of payments which uses customs data for the trade balance and settlements data for the rest of the items (see Table 4).

13. However, the relaxation of manning requirement for Greek ships and the crises in Asia should have a negative effect on the invisibles balance.

14. *Annex I* contains the revealed comparative advantages for the Greece's main exports at a SITC 3 digit level and a note on the Aiginger price *versus* quality index.

15. EU regulations permit garments manufactured outside EU countries which are returned for finishing and to be exported as EU products. OECD trade data reveal a brisk acceleration in Greece's imports of articles of apparels and clothing accessories from CEECs beginning in 1993, paralleled by a steady rise in Greece's exports of textile yarn and related products to these countries.

16. The increase in tourism receipts was not commensurate with arrivals, since tourists are spending less in real terms. Specifically, arrivals in 1997 rose by an estimated 8 per cent, and tourism receipts per arrival rose by 2 per cent in drachma terms (though balance of payments data on a settlements basis likely underecord tourism receipts as some are maintained in the underground economy).

17. For the most part, the difference between the two sets of trade data is due to an underestimation of exports by some 50 per cent by the settlements data compared with the customs data. For an explanation of the factors leading to the difference in exports, refer to OECD (1997*b*) *Economic Survey of Greece.*

18. The import income elasticity in Greece has been low in recent years (averaging about 1½), due to the significant contribution of construction activity to output growth.

19. For international comparison on import penetration by industry, see OECD (1997), *STI Scoreboard.*

20. The OECD presentation of the balance of payments attempts to correct for foreign-currency denominated loans provided by domestic banks. The settlements basis balance of payments

is based on the currency of transactions (drachma *versus* foreign exchange) rather than a residency/non-residency criterion. The correction does not affect overall private capital flows, just the composition between the items ''entrepreneurial capital'' and ''other capital''.

21. The interest service ratio overstates the debt burden since it is based on settlements data for exports, which are smaller than customs basis data.

22. Monetary policy decisions are taken by a Monetary Policy Committee (MPC), comprising 6 individuals selected by Government and approved by Presidential Decree for staggered 6 year terms. The MPC includes the Governor and the two Deputy Governors, as well as three other individuals whose sole employment is to sit on the MPC. The Bank of Greece will issue semi-annual reports on monetary policy.

23. The respective Maastricht criteria are an average annual inflation rate – according to the EU's harmonised index – of no more than 1½ percentage points above the average of the three best performing countries in the European Union with respect to inflation, and average annual interest rates on 10-year bonds of no more than 2 percentage points above the comparable average of the same three best performing countries with respect to inflation.

24. The Bank of Greece did not adopt target ranges for either M3, M4, or total credit, as it had traditionally in the past. These were viewed to be incompatible with the new monetary framework, especially as the stability of monetary aggregates in Greece had decreased due to increased international capital mobility and continuing financial innovations. However, the Bank of Greece did present indicative 1998 projections for the growth of M3 (6-9 per cent) and total credit (4-6 per cent) in view of their useful role, under certain conditions, as indicators of inflation developments.

25. Additional policies that aimed to discourage international capital inflows include the reduction of two taxes which discriminated against domestic drachma-denominated credits. The tax placed on the value of all domestic loans (EFTE) was reduced from 4 per cent to 3 per cent and also required for domestic foreign-currency denominated loans, while the base of a 1.2 per cent tax placed on loan interest (which favours lower-interest, foreign-currency denominated credits), was changed to the value of the loan. Finally, in order to facilitate capital outflows and bring reserve requirements applied to foreign currency deposits in line with those applied to domestic deposits, the reserve requirements on foreign currency deposits were reduced for residents and for non-residents, excluding Greek seamen and workers abroad, from 70 per cent to 12 per cent. Reserve requirements on deposits in foreign currency by Greek seamen and workers abroad were reduced by a smaller extent (from 70 per cent to 60 per cent).

26. The Bank of Greece clearly stated that an important assumption for achieving its 1998 indicative monetary targets was that 70 per cent of the financing needs of the public sector would be met through non-bank financing.

27. The Government has borrowed in external markets over US$5 billion during the first half of 1998 (Dr 1.7 trillion). However, the Government has committed to use these funds only to replace maturing external debt.

28. The source of the sharp reduction in interest payments was four-fold. First, interest rates fell by 240 basis points, with the interest rate on the benchmark 12-month Treasury bill falling

from 12.8 per cent to 10.4 per cent. Second, a large share of the gross financing needs were met through the issuance of bonds rather than Treasury bills; this switch pushes interest payments one year forward and takes advantage of a downward sloping yield curve. Third, towards the latter half of the year when the international market turbulence pushed up domestic interest rates, large recourse was made to external financing on which interest rates are much lower. Finally, the issuance of two-year zero-coupon bonds reduced interest payments by the equivalent of almost 0.5 per cent of GDP. It should also be noted that for a third year in a row about Dr 200 billion of interest payments (about 0.6 per cent of GDP) were rescheduled over a 7 year period; 1997 is expected to be the last year such a transaction will occur.

29. Specifically, the state undertakes a capital increase in a public enterprise and, thus, obtains an asset (equity) equivalent in value to the resources injected into the enterprise. To the extent that the budget receives EU funds to co-finance the specific public enterprise project, the budget balance improves by an amount equal to the EU financial support. Without this technique, the budget balance would deteriorate by the domestic share of the project's financing.

30. These expenditures amount to Dr 83 billion and include, *inter alia*, payments for judicial decisions awarding retroactive pay to judges and civil servants, and EU refusal for reimbursement of agricultural subsidies. According to Eurostat accounting principles, the payments have been included in the budgets in which the liability was incurred, although the payment was made in 1997.

31. On a December to December comparison, the increase in Government employees was much lower, reflecting the gradual effect of the hiring policy introduced in 1997. Specifically, the recruitment policy replaces only one in five retiring civil servants, with the exception of education, health, and the security forces (accounting for one-half of the total) where the ratio is 1:1.

32. These ''quasi-fiscal'' financial operations included, *inter alia*, the equity acquisitions (Dr 420 billion), the payments debited to past budgets (Dr 83 billion), capital injections into state-controlled banks and public enterprises (Dr 136 billion), and the clearance of arrears to public hospitals (Dr 174 billion).

33. The Government is also planning to set up a Treasury Department in 1999 to be responsible for debt management.

34. The drachma-denominated privatisation certificates have three-year maturities while the euro-denominated ones have five-year maturities. The drachma-denominated ones are zero-coupon instruments and will receive a 5 per cent discount when exchanged into equity. Furthermore, the Government has committed 40 per cent of all future public offerings of public enterprises to holders of these certificates.

35. Prior to 1998, pensions were indexed to civil service cost-of-living increases. However, 1998 legislation permitted the Government to set pensions ''in line'' with Government incomes policy. For 1998 – with income policy for civil servants set at 2.5 per cent – cost-of-living adjustments for pensions of private sector employees (IKA) was set at 3.5 per cent for the component of a pension up to Dr 200 000 per month and 2.5 per cent for the component of a pension exceeding Dr 200 000 per month. With most pensions below this

limit, in effect, most pensions should increase by 3.5 per cent. Payments for civil servant pensions will also increase by more than civil service wage indexation due to the introduction of new civil service pension scales in August 1997 (though outlays occurred in 1998). The increase in civil servants' pensions was seen as a correction of an inequity arising from the 1997 introduction of a new pay scale for civil servants. Pensions increased between 5 and 15 per cent.

36. The impact of the measures is reduced by the indexation of tax brackets by 5.7 per cent – their first change in five years.

37. Other important measures include: *i)* raising the corporate tax rate on unlisted incorporated enterprises from 35 per cent to 40 per cent (while for listed ones and un-incorporated ones the rate is 35 per cent); *ii)* raising the withholding tax on newly issued government securities from 7½ to 10 per cent (and reducing the gap with respect to the withholding tax on other interest earning instruments which is 15 per cent); and *iii)* introducing a programme for closing outstanding tax cases (1993-97) which is expected to yield about Dr 100 billion.

38. The benchmark 12-month Treasury bill rate will exceed the 9 per cent on average assumed in the budget by nearly 3 percentage points, and the Government has made a concerted effort to finance the budget with non-bank domestic financing.

39. The main sources of privatisation receipts during 1998 have been the sale of equity in OTE (Dr 430 billion) and Hellenic Petroleum (Dr 35 billion), and the sale of the Bank of Crete (Dr 22 billion).

40. The programme was presented by the Government in June and was approved by ECOFIN in mid-October 1998.

41. The cost of living adjustment for state pensions is expected to be about 3 per cent, though similar to 1998, the adjustment will be lower for higher level pensions.

42. An additional monetary stimulus that could occur in the future would come from the need to reduce required reserve ratios from the current high levels to those of other EMU countries.

43. The decomposition of the 13 per cent of GDP improvement in the general government deficit between 1990 and 1998 is as follows: tax revenues (4 per cent of GDP), non-tax revenues (2 per cent of GDP), primary current expenditure (1 per cent of GDP), interest payments (1 per cent of GDP), and net capital expenditure (5 per cent of GDP, with a reduction in the take-over of government guaranteed debt accounting for 3 per cent of GDP and the introduction of equity acquisitions for nearly 2 per cent of GDP). During this period, the debt to GDP ratio increased from 80 per cent of GDP to almost 110 per cent of GDP.

44. The new tax system would raise all tax brackets (with the higher ones proportionally less than the lower ones) and reform the system of deductions. The reform would serve to correct for the significant bracket creep that occurred during the period 1992-97. The reform would also reduce personal income tax rates which are already relatively low. Average tax rates on personal income are currently below 7 per cent, reflecting the fact that the income of the average production worker faces a marginal income tax of 15 per cent. Nevertheless, the tax wedge is close to 50 per cent due to the high social security contributions. Regarding the distribution of the tax burden, the 8 per cent of the tax payers with declared incomes above Dr 7 million account for 27 per cent of declared income and pay close to double that share of total personal income taxes, and among categories of workers, self-employed workers (to

whom the presumptive taxation system applies) account for 37 per cent of total personal income tax and 27 per cent of declared income.

45. Tax exemptions on personal income tax reduce tax collection by about 2½ per cent of GDP, with the largest exemptions being the zero rate for the first Dr 1 million earnings (Dr 1.3 million for salary-earners and pensioners), owner-occupancy deductions, and deductions for social security payments. Regarding VAT, it has two reduced rates (4 per cent and 8 per cent), which account for most of the revenue losses (about 1 per cent of GDP).

46. See Box 1 of the OECD (1997*b*) *Economic Survey of Greece.*

47. The law also introduces labour inspectors whose role will be to verify the implementation of labour regulations, including the use of illegal immigrants.

48. The draft legislation stipulates that the 10 hour limit would be possible only in case of objective or technical reasons, but it is not clear how constraining these requirements will be. In the event, the law also permits, following an agreement with workers, working time to be expanded to 9 hours per day at normal pay for up to three months, if these hours are made up in the following six months.

49. Further discrimination against smaller firms, which usually lack a collective agreement, arises from the fact that the Organisation of Mediators and Arbitrators (OMED) will only review disagreements over the terms of a collective agreement (*e.g.* working hours).

50. For a white collar worker in manufacturing, the ratio of average earnings to the entry level minimum wage (for an unmarried worker) is 3.0, and the minimum wage can increase by 40 per cent for a married worker with 10 year's experience.

51. The programme provides for employment subsidies of Dr 4 000 per day (about two-thirds of the entry level minimum wage) for 18 to 22 months or a lump sum of Dr 1.2-1.8 million for approved self-employment activities.

52. The objective of a third employment subsidy programme is not to provide work experience but solely to help provide employment to the aged unemployed with little hope of re-training. The programme will cover 2 000 unemployed above the age of 58 in 1998 (and 8 000 in a three-year period) until they qualify for an old-age pension. The employment subsidy covers all social security contributions and will be financed from a special fund (LAEK) financed by social security contributions. The annual cost is Dr 22 billion.

53. The contribution of the CSF II through the European Social Fund to active labour market polices and human resource development are 2 583 million ECU during 1994-99 (equivalent to about Dr 850 billion). So far the vast majority of the funds has not been used.

54. See OECD (1996) *Economic Survey of Greece*, Chapter III, Implementing the Jobs Strategy.

55. The major elements of the 1996 labour market reform include the introduction of multi-use vouchers for the unemployed to be applied to training; employment subsidies; or retained as an unemployment benefit, and the introduction of employment cards, which would henceforth be a requirement for eligible workers. Both aspects have started to be implemented since April 1998.

56. Layoff limits are 5 workers per month for firms with 20-50 employees and no limit exists for smaller firms.

57. The statutory minimum wage in Greece is fixed by collective bargaining.

58. The 1998 OECD *Employment Outlook* reports on findings of negative impacts of minimum wages on youth employment in several OECD countries.

59. For a description of the Greek pension system, see the OECD (1997*b*) *Economic Survey of Greece.*

60. The package of measures also includes raising the limit on the share of liquid assets that pensions funds can invest in the stock exchange from 20 per cent to 23 per cent beginning in 2001; the remainder will remain invested in government bonds.

61. There are 210 000 women eligible for this benefit with a budgetary cost of Dr 130 billion.

62. Women without underage children who entered the labour force after 1992 have the same retirement age as men (*i.e.* 65).

63. The explanation for why high statutory replacement rates coincide with low actual pensions is that average contribution periods are short (but this generally does not affect the pension base which is determined by the salary over the last few years of one's career).

64. The EC has questioned whether this limit constitutes a constraint on trade.

65. The charges applied to drug prices to support the social security funds of doctors and pharmacists and the National Drug Institute were reduced from 6.5 per cent, 2 per cent, and 5 per cent, respectively, to 4.6 per cent and 1.5 per cent and 1 per cent. Moreover, the first two will be based on the ex-factory price rather than the wholesale price.

66. The regulated profit margins for both the pharmacist and the wholesaler (35 per cent and 8 per cent, respectively) will be calculated on the (lower) ex-factory price rather than the wholesale price. The domestic producers' profit margin was reduced from 12.5 per cent to 8.5 per cent of the ex-factory price and the importers' profit margin (12.5 per cent) was eliminated.

67. NBG's regulatory capital has been increased since 1996 by: *i)* the transformation of a restructuring bond from the Government into a 5-year mandatory convertible bond (Dr 94.2 billion) counting for Tier I capital; *ii)* a 10-year subordinated bond issued in June 1997 (US$200 million) counting for Tier II capital; and *iii)* a Dr 80.4 billion rights issue.

68. The funding for OEK comes from a 1.75 per cent levy on workers' wages, and lending is primarily based on the income criteria of salaried workers (though in the past they were often made to employees of public enterprises). Prior to the introduction of new measures in November 1996, mortgage repayments on OEK sponsored loans were below 10 per cent of amounts due. Civil servants also receive subsidised housing, but their loans come from the highly profitable postal bank and the Consignment and Loan Fund, which are both exempt from required reserve requirements.

69. ETBA's overall balance sheet has also shrunk by about two-thirds, partly due to the sale of a substantial portfolio in Government securities (Dr 500 billion) that had been financed by bond issues of the bank (at a large loss in net interest income to ETBA).

70. Operating profits have improved due to a healthier loan portfolio, the repayment of its expensive Dr 1 trillion bond issues, and a 40 per cent reduction in personnel (to less than 500) through early retirements and transfers to regional governments. Total recapitalisation will have amounted to approximately Dr 800 billion during the period 1993-2000.

71. The most recent write-off of Dr 180 billion of loans to farm collectives is currently under investigation for violation of EC regulations.

72. Agricultural sector credits for cultivation and long-term loans are subsidised by 1 to 2 percentage points through a 1.2 per cent levy on the outstanding value of all other credits.

73. As part of the final agreement with the unions, the Government made a condition of sale the maintenance of the very generous labour regulations (including the retirement benefits), as well the retainment of all staff with more than two years work experience.

74. The four sales comprised: *i)* 37 per cent of the shares of the Bank of Macedonia-Thrace which were sold by a group of state-controlled banks to the privately owned Piraeus Bank in April 1998 for Dr 27.3 billion; *ii)* 33 per cent of the shares of the General Bank which were sold by the Greek Army's pension fund to the Interamerican insurance group and other international investors in April 1998 for Dr 14 billion; *iii)* 97 per cent of the shares of the Bank of Crete were sold by the Greek state to the privately-owned EuroBank in July for Dr 93 billion; and *iv)* 51 per cent of the shares of the Bank of Central Greece were sold in August by the ATE to the privately-owned Egnatia Bank.

75. The state also controls the specialised financial institutions, which are grouped independently of the commercial banks: the ATE, ETBA, the Postal Bank, and the Consignment and Loan Fund.

76. A special tax to support the budget is placed on loan interest (3 per cent), however it is paid by the borrower, and thus has no direct impact on the intermediation spread.

77. Other factors explaining the large intermediation spread are the high transaction costs linked to bankruptcy proceedings, and, more generally, large delays incurred by banks in collecting on bad credits through the judicial system.

78. The draft stock market reform presented to Parliament in August 1998 includes: *i)* replacing the requirement that a company post profits for the 3 years prior to entry by the requirement that it publish accounts for the past 3 years; *ii)* raising the minimum equity requirement to Dr 1.5 billion; and *iii)* eliminating the need for a newly entering firm to commensurately undertake a capital increase.

79. OECD (1998*a*), *The OECD Report on Regulatory Reform,* Volume I and II, Paris and OECD (1998*b*), ''International Energy Agency'', *Energy policies of IEA countries – Greece – 1998 review.*

80. The most costly operations have been related to loans for the following enterprises: Hellenic Shipyards (credits of Dr 100 billion, equivalent to 0.3 per cent of GDP, the Chalkidas cement company (credits of Dr 200 billion, equivalent to 0.6 per cent of GDP), and the Pieriki-Patraíki textile mills (credits of Dr 100 billion equivalent to 0.3 per cent of GDP).

81. Hellenic Shipyards survives on public contracts, most importantly with the Navy for the construction of frigates. In 1996, 49 per cent of the company was ''sold'' to the employees, however, management was contracted out to an international operator (Brown and Root).

82. An exception is the dividend from the national lottery organisation, which is considered a public enterprise.

83. KEPE (1996), *Inter-entity Arrears of the Public Sector,* Athens.

84. Excess personnel in public enterprises has also been the result of judicial decisions which have provided indefinite contracts to workers on temporary contract.

85. The age limits are slowly being raised. They will remain generous for those hired prior to 1982 (*e.g.* a pension at age 55 with 25 years of contributions for men), but be less generous for those hired after 1982, for whom the retirement age is gradually raised from 60 to 65 over a 10 year period. The regulations are more relaxed for women, especially those with under-age children. For those hired after 1992, pension benefits are identical to those offered by the state to private-sector employees.

86. See the OECD (1996) *Economic Survey of Greece.*

87. Employers' contributions are almost double those in the private sector (28 per cent *versus* 16 per cent).

88. The implicit pension liabilities are an acute problem for the Public Power Corporation (DEH), as the pension system is integrated into the operating accounts (*i.e.* it is not a separate fund) and employer contributions for many years were used to fund infrastructure projects. In 1997, DEH had to contribute Dr 50 billion to cover pension payments. In the case of OTE, its pension fund has been burdened with the payment of pensions of a large group of employees of ELTA and OSE. It suffered a deficit of Dr 28 billion in 1997 despite receiving a Dr 10 billion injection from the parent company.

89. To meet the end year deadline, most of the contracts were awarded prior to the signing of detailed contracts (including cost or product specifications). Ironically, despite the effort to distribute the contracts to domestic firms, with a view to job creation, it remains unclear whether the domestic firms in all cases have the required technological capabilities and may be forced to resort to imports.

90. An often cited example in Greece is the very good economic performance of the Neorion Shipyards which have doubled employment and become a mainstay on the island of Syros since their privatisation in 1994. They have since participated in the purchase of the Eleusis Shipyards in 1997.

91. Regarding two out of the four highly loss-making military-equipment manufactures, the Government is considering shifting their activities more towards commercial manufacturing and selling minority shares to private investors. For example, Hellenic Aviation Industry (EAB) may maintain commercial as well as military aircraft, and the tank manufacturer (ELBO) could focus more on vehicle construction (it already produces buses for urban transport in joint ventures).

92. OTE shares were sold in three public offerings and one market placement.

93. The other enterprises include the Greater Athens Water and Sewer Company (EYDAP), the Corinth Canal, the Thessaloniki International Fair, the Athens Port Organisation, the Thessaloniki Port Organisation, the Thessaloniki Water Organisation, the Thessaloniki Sewer Organisation, Olympic Catering, the Horse Racing Organisation, and the Athens Stock Exchange.

94. In fact, this limit stood at 25 per cent and was raised to 49 per cent prior to the November 1998 sale of a 10 per cent tranche of OTE.

95. The degree of segmentation is diminished by the fact that refineries can own marketing companies and marketing companies are permitted to own or lease the land and installations of petrol stations.

96. In addition, since 1993, marketing companies were prohibited from issuing recommended prices to retail dealers.

97. The State retains the right to introduce price ceilings on gasoline in areas where it believes monopolistic conditions prevail. These ceilings have been usually introduced during peak demand periods of the summer tourist season in areas outside of Athens and Thessaloniki.

98. The existing refineries' storage capacity are not a constraint to such practices. HP's refined product storage capacity alone is equivalent to about 100 days of total domestic sales.

99. Over 1995-97, HP has upgraded significantly its refinery capabilities through a Dr 60 billion investment programme, which included raising its: *i)* conversion capacity; *ii)* gasoline octane possibilities; and *iii)* desulpherisation capacity. Thus, its 125 thousand barrels per day Athens refinery (Asporpyrgos) was transformed into the most modern refinery in Greece as well as one of the largest and complex refineries in Europe.

100. For example, Greece's third refinery, the simple topping Petrola refinery owned by the Latsis group would have to make large investments to cover the costs of complying with EC Directive on environmental standards which requires a reduction of sulphur in oil products in 2000.

101. The 1987 bilateral agreement with Russia was revised in 1994 due to constructions delays on the gas pipeline in Greece. As compensation, the revised agreement permitted a Gazprom-Greek joint venture, to construct 5-8 branches of the network (depending on the number built) and a lignite fired plant at Florina. These contracts would be ''priced competitively'' but without competitive bidding procedures. In addition, Prometheus would be the sole seller for Greek demand exceeding the ''take or pay'' contractual limits. These reach a maximum of 3 bcm per year in 2016. Regarding the Algerian ''take or pay'' agreement, the contracted volume is for 0.6 to 0.7 bcm per year of LNG. No revisions have been made due to Greece's delays in taking delivery.

102. EU assistance is made through the REGEN, INTERREGEN and trans-European Networks programmes. The REGEN Programme was launched in 1990, *inter alia*, to enable natural gas to be introduced in the so called ''Objective 1 regions''.

103. Initially distribution was to have been developed jointly by the municipal authorities and the DEPA subsidiaries. However, the local authorities subsequently proved to lack the necessary expertise to participate.

104. Under the revised bilateral agreement between Russia and Greece, natural gas supplies above the contractual limit will be supplied solely by a joint venture (Prometheus) between a private Greek company and the Russian gas supplier (Gazprom). To circumvent these terms and to diversify supply, Greece has applied to use EU funds to construct a natural gas pipeline connection with Italy. An agreement was signed in September 1998 between DEPA and Italy's ENI to construct a 180 kilometer pipeline with an annual capacity of 5 bcm.

105. DEPA assumes part of the cost of converting energy equipment for industries, through rebates on gas prices for a five-year period. In addition, households and small enterprises can deduct from taxable income 75 per cent of the expenses to switch to natural gas.

106. Prices applying to households and commercial activities will result from the contracts negotiated by DEPA and the private investors of the gas supply companies (GSCs). The law allows for differences in prices across the 3 regions where gas distribution companies (GDCs) are present, to reflect variation in the costs of developing the network.

107. The agreement stipulates that DEPA would reimburse DEH, if its profits (after depreciation allowance) exceed 8 per cent of total income.

108. The 1995 natural gas law required the establishment of an independent ''Board of Energy Planning and Control'' by December 1997. However, as of September 1998, the Board is not operating, and in any event, its functions would be mostly of an advisory nature. Final decisions would remain with the Ministry of Development, which is also the majority owner of the main incumbents in the energy sectors.

109. Since the mid-1980s, a series of laws have aimed at fostering private participation in power generation. However, rather than opening up the electricity market in any major way and fostering private participation in power generation, they mainly set out exceptions to DEH's exclusive rights. Most importantly, a 1996 law allowed for a consortium to establish a co-generation station, with the use of the national grid to transport electricity to consortium members if they are located within 10 km of the station. In addition, a 1994 law permitted small private generators based on renewables. As a result of the restrictions to entry, the only important generating capacity not owned by DEH are a few industrial combined heat and power plants (the two main refineries and the state-controlled sugar company). It is worth mentioning that in 1990, the possibility for independent power producers to produce electricity for the grid was recognised, but was conditional upon DEH's consent. This provision was subsequently abolished.

110. Available capacity equalled peak-load demand in 1995. However, the demand for electricity has been growing rapidly, especially as the use of air conditioning is becoming more prevalent. Regarding security of supply, there are insufficient lines connecting the main consumption centre of Athens to the north of Greece where most of the generation occurs.

111. The European Union has committed ECU 117 million for interconnecting the electricity networks of Italy and Greece, through an undersea cable linking Oporto Badisco in Italy to Aetos in Greece.

112. The comparison uses International Energy Agency price calculations, which are based on revenues rather than listed prices, and are considered better estimates of prices as they capture the effects of discounts. The price comparisons are also adjusted for the recent real appreciation of the drachma by converting to a common price through an application of PPP exchange rates. Excluding the adjustment for PPP, electricity prices are near the OECD average for households and about 15 per cent below the OECD average for industry.

113. The aluminium company consumes roughly one-quarter of total industrial demand. Adjusting for this factor alone – and ignoring the low price received by the nickel producer (Larko) – would increase the average price of the remainder of the sector by 12 per cent.

114. Other uncompensated public service obligations include low electricity prices for certain categories of consumers (*e.g.* agriculture and families with many children) reflecting social and development policies.

115. The adjusted timetable applying to Greece defines eligible consumers as those consuming more than 40 Gwh per year from February 2001 (covering approximately 25 per cent of the market), and increasing to those consuming more than 20 Gwh per year from 2002 (covering approximately 28 per cent of the market), and then to those consuming more than 9 Gwh per year in 2005 (covering approximately 33 per cent of the market).

116. Lignite-fired plants will remain inexpensive as all but one unit of DEH's lignite-fired plants are exempted from reducing sulphur emissions under the EC Directive for large combustion plants, as they were built before the 1978 cut off date contained in the directive.

117. Pension contributions have been used to fund DEH's past investment, and implicit pension liabilities are reported to amount to Dr 1.5 trillion, and would roughly double the liabilities reported in DEH's balance sheet.

118. Stranded costs are those unamortised costs of prior investments and other decisions which were scheduled for recovery through previous state-owned monopoly rules but would not be recovered under competition.

119. At the same time, the EC granted a small additional implementation period to October 1997, compared with the original date of July 1996, with respect to the lifting of restrictions on the provision of the so-called ''already liberalised telecommunication services'' (*i.e.* leasing access to OTE's network for services such as data transmission through high capacity bandwidth lines (ISDN) and other value added services). The Greek Government has passed the conforming legislation in 1998.

120. The two GSM (digital) licences each cost Dr 30 billion (US$150 million) and provided exclusivity in the market for mobile phones for 8 years. In return for the high licence fee, the two operators received favourable interconnection charges; for mobile to fixed domestic connections, the charge is 5 per cent of airtime revenues and for fixed to mobile domestic connections, the charge is 96 per cent of the OTE tariffs. As a result, the mobile operators pay about 7 per cent of revenues to OTE which is about one-third the comparable share for other mobile telephone companies.

121. The CosmOTE PCN-1 800 licence (for Dr 14.2 billion) has resulted in a law-suit by the other mobile operators citing a breach of the exclusivity clause in their licensing agreement. The authorities' response has been that CosmOTE received a different type of licence than the GSM licenses awarded to the other two operators.

122. The weights in the CPI basket are based on a household expenditure survey. Since revenues from trunk and international calls exceed that of local calls, the rebalancing should have a dampening effect on average telephone prices. Obviously, a reduction of only the more expensive calls would have an even larger downward influence on the CPI.

123. OTE has requested the regulatory authorities to increase interconnection charges for mobile phones, which are very low compared to other mobile operators (and based on a share of mobile company call revenues). However, OTE is having difficulties substantiating its claim due to the lack of a detailed accounting system to determine the cost of supplying fixed-line interconnections. In the event, the EC is pushing for cost-based interconnection charges for the EU. These would bring currently prevailing interconnection charges in the EU down towards the levels existing in Greece.

124. Lack of funds for infrastructure investment could be partly attributed to the use of part of OTE's profits prior to 1992 to finance the operating deficit of the Greek Post (ELTA).

125. The EU's contribution to the Crash programme was originally envisaged to be about 50 per cent of the programme's ECU 260 billion cost, but it was subsequently reduced to ECU 71 billion due to the authorities' inability to implement the planned measures on schedule.

126. Based on three indicators of service quality, OTE's performance has made great strides since 1990. The time required to obtain a phone has fallen from 47 months in 1990 to two weeks in 1997 and is targeted to fall to three days by 2000. The failures to connect per 1 000 calls has fallen from 55 in 1990 to 31 in 1997, and is targeted to decline to 20 by the year 2000. Finally, repair service response time within 24 hours has risen from 60 per cent in 1990 to 70 per cent in 1997, and is targeted to rise to 95 per cent in 2000.

127. The early retirement programme will be quite expensive. It is expected that staff will be reduced by 5 000 at a cost of Dr 81 billion. However, after taking into account salary savings, the retirement programme will be financially beneficial for OTE.

128. OTE has made the following foreign direct investments during the past two years: the purchase of 20 per cent of the Serbian telecommunications operator Telekom Serbija for $400 million; the purchase of 90 per cent of the Armenian operator (ArmentTel) for $142 million and the purchase of 35 per cent of the Romanian telecommunications operator (RomTelecom) for $675 million. In addition, OTE or its subsidiary Hellascom, have undertaken much smaller operations in Jordan, the Ukraine, Lithuania, and Georgia. Finally, OTE is currently negotiating the purchase of a large share of the Moldavian telecommunications operator.

129. The EC has ruled that the monopoly provided to OTE and ERT for cable television is a constraint on the development of auxiliary markets, and the Greek authorities envisage appropriate legislative changes in 1998, at the time of passage of a telecommunications law which harmonises Greek law fully with the requirements of the EC and unifies current legislation under one framework.

130. Even with conveyance loss of about 10 per cent, the system provides a safe yield of 530-580 million cubic meters per year; 460 million cubic meters by the low cost gravity system of Mornos, and another 140 million cubic meters by the Ylikis system. Current consumption is 285 million cubic meters per year and demand is forecast (under the assumption of constant real prices) to be between 400-535 million cubic meters per year by 2010. Peak demand could be equivalent to 657 million cubic meters on an annualised basis, but for short periods that could be met through storage facilities.

131. Since 1992, EYDAP basic water prices have not been raised, excluding adjustments for the introduction of VAT, the inclusion of a long-standing surcharge for the financing of the Mornos projects into the tariff in 1995, and the introduction of a minimum consumption level so as to raise the payments from offices.

132. These estimates were made by the consultant group, Knight Piésold, in *EYDAP Master Plan: Water Division*, December 1996.

133. The consultant's reports suggest only a 75 per cent increase in the price of water charged to local authorities. This would not result in any significant price increase to the final consumer, assuming the mark-up is commensurately reduced.

134. Another source of revenue loss was the Government's suppression of a special contribution (two-thirds of a 3 per cent tax on construction sites), which provided about 10 per cent of EYDAP's revenue. As EYDAP expects to receive an equivalent sum as a transfer from the Budget, it includes this amount in its accounts.

135. This debt has been accumulated since 1992, when EYDAP received a capital increase and the Government took over its outstanding debt (both operations totalling Dr 130 billion).

136. The Government is also reconsidering the appropriate industry structure, and whether the sewerage and water functions should be separated. Economies of scope on pipeline operations and co-ordination of the placement of infrastructure investment – so as to avoid externalities regarding effluent control – suggest a preference for a joint company, and the Government announced in September 1998 that it would merge the Thessaloniki water and sewerage companies.

137. In view of the long life of the infrastructure projects in this sector (75-100 years), two-tier pricing, comprising a fixed and usage related charge is not considered appropriate.

138. The Olympic Group comprises Olympic Airways, Olympic Aviation, Olympic Tourism, and Olympic Catering.

139. The domestic market was opened in 1992, just prior to the timetable contained in the EC directives on air transport liberalisation (EC/2408/92).

140. Handling charges amount to 10 to 20 per cent of total operating costs on a typical two hour intra-European flight (see OECD 1997*c*). Self-handling is permitted, but is not an economically efficient alternative.

141. See the Official Journal of the European Communities (1994).

142. The early retirement programme has been problematic. Its cost has been considered by the EC as unauthorised public assistance to the carrier (Dr 20 billion) and it has resulted in the loss of many qualified personnel (1 500).

143. On the other hand, operating costs are decreased by low depreciation costs arising from the old age of the fleet (an average age of 16 years compared with an IATA average of 11 years).

144. The Dr 43 billion improvement in OA's net result is calculated from a ''no-action'' scenario. Controlling labour cost developments is the main element (expected to contribute for 60 per cent of the improvement). It includes a three-year wage freeze from the end-1997 level (*i.e.* after a 50 per cent increase over the previous 2 years), the introduction of more flexible working hours (with a maximum of 10 working hours per day and 48 per week), the re-examination of certain non-wage costs (in particular the food allowance). An increase in revenues in drachma, mainly arising from the March 1998 devaluation, is projected to contribute for 20 per cent of net result improvement (most handling and international flights are invoiced in foreign currencies). The renewal of the fleet, the redesigning of the route network and the change in OA's marketing policy, would contribute the remainder.

145. However, a consortium containing Olympic has received the tender to operate the (monopoly) hydrant fuelling system at the new Spata airport.

146. In addition, many airlines have preferred the flexibility offered by leasing compared to purchasing new planes. Since OA is about to embark on a large new aircraft acquisition programme (Dr 1.2 trillion) it may wish to consider switching to a leasing programme instead.

147. The new airport is being constructed under a BOT operation, and fee setting will be the responsibility of the private-sector operator (rather than the Civil Aviation Service). Thus, its airport charges are likely to be more expensive than the current one since it has to recover the cost of the investment.

148. Passenger and cargo were both near historical highs in 1987. Since then, passenger demand fell, but then almost returned to its 1987 level (1 930 million kilometre passengers in 1997 *versus* 1 973 million in 1987). However, cargo demand in 1997 has fallen to about one-half the 1987 level (331 million kilometre tonnes in 1997 *versus* 599 million in 1987).

149. Passenger rates are the lowest in the EU (about Dr 7 per kilometre passenger *versus* Dr 40 in the United Kingdom for British Rail, Dr 26 in Germany for DB, Dr 33 in Sweden for SJ, and Dr 13 for Renfe in Spain). Cargo rates are closer to rates in other countries.

150. Under the 1972 legislation, the State is responsible for all operating losses of OSE.

151. Labour costs are also pushed up by a plethora of bonuses and overtime payments, which appear to form a disproportionate share of compensation, and by generous pensions, for which the average retirement age is 56.

152. The new management also envisages complying fully with all EC directives for rail transportation in 1998, including the separation of the accounts for operating and infrastructure and the creation of a special account for past losses (91/440), and the specification of conditions and terms for third party access (95/18 and 95/19).

153. Until 1999, funding will be from the CSF II and the Cohesion Fund (with 60 per cent and 85 per cent participation, respectively, of EU funds). This investment programme will add 100 km of double lines to the 300 currently existing (raising them to 25 per cent of the network with ordinary width), reduce metric gauge lines by 10 per cent (887 km to 807 km), increase the electrification of the network to 400 km of double rail from zero and from 76 km to 196 km on single rail, double high speed track capacity (above 200 km per hour) to 200 km of track, and almost double automatic signalling to 840 km of track.

154. A subsidiary of OSE was recently created (ERGOSE) and made responsible for the infrastructure construction.

155. Of OSE's 18 lines, the business plan considers 9 not to be viable, three to have the potential to be profitable, and another 6 to be questionable. On some of the non-viable lines, capacity utilisation is in single digits.

156. In fact, the OASA has three subsidiaries, ETHEL (buses), HSAP (the one rail subway line from Athens to Piraeus), and HLPAP (trolley buses).

157. Inter-city bus service is offered by private operators under the loose control of an umbrella organisation (KTEL).

158. The government guarantees to banks which have fallen due have been paid and the claims converted to those of the Government on OASA. However, it is a debt that can never be repaid. About half the debt is due to the cost of re-nationalising the bus company in 1993 following its privatisation in 1992.

159. Losses also arose from the fact that during most of the 1980's, urban travel was free of charge between 5.00 and 8.00 am. This practice was discontinued in the 1990s. Another source of revenue loss is ticket payment evasion, which has reportedly become more acute as the frequency of ticket checks has declined. In Thessaloniki, where ticket prices were also Dr 75 in 1996, revenue per passenger was Dr 63 *versus* Dr 47 in Athens.

160. Additional savings in personnel and vehicles could be made if the number of kilometres and length of time it took buses and trolleys to go to and from their respective routes were reduced through a more efficient allocation of depots and design of routes.

161. Data for the private company operating the buses in Thessaloniki are sparse. However, its operating costs per passenger are higher than for the bus subsidiary of OASA. This fact may reflect that the franchise contract is based on a guaranteed rate of return of about 15 per cent, which may not be providing appropriate cost efficiency incentives.

162. Though ELTA does not receive a specific payment for providing a public service obligation, its financing needs were covered by a transfer from OTE until 1994, when this practice was terminated. ELTA also receives transfers from the Postal Bank.

163. The EC Directive requires the elimination of all subsidies, as well as recourse to cross-subsidies. In the event the network is considered to be expensive, it permits the introduction of a fee to be placed on all market participants in order to finance public service obligation.

164. ELTA's non-bank liabilities include social security contributions to OAED (Organisation for Manpower Development), and arrears to DEH.

165. OECD (1998*a*), *The OECD Report on Regulatory Reform.*

166. Some enterprises are assumed to continue to receive subsidies to finance their public service obligations. Their profit margins are negative before they receive these subsidies.

167. The second-round multiplier effects are based on the results of similar exercises undertaken for the larger OECD countries. This assumption is not expected to greatly influence the estimated outcome.

Glossary of acronyms

ALMP	Active labour market policies
ATE	Greek Agricultural Bank
BOT	Built-Operate-Transfer
CEECs	Central and Eastern European Countries
CSF II	Community Support Framework II
DEH	Public Power Corporation
DEKA	Public Enterprise for Liquid Assets
DEPA	Public monopoly in natural gas
EAB	Hellenic Aviation Industry
EKEPIS	National Centre for Certification
ELBO	Tank manufacturer
ELTA	Greek Post
ERGOSE	Infrastructure subsidiary of Greek Rail
ERT	National Radio and Television Company
ETBA	Hellenic Bank for Industrial Development
EThEL	Athens bus corporation
EYDAP	Greater Athens water and sewerage company
FYROM	Former Yugoslav Republic Of Macedonia
GDCs	Gas Distribution Companies
GSCs	Gas Supply Companies
GSM	Global system for mobile communication
HDAT	Electronic secondary market for government securities on Athens Stock Exchange
HLPAP	Athens Trolleys
HP	Hellenic Petroleum
HSAP	Old Athens subway
IATA	International Air Transport Association
IKA	Social insurance institute
IRO	Industrial Restructuring Organisation
KEK	Vocational training programmes
KTEL	Private inter-city bus service
MPC	Monetary Policy Committee
NBG	National Bank of Greece
NLG	Natural liquified gas
NAPE	National Action Plan for Employment
OA	Olympic Airways
OAED	Organisation for manpower development
OASA	Group of public enterprises involved in urban transport in Athens area
OATh	Thessaloniki Sewerage Organization
OEK	Organisation for Workers Housing
OMED	Organisation of Mediators and Arbitrators
OSE	Greek Rail
OTE	National Telephone Company
OYTh	Thessaloniki Water Organization
PPP	Purchasing power parity
RCA	Revealed comparative advantage
RTK	Revenue tonne kilometer
TSMEDE	Pension fund for engineers

Bibliography

Aiginger, K. (1995), ‘‘The use of unit values for evaluating the competitiveness of nations’’, WIFO Institute, Vienna.

Aiginger, K. (1997), ‘‘The use of unit values to discriminate between price and quality competition’’, Cambridge, Journal of Economics, No. 21, pp. 571-592.

European Communities (1994), Official Journal of the European Communities, No. L273/22, October 25.

KEPE (1996), Inter-entity Arrears of the Public Sector, Athens. In Greek.

OECD (1995), *Economic Survey of Greece.*

OECD (1996), *Economic Survey of Greece.*

OECD (1997*a*), Science, Technology and Industry, *Scoreboard of indicators, 1997,* Paris.

OECD (1997*b*), *Economic Survey of Greece.*

OECD (1997*c*), *The Future of International Air Transport Policy: Responding to Global Change.*

OECD (1998*a*), *The OECD Report on Regulatory Reform.*

OECD (1998*b*), International Energy Agency, *Energy policies of IEA countries – Greece – 1998 Review.*

Annex I

The Aiginger Index and revealed comparative advantages

Karl Aiginger developed an indicator to assess whether the external performance of a country depends on pure price competitiveness (*i.e.* low costs) or on non-price competitiveness (*i.e.* quality or innovation).* The methodology is based on the unit value of exports and imports, and enables to discriminate between manufactured good markets in which the quantity traded depends more on price competitiveness and those markets in which the quantity traded reflects more quality competition.

Price competition dominates in those markets where products are homogeneous, and for which the production technique is not country-specific. For these products, unit values should reflect average costs and countries with lower costs should be net exporters, while countries with higher costs should be net importers. Therefore, in markets where price competition dominates, when the unit value of exports exceeds (inversely, is lower than) the unit value of imports, the country should show a trade deficit (inversely, a surplus) for this product in volume terms.

On the other hand, when quality and product innovation are the important competitive factors, a higher unit value will reflect the ability to set prices. A country will be a net exporter despite the fact that it has higher unit values, because quality matters. Therefore, in product markets where quality competition dominates, when unit value of exports exceeds (inversely, is lower than) unit value of imports, the country should show a trade surplus (inversely, a deficit) for this product in volume terms.

Based on these assumptions, Aiginger built an index based on the observation of unit values as well as of flows of manufactured export and import across a sample of 18 countries (the United States, EU-12, Japan, Canada, the Czech Republic, Hungary and Poland) for year 1993. The index measures the degree to which a products competes on price or on quality. It ranges from 1 to 166, with the higher the value of this indicator, the higher the degree of price competition.

The Table A1 provides Greece's top export products (accounting for a cumulated 70 per cent of total merchandise exports) and the value of the Aiginger index for

* Aiginger K. (1997), ''The use of unit values to discriminate between price and quality competition'', Cambridge Journal of Economics, No. 21, pp. 571-592.

Table A1. **Detailed structure of Greek exports, 1995**

Main comparative advantages

SITC 3 code	Merchandise exported	RCA[1]	Quality ranking Aiginger Index	Share of exports	Cumulative share
845	Articles of apparel, of textile fabrics, n.e.s.	5.91	159	6.58	6.58
421	Fixed vegetable fats and oils, crude, refined, fractional	5.26	–	5.50	12.08
334	Petroleum oils or bituminous minerals > 70% oil	4.28	–	5.91	18.00
844	Women's clothing, of textile, knitted or crocheted	4.11	163	4.42	22.41
057	Fruits and nuts (excluding oil nuts), fresh or dried	3.83	–	4.40	26.82
263	Cotton	3.52	–	3.60	30.42
058	Fruit, preserved, and fruit preparations (no juice)	3.29	–	3.37	33.79
684	Aluminium	3.05	154	3.98	37.77
661	Lime, cement, fabricated construction material (excluding glass, clay)	3.02	126	3.07	40.84
121	Tobacco, unmanufactured; tobacco refuse	2.44	–	2.82	43.66
056	Vegetables, roots, tubers, prepared, preserved, n.e.s.	2.13	–	2.40	46.06
842	Women's clothing, of textile fabrics	1.87	140	2.49	48.55
931	Special transactions and commodities not classified	1.81	–	1.83	50.38
651	Textile yarn	1.53	108	2.42	52.80
848	Articles of apparel, clothing accessories, excluding textile	1.28	163	1.65	54.45
773	Equipment for distributing electricity, n.e.s.	1.09	146	1.21	55.66
285	Aluminium ores and concentrates (including alumina)	0.89	–	0.89	56.55
673	Flat-rolled products, iron, non-alloy steel, not coated	0.78	117	1.37	57.92
034	Fish, fresh (live or dead), chilled or frozen	0.78	–	1.07	58.99
278	Other crude minerals	0.74	–	0.84	59.83
054	Vegetables; roots and other edible vegetable products	0.73	–	1.18	61.00
674	Flat-rolled products, iron, non-alloy steel, coated, clad	0.73	117	1.24	62.24
112	Alcoholic beverages	0.72	–	1.74	63.98
843	Men's or boy's clothing, of textile, knitted, crocheted	0.57	159	0.71	64.70
042	Rice	0.50	–	0.54	65.23
714	Engines and motors, non-electric; parts, n.e.s.	0.43	57	0.73	65.97
658	Made-up articles, of textile materials, n.e.s.	0.32	159	0.66	66.62
122	Tobacco, manufactured	0.30	–	0.88	67.50
792	Aircraft and associated equipment; spacecraft, etc.	0.24	24	0.63	68.13
682	Copper	0.23	96	1.45	69.58
211	Hides and skins (except furskins), raw	0.23	–	0.33	69.91
692	Metal containers for storage or transport	0.21	96	0.39	70.30

1. Revealed comparative advantage. The RCA indicator is calculated as $(Xi/X - Mi/M) \times 100$, where Xi(Mi) and X(M) are exports (imports) of manufactured goods i and exports (imports) of total manufactured goods respectively.

Source: OECD, *Foreign Trade Statistics.*

manufactured goods. The products are ranked by the value of their revealed comparative advantage (RCA). This indicator of competitive position by industry has the advantage of being independent of both the degree of trade openness of the country and its overall trade performance. Furthermore, for any level of product classification, the RCA is bound

between 100 and –100 and the sum of RCAs is equal to zero, thus facilitating international comparisons. Also provided in the table are the share in total Greece's merchandise exports for each item listed and cumulative shares. For the case of Greece, one notes that manufactured products with a significant competitive advantage, as represented by their RCAs, are mostly competitive on prices rather than quality.

Annex II

Calendar of main economic events

1997

June

Second offering of OTE's shares (12 per cent of the share capital, raising Dr 298 billion).

Privatisation of Elefsina Shipyards, raising Dr 30 billion.

The European Commission endorsed a government plan to grant Scaramanga (shipyards) investment-related State assistance.

Reduction of regulated profit margins in the drug industry (pharmacists, wholesalers, domestic producers and importers). Several charges placed on pharmaceuticals to finance the National Drug Institute and the pensions of medical personnel were also cut.

Government issued first 10-year fixed rate bonds.

July

Liberalisation of foreign exchange deposits for Greek residents and reduction of required reserve requirement on these deposits from 70 per cent to 12 per cent (similar to that on drachma deposits).

Legislation introducing primary and secondary market for government bonds was passed.

September

Greece awarded organisation of 2004 Olympic Games.

The tax placed on the value of all domestic loans (EFTE) was reduced from 4 per cent to 3 per cent and also required for domestic foreign-currency denominated loans, while the base of a 1.2 per cent tax placed on the loan interest (which favours lower-interest, foreign-currency denominated credits), was changed to the value of the loan.

October

Bank of Greece temporarily raises overnight rate (on marginal increases in deposits with the central bank) to 170 per cent.

Government reduced excise tax on heating oil by 33 per cent.

November

Social partners agreed on a medium term strategy for reforming the labour market, resulting in the ‘‘Pact of confidence established between the Government and social partners on the threshold of the year 2000’’.

December

The Parliament approved the 1998 Budget, targeting a decline in the central government deficit from 6.2 per cent of GDP in 1997 to 4.4 per cent of GDP in 1998, compatible with the convergence plan target of a general government deficit of 2.4 per cent of GDP.

Parliament approved legislation granting Central Bank independence and making price stability its primary objective.

The Government distributed without competitive bidding, large supply contracts, to domestic producers worth Dr 1.3 trillion (equivalent to 3½ per cent of GDP), just ahead of the EC Directive (EC 9338/1994) deadline requiring open competitive bidding.

1998

January

Process of registration of illegal immigrants commences.

February

Legislation introduced that required labour regulations in loss-making public enterprises to be made more flexible. The Government would act unilaterally if negotiation with public sector unions fail to yield agreement within six months.

Moody placed Greece’s long-term credit ceiling for debt and bank deposits on watch for a possible downgrade.

March

Entry of the drachma into the European exchange rate mechanism (ERM). The central rate is fixed at 357 drachma per ECU (a 12.3 per cent devaluation against the latest market rate). Following the entry of the drachma into the ERM, the Government

announced a number of accompanying measures aimed at containing inflation, strengthening fiscal adjustment, and accelerating structural reform.

Two presidential decrees on natural gas were enacted adapting Greek to EU legislation, especially the provisions of EU Directive 93/38 on the prospective status governing Greek state suppliers.

The Duty Free Shops were listed on the Athens Stock Exchange through an initial public offering for 20 per cent of the capital.

The National Action Plan for Employment is presented to the European Commission.

April

Introduction of a positive list of pharmaceuticals which limit drug reimbursement by the social security system to some 3 000 drugs.

Sale of 37 per cent of the shares of the Bank of Macedonia-Thrace by a group of state-owned banks to the privately-owned Piraeus bank for Dr 27.3 billion.

Sale of 33 per cent of the shares of the General Bank by the Greek Army's pension fund to the Interamerican insurance group and other institutional investors for Dr 14 billion.

Bank of Greece issues its first monetary policy report following independence and announces a move towards inflation targeting.

Capodistrias Law passed which reduces the number of local governments significantly.

May

Social partners reached a two-year collective agreement, including, in particular, minimum wage increases for 1999 and 2000.

Approval of a restructuring plan for the air carrier Olympic Airways (OA).

Government issued first 15-year fixed-rate bonds.

June

The Government presented a labour bill aimed at increasing flexibility of the labour market.

The European Commission agreed to raise the share of EU contribution to projects under the Community Support Framework (Delors II package) from 66.3 to 69.2 per cent.

First regional tax audit centers began operations using TAXIS (Tax information system).

July

The Bank of Greece lowered the reserve requirement on foreign currency deposits held by Greek seamen and foreign workers at commercial banks from 70 to 60 per cent.

Further opening of the Greek air traffic market to competition, including the removal of any constraints on charter services and full access to the Greek islands, as well as the elimination of Olympic Airways' monopoly on ground handling.

Sale of 97 per cent of the shares of the Bank of Crete by the Greek State to the privately-owned EuroBank for Dr 93 billion.

Public offering for 23 per cent of Hellenic Petroleum's share capital, which raised Dr 90 billion.

The European Commission unfroze the State aid package for Olympic Airways, and Dr 14 billion was disbursed.

Attempt to sell 80 per cent of Hellenic Duty Free Shops was unsuccessful.

Approval of a restructuring programme for the Greek Post (ELTA).

August

Sale of 51 per cent of the shares of the Bank of Central Greece by the ATE to the privately-owned Egnatia Bank.

The take-over call for 51 per cent of Ionian bank shares issued by the Commercial Bank of Greece proved inconclusive.

September

Tendering of a 67 per cent stake of Hellenic Duty Free Shops (for the second time). This was also unsuccessful.

The EU Economy and Finance Minister Council has approved Greece's application for including the drachma into the ERM-2 as of January 1, 1999.

Approval of a five-year restructuring programme for the National Broadcasting Corporation (ERT).

October

ECOFIN approved the new Convergence Plan for the period 1998 to 2001.

A bill was passed widening the access to the Greek stock market to new enterprises. It replaced the obligation of 5 consecutive years of profits by the mandatory publication of balance sheets for 3 consecutive years prior the date of application for listing.

The Government announced a tax cut on heating oil and on gasoline.

Merger of the NBG with the National Mortgage Bank of Greece.

Approval of a restructuring plan for the rail transport company (OSE).

Approval of the business plan for Athens Urban Transport Authority (OASA).

The Government issued tax free bonds (if held to maturity) and bonds exchangeables into shares of public enterprises on favourable terms.

November

1999 Budget presented to Parliament. Deficit target is 1.9 per cent of GDP.

The Government announced a tax cut on cars.

December

Standard and Poors upgrades Greece's credit rating on external debt from BBB- to BBB.

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

OECD PUBLICATIONS, 2, rue André-Pascal, 75775 PARIS CEDEX 16
PRINTED IN FRANCE
(10 98 16 1 P) ISBN 92-64-16002-7 – No. 50449 1998